Introduction to Data Science 1MS041

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Preface

These lecture notes were developed for the course Introduction to Data Science. They provide the probability, statistics, and linear algebra needed for the data-science methods studied in the course. The notes assume little prior probability theory, but they do assume comfort with elementary mathematical notation and arguments.

A central distinction throughout the notes is the distinction between a statistical model and an estimation procedure. The model states assumptions about the data. The procedure is the rule or algorithm used to construct an estimate or prediction from observed data. For example, a linear predictor may be useful even when the unknown regression function is not exactly linear. Its performance must then be evaluated as an approximation rather than justified by assuming that the truth belongs to the fitted class.

Concentration inequalities provide finite-sample probability bounds in many of the later chapters. We use them to study estimation error, held-out model evaluation, random projection, and other data-science calculations. The emphasis is on explicit assumptions and quantitative conclusions that can be checked in concrete models.

How to use the notes in the course. Each numbered chapter begins with a course guide that states the lecture in which the material is taught, the exact required sections, the prerequisites, and the optional or advanced material. Required material is part of the core course and may be used in assignments and assessment. Optional material gives examples or extensions that are not assumed later. Advanced material is outside the assessed core unless the course guide explicitly names a result as required. The lecture calendar and these chapter guides use the same 15-lecture sequence.

Each numbered content chapter ends with exercises. Exercises marked "computational" use only the Python tools introduced by that point in the course notebooks. Optional exercises may use material from an optional section and are not part of the required route through the chapter.

Topics

- Axiomatic probability: Chapters 1 and 2. These chapters introduce probability models, random variables, distributions, and expectations.

- Concentration of measure: Chapter 3. This chapter develops elementary tail bounds, modes of convergence, the law of large numbers, and the central limit theorem.

- Risk: Chapter 4. This chapter formulates regression, classification, and parameter estimation as risk-minimization problems.

- Fundamentals of estimation: Chapter 5. This chapter introduces point estimators, bias, standard error, mean squared error, and consistency.

- Random-variable generation: Chapter 6. This chapter introduces pseudorandom generators and methods for sampling from specified distributions.

- Finite Markov chains: Chapter 7. This chapter models dependent sequential observations on a finite state space and studies simulation, hitting times, stationarity, and reversibility.

- Pattern recognition: Chapter 8. This chapter uses linear classifiers, the perceptron, and kernels to introduce the geometry and algorithms behind classification.

- Model evaluation: Chapter 9. This chapter separates fitting from evaluation. It develops held-out guarantees, classification and regression metrics, cross-validation, and the role of stability when validation data are reused.

- High-dimensional data: Chapters 10 and 11. These chapters study high-dimensional geometry, random projection, singular value decomposition, principal component analysis, and K-means clustering.

- Complexity and a priori bounds: Chapter 12. This optional chapter develops finite-class and VC-theory guarantees that do not rely on an independent test set. It is not part of the current assessed core.

Contents

List of Figures

Chapter 1

Probability Model

Course guide

Scheduled lecture(s): Lecture 1 (and review in Lecture 10)
Required for those lectures: Sections 1.1–1.3 and the core exercises in Section 1.5
Prerequisites: None beyond elementary set notation
Optional or advanced: Section 1.4 and the exercise marked optional are optional.

1.1 Experiments

Ideas about chance and random behaviour arose from games of chance long before they were studied mathematically. In the middle of the seventeenth century, correspondence between Fermat and Pascal helped establish the mathematical study of probability.

Definition 1.1 (Experiment, outcome, and event)**.** *An* **experiment** *is a procedure with distinct, well-defined possible results. A possible result is an* **outcome***. The set of all outcomes is the* **sample space** *and is denoted by* Ω*. Thus an outcome is an element* $\omega \in \Omega$*.*

We begin with finite sample spaces. In this setting an **event** *is a subset* $A \subseteq \Omega$*. If the observed outcome is* ω*, then* A *occurs exactly when* $\omega \in A$*. The outcome* ω *and the singleton event* $\{\omega\}$ *are different objects: the first is an element of* Ω*, while the second is a subset of* Ω*. The singleton* $\{\omega\}$ *is called a* **simple event***.*

For $n \in \mathbb{N}$ *with* $n \geq 2$*, events* $A_1, \ldots, A_n$ *are* **mutually exclusive***, or* **pairwise disjoint***, if no two can occur simultaneously. Equivalently,* $A_i \cap A_j = \emptyset$ *whenever* $i \neq j$*.*

Example 1.2. *Some standard experiments and their sample spaces are:*

- *inspecting a light bulb,*

$$\Omega = \{\textsf{\textit{Defective}}, \textsf{\textit{Non-defective}}\};$$

- *tossing a coin once,*

$$\Omega = \{\texttt{H}, \texttt{T}\};$$

- *rolling a six-sided die,*

$$\Omega = \{1, 2, 3, 4, 5, 6\}.$$

For the die, the events

$$A = \{1, 3, 5\}, \qquad B = \{2, 4, 6\}$$

are the events that the result is odd and even, respectively. The simple events are $\{1\}, \dots, \{6\}$.

Definition 1.3 (Trial and product experiment)**.** *A* **trial** *is one performance of an experiment and produces one outcome.*

*Let $n \in \mathbb{N}$ with $n \geq 1$. An n-***product experiment** *consists of performing n trials and recording the ordered tuple of outcomes. If the original experiment has sample space Ω, then the product experiment has sample space Ω^n. Forming a product experiment does not by itself assert that the trials are independent.*

Example 1.4 (Tossing a coin n times)**.** *Suppose that we toss a coin n times and record* `H` *for Heads and* `T` *for Tails. For $n = 3$ the sample space is*

$$\begin{aligned}\Omega &= \{\mathtt{H}, \mathtt{T}\}^3 \\ &= \{\mathtt{HHH}, \mathtt{HHT}, \mathtt{HTH}, \mathtt{HTT}, \mathtt{THH}, \mathtt{THT}, \mathtt{TTH}, \mathtt{TTT}\}.\end{aligned}$$

The event that at least two Heads occur is

$$A = \{\mathtt{HHH}, \mathtt{HHT}, \mathtt{HTH}, \mathtt{THH}\}.$$

The event that no Heads occur is $B = \{\mathtt{TTT}\}$. The event B contains one outcome, so it is a singleton event; it is not itself the outcome `TTT`*.*

1.2 Probability

Probability is a mathematical model for uncertainty. Relative frequencies motivate the model, while the axioms below specify the mathematical object we will use.

Idea 1.5 (Relative frequency)**.** *For one coin toss, let $H = \{\mathtt{H}\}$ and $T = \{\mathtt{T}\}$ be the Heads and Tails events. After n tosses, let $N(A, n)$ be the fraction of trials on which the event A occurred. More precisely,*

$$N(A, n) := \frac{\#\{1 \leq j \leq n : A \text{ occurs on trial } j\}}{n}. \tag{1.1}$$

For example, 487 Heads in 1000 tosses gives $N(H, 1000) = 0.487$. Since exactly one of H and T occurs on each toss,

$$N(H, n) + N(T, n) = 1.$$

If the experiment is repeated under the same conditions and the trials do not influence one another, relative frequencies often settle near stable values. This motivates assigning a number $\mathbb{P}(A)$ to each event and expecting

$$N(A, n) \approx \mathbb{P}(A) \qquad \text{when } n \text{ is large.}$$

This is motivation, not an assertion about every possible sequence of outcomes. The strong law of large numbers in Definition 3.26 gives a precise almost-sure convergence statement under independent and identically distributed assumptions.

1.2.1 Collections of events

For every finite or countable example in the required part of this chapter, we use the collection

$$2^{\Omega} := \{A : A \subseteq \Omega\}.$$

Thus every subset of Ω is an event. The following definition lets us use the same notation for more general sample spaces. Students are not expected to construct complicated event collections in this course.

Definition 1.6 (Event collection (σ-algebra)). *A collection $\mathcal{F}$ of subsets of Ω is a* **σ-algebra**, *or* **σ-field**, *if*

1. $\Omega \in \mathcal{F}$;
2. $A \in \mathcal{F}$ *implies* $A^c := \Omega \setminus A \in \mathcal{F}$;
3. $A_1, A_2, \ldots \in \mathcal{F}$ *implies*

$$\bigcup_{i=1}^{\infty} A_i \in \mathcal{F}.$$

The members of $\mathcal{F}$ are called events.

The point of these rules is that the usual set operations produce events again. Indeed, $\emptyset = \Omega^c \in \mathcal{F}$, and De Morgan's laws show that countable intersections belong to $\mathcal{F}$. In particular, if $A, B \in \mathcal{F}$, then $A \cap B$, $A \cup B$, and $A \setminus B = A \cap B^c$ are events.

Definition 1.7 (Probability). *Let Ω be a sample space and let $\mathcal{F}$ be a σ-algebra on Ω. A* **probability measure** *is a function $\mathbb{P} : \mathcal{F} \to [0, 1]$ such that*

1. $\mathbb{P}(\Omega) = 1$;
2. *if $A_1, A_2, \ldots \in \mathcal{F}$ are pairwise disjoint, then*

$$\mathbb{P}\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} \mathbb{P}(A_i).$$

The triple $(\Omega, \mathcal{F}, \mathbb{P})$ is called a **probability triple**, *or probability model.*

For the finite calculations in this chapter, the second axiom says that the probabilities of mutually exclusive events add. Its countable form is needed later for random variables with countably many possible values.

Example 1.8 (A finite probability model). *Let $m \in \mathbb{N}$ with $m \geq 1$, set $\Omega = \{\omega_1, \ldots, \omega_m\}$, and choose numbers $p_1, \ldots, p_m \geq 0$ satisfying $\sum_{i=1}^{m} p_i = 1$. On $\mathcal{F} = 2^{\Omega}$, define*

$$\mathbb{P}(A) := \sum_{\{i : \omega_i \in A\}} p_i.$$

This gives a probability model. For a fair die, $p_i = 1/6$, so

$$\mathbb{P}(\{1, 3, 5\}) = \frac{3}{6} = \frac{1}{2}.$$

1.2.2 Consequences of the probability axioms

Lemma 1.9. *Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability triple. Then*

1. *For every $A \in \mathcal{F}$,*
$$\mathbb{P}(A^c) = 1 - \mathbb{P}(A).$$
In particular, $\mathbb{P}(\emptyset) = 0$.

2. *For $A, B \in \mathcal{F}$,*
$$\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B).$$

3. *For $A, B \in \mathcal{F}$,*
$$\mathbb{P}(A \cup B) \le \mathbb{P}(A) + \mathbb{P}(B).$$

Proof. Apply countable additivity to the pairwise disjoint sequence

$$\Omega, \emptyset, \emptyset, \ldots.$$

Since all probabilities are nonnegative,

$$1 = \mathbb{P}(\Omega) = \mathbb{P}(\Omega) + \mathbb{P}(\emptyset) + \mathbb{P}(\emptyset) + \cdots$$

implies $\mathbb{P}(\emptyset) = 0$. Consequently, if C and D are disjoint, countable additivity applied to $C, D, \emptyset, \emptyset, \ldots$ gives

$$\mathbb{P}(C \cup D) = \mathbb{P}(C) + \mathbb{P}(D). \tag{1.2}$$

Since A and A^c are disjoint and $A \cup A^c = \Omega$, (1.2) gives

$$1 = \mathbb{P}(\Omega) = \mathbb{P}(A) + \mathbb{P}(A^c),$$

which proves the complement formula.

Both unions in

$$B = (B \setminus A) \cup (A \cap B), \qquad A \cup B = A \cup (B \setminus A)$$

are disjoint. Hence

$$\mathbb{P}(B) = \mathbb{P}(B \setminus A) + \mathbb{P}(A \cap B), \qquad \mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B \setminus A).$$

Eliminating $\mathbb{P}(B \setminus A)$ proves the two-event addition formula. Finally, the two-event union bound follows because $\mathbb{P}(A \cap B) \ge 0$. □

Lemma 1.10 (Finite union and addition rules)**.** *Let $n \in \mathbb{N}$ with $n \ge 1$, and let $A_1, \ldots, A_n \in \mathcal{F}$.*

1. *If $A, B \in \mathcal{F}$ and $A \subseteq B$, then $\mathbb{P}(A) \le \mathbb{P}(B)$.*

2. *The* **union bound** *is*
$$\mathbb{P}\left(\bigcup_{i=1}^{n} A_i\right) \le \sum_{i=1}^{n} \mathbb{P}(A_i).$$

3. If $A_1, \dots, A_n$ are pairwise disjoint, then

$$\mathbb{P}\left(\bigcup_{i=1}^{n} A_i\right) = \sum_{i=1}^{n} \mathbb{P}(A_i).$$

Proof. If $A \subseteq B$, then $B = A \cup (B \setminus A)$ is a disjoint union. Therefore,

$$\mathbb{P}(B) = \mathbb{P}(A) + \mathbb{P}(B \setminus A) \geq \mathbb{P}(A).$$

For the union bound, the case $n = 1$ is immediate. If the bound holds for n, then the two-event union bound gives

$$\begin{aligned}\mathbb{P}\left(\bigcup_{i=1}^{n+1} A_i\right) &= \mathbb{P}\left(\left(\bigcup_{i=1}^{n} A_i\right) \cup A_{n+1}\right) \\ &\leq \mathbb{P}\left(\bigcup_{i=1}^{n} A_i\right) + \mathbb{P}(A_{n+1}) \\ &\leq \sum_{i=1}^{n+1} \mathbb{P}(A_i).\end{aligned}$$

This proves the bound by induction. If the events are pairwise disjoint, apply countable additivity to $A_1, \dots, A_n, \emptyset, \emptyset, \dots$ to obtain the last formula. □

1.3 Conditional Probability

Conditional probability describes how probabilities change when partial information about the outcome is available.

Example 1.11 (Restricting a die roll)**.** *Roll a fair die and let*

$$A = \{4, 5, 6\}, \qquad B = \{2, 4, 6\}.$$

Suppose that we are told that B occurred, so the result is even. Of the three equally likely outcomes in B, two are also in A. The conditional probability of A given B is therefore $2/3$. In terms of the original probability model,

$$\frac{2}{3} = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)} = \frac{2/6}{3/6}.$$

Definition 1.12 (Conditional probability)**.** *Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability triple and let $A, B \in \mathcal{F}$ with $\mathbb{P}(B) > 0$. The* **conditional probability** *of A given B is*

$$\mathbb{P}(A \mid B) := \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}. \tag{1.3}$$

For a fixed conditioning event, conditional probability is itself a probability measure.

Lemma 1.13. *Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability triple and let $B \in \mathcal{F}$ satisfy $\mathbb{P}(B) > 0$. Then*

$$Q(A) := \mathbb{P}(A \mid B), \qquad A \in \mathcal{F},$$

defines a probability measure on $(\Omega, \mathcal{F})$.

Proof. Since $A \cap B \subseteq B$, monotonicity gives $0 \le Q(A) \le 1$. Moreover,

$$Q(\Omega) = \frac{\mathbb{P}(B)}{\mathbb{P}(B)} = 1.$$

If $A_1, A_2, \dots$ are pairwise disjoint, then $A_1 \cap B, A_2 \cap B, \dots$ are pairwise disjoint and

$$\begin{aligned} Q\left(\bigcup_{i=1}^{\infty} A_i\right) &= \frac{\mathbb{P}\left(\bigcup_{i=1}^{\infty}(A_i \cap B)\right)}{\mathbb{P}(B)} \\ &= \sum_{i=1}^{\infty} \frac{\mathbb{P}(A_i \cap B)}{\mathbb{P}(B)} = \sum_{i=1}^{\infty} Q(A_i). \end{aligned}$$

□

Thus the usual probability identities may be used under a fixed conditioning event of positive probability.

Example 1.14 (A medical test)**.** *Let D be the event that a person has a disease, and let $+$ denote a positive test result. Suppose the joint probabilities are*

	D	D^c
$+$	0.009	0.099
$-$	0.001	0.891

The test is positive for 90% of people with the disease because

$$\mathbb{P}(+ \mid D) = \frac{0.009}{0.009 + 0.001} = 0.9.$$

On the other hand, after observing a positive result,

$$\mathbb{P}(D \mid +) = \frac{0.009}{0.009 + 0.099} = \frac{1}{12} \approx 0.083.$$

The difference occurs because the second calculation also includes the much larger group of people without the disease who test positive.

1.3.1 Bayes' theorem

Bayes' rule reverses the order of conditioning.

Proposition 1.15 (Bayes' rule)**.** *Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability triple and let $A, B \in \mathcal{F}$ with $\mathbb{P}(A) > 0$ and $\mathbb{P}(B) > 0$. Then*

$$\mathbb{P}(A \mid B) = \frac{\mathbb{P}(A)\,\mathbb{P}(B \mid A)}{\mathbb{P}(B)}.$$

Proof. By the definition of conditional probability,

$$\mathbb{P}(A \cap B) = \mathbb{P}(A)\,\mathbb{P}(B \mid A). \tag{1.4}$$

Dividing (1.4) by $\mathbb{P}(B)$ gives the result. □

To compute the denominator in Bayes' rule, we split the sample space into mutually exclusive cases.

Theorem 1.16 (Total probability)**.** *Let* $(\Omega, \mathcal{F}, \mathbb{P})$ *be a probability triple and let* $k \in \mathbb{N}$ *with* $k \geq 1$*. Suppose that* $A_1, \ldots, A_k \in \mathcal{F}$ *satisfy*

$$\bigcup_{h=1}^{k} A_h = \Omega, \qquad A_i \cap A_j = \emptyset \quad (i \neq j).$$

Then, for every $B \in \mathcal{F}$*,*

$$\mathbb{P}(B) = \sum_{h=1}^{k} \mathbb{P}(B \cap A_h).$$

If, in addition, $\mathbb{P}(A_h) > 0$ *for every* $h \in \{1, \ldots, k\}$*, then*

$$\mathbb{P}(B) = \sum_{h=1}^{k} \mathbb{P}(B \mid A_h)\, \mathbb{P}(A_h). \tag{1.5}$$

Proof. The events $B \cap A_1, \ldots, B \cap A_k$ are pairwise disjoint, and

$$B = B \cap \Omega = \bigcup_{h=1}^{k} (B \cap A_h).$$

Finite additivity gives the first displayed identity. Under the additional positivity assumption, the identity

$$\mathbb{P}(B \cap A_h) = \mathbb{P}(B \mid A_h)\, \mathbb{P}(A_h)$$

gives (1.5). □

The positivity assumption is needed exactly where the conditional probabilities $\mathbb{P}(B \mid A_h)$ occur. The intersection formula remains valid when some partition events have probability zero.

Theorem 1.17 (Bayes' theorem)**.** *Let* $(\Omega, \mathcal{F}, \mathbb{P})$ *be a probability triple and let* $k \in \mathbb{N}$ *with* $k \geq 1$*. Suppose that* $A_1, \ldots, A_k \in \mathcal{F}$ *are pairwise disjoint, their union is* Ω*, and* $\mathbb{P}(A_j) > 0$ *for every* $j \in \{1, \ldots, k\}$*. Let* $B \in \mathcal{F}$ *satisfy* $\mathbb{P}(B) > 0$*. Then, for every* $i \in \{1, \ldots, k\}$*,*

$$\mathbb{P}(A_i \mid B) = \frac{\mathbb{P}(B \mid A_i)\, \mathbb{P}(A_i)}{\sum_{j=1}^{k} \mathbb{P}(B \mid A_j)\, \mathbb{P}(A_j)}. \tag{1.6}$$

Proof. Bayes' rule gives

$$\mathbb{P}(A_i \mid B) = \frac{\mathbb{P}(B \mid A_i)\, \mathbb{P}(A_i)}{\mathbb{P}(B)}.$$

Substitute the expression for $\mathbb{P}(B)$ from Definition 1.16. □

The probabilities $\mathbb{P}(A_i)$ are often called prior probabilities, while $\mathbb{P}(A_i \mid B)$ are called posterior probabilities. These names describe which information is treated as observed in a particular model.

1.3.2 Independence

Example 1.18 (Two independent coin tosses)**.** *Toss a fair coin twice and give each of the four outcomes* HH, HT, TH, TT *probability* $1/4$*. Let* A *be the event that the first toss is Heads and* B *the event that the second toss is Heads. Then*

$$\mathbb{P}(A \cap B) = \frac{1}{4} = \frac{1}{2} \cdot \frac{1}{2} = \mathbb{P}(A)\,\mathbb{P}(B).$$

This equality is the mathematical definition of independence.

Definition 1.19 (Independence of two events)**.** *Events* $A, B \in \mathcal{F}$ *are* **independent** *if*

$$\mathbb{P}(A \cap B) = \mathbb{P}(A)\,\mathbb{P}(B). \tag{1.7}$$

When the conditioning event has positive probability, independence is equivalent to saying that conditioning does not change the probability.

Lemma 1.20. *Let* $A, B \in \mathcal{F}$ *and suppose that* $\mathbb{P}(B) > 0$*. Then* A *and* B *are independent if and only if*

$$\mathbb{P}(A \mid B) = \mathbb{P}(A).$$

Proof. By Definition 1.12,

$$\mathbb{P}(A \mid B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}.$$

Therefore $\mathbb{P}(A \mid B) = \mathbb{P}(A)$ is equivalent to $\mathbb{P}(A \cap B) = \mathbb{P}(A)\,\mathbb{P}(B)$. $\square$

Independence is a property of the probability model; it is not a statement about one event physically causing another. If $\mathbb{P}(A) > 0$ as well, symmetry gives $\mathbb{P}(B \mid A) = \mathbb{P}(B)$.

Definition 1.21 (Mutual independence)**.** *Let* $n \in \mathbb{N}$ *with* $n \geq 1$*. Let* I *be either* $\{1, \ldots, n\}$ *or* $\mathbb{N}$*, and let* $(A_i)_{i \in I}$ *be a family of events. The family is* **mutually independent** *if, for every finite set* $J \subseteq I$ *with* $|J| \geq 2$*,*

$$\mathbb{P}\left(\bigcap_{j \in J} A_j\right) = \prod_{j \in J} \mathbb{P}(A_j).$$

When $I = \mathbb{N}$*, we also call* $(A_i)_{i \in \mathbb{N}}$ *an independent sequence.*

A family is pairwise independent if the displayed identity is required only for sets J containing two indices. Pairwise independence does not imply mutual independence.

Example 1.22 (Pairwise but not mutually independent)**.** *Draw one ball uniformly from an urn containing balls labelled* $1, 2, 3, 4$*, so* $\mathbb{P}(\{j\}) = 1/4$ *for every* j*. Let*

$$A = \{1, 2\}, \qquad B = \{1, 3\}, \qquad C = \{1, 4\}.$$

Each event has probability $1/2$*, and each pair has intersection* $\{1\}$*. Hence*

$$\mathbb{P}(A \cap B) = \mathbb{P}(A \cap C) = \mathbb{P}(B \cap C) = \frac{1}{4},$$

so the events are pairwise independent. However,

$$\mathbb{P}(A \cap B \cap C) = \frac{1}{4} \neq \frac{1}{8} = \mathbb{P}(A)\,\mathbb{P}(B)\,\mathbb{P}(C).$$

Thus they are not mutually independent.

1.4 Optional note: infinite and continuous sample spaces

The required examples above use finite sample spaces. The same definition of probability also covers countably infinite and continuous models.

If $\Omega = \{\omega_1, \omega_2, \ldots\}$ is countable, we may take $\mathcal{F} = 2^{\Omega}$. Given numbers $p_i \geq 0$ with $\sum_{i=1}^{\infty} p_i = 1$, define

$$\mathbb{P}(A) = \sum_{\{i:\omega_i \in A\}} p_i.$$

Countable additivity guarantees that this model behaves consistently for events containing infinitely many outcomes.

For a continuous sample space such as $\mathbb{R}^d$, the standard event collection is denoted by $\mathcal{B}(\mathbb{R}^d)$ and is called the **Borel σ-algebra**. It is the smallest σ-algebra containing all open subsets of $\mathbb{R}^d$. In particular, it contains all open and closed sets, intervals, rectangles, and half-spaces. In this course, $\mathcal{B}(\mathbb{R}^d)$ is treated as part of the given probability model. Students are not expected to construct it or characterize all of its members. Continuous probability models and densities are introduced in the next chapter.

1.5 Exercises

Exercise 1.23 (Events in a finite model). *Roll a fair six-sided die. Let*

$$A = \{1, 2, 3\}, \qquad B = \{2, 4, 6\}.$$

1. *Write $A \cap B$, $A \cup B$, and $A \setminus B$ as sets of outcomes, and compute their probabilities.*

2. *Verify the two-event addition formula for A and B.*

3. *Compute $\mathbb{P}(A \mid B)$ and $\mathbb{P}(B \mid A)$. Are A and B independent? Justify the answer from the definition.*

Exercise 1.24 (Complements and independence). *Let A and B be independent events. Prove directly from Definition 1.19 that A^c and B are independent. Deduce that A^c and B^c are independent. Your proof must also cover the cases $\mathbb{P}(A) = 0$ and $\mathbb{P}(B) = 0$; do not divide by either probability.*

Exercise 1.25 (Total probability and Bayes' theorem). *A factory uses three machines. Machines 1, 2, and 3 produce proportions 0.50, 0.30, and 0.20 of all items, respectively. Their defective rates are 0.01, 0.02, and 0.05. An item is drawn uniformly from the day's production.*

1. *Specify a partition of the sample space and state every probability supplied by the model.*

2. *Compute the probability that the selected item is defective.*

3. *Given that it is defective, compute the probability that it was produced by machine 3.*

4. *Identify the positivity assumptions that make each conditional probability in your calculation well defined.*

Exercise 1.26 (Computational relative frequencies)**.** *Use Python with* `numpy` *and* `matplotlib`*. Set the seed with* `numpy.random.seed(2026)`*, and use* `numpy.random.randint` *to simulate* 10 000 *independent fair die rolls. For each*

$$n \in \{10, 30, 100, 300, 1000, 3000, 10000\},$$

compute the relative frequency of the event $A = \{1, 2\}$*. Plot the relative frequencies against* n*, and add a horizontal line at* $\mathbb{P}(A) = 1/3$*. Report the largest absolute error among the displayed sample sizes. Explain why the plot is evidence about this simulated sequence rather than a proof that every outcome sequence converges.*

Exercise 1.27 (Optional: a countable model)**.** *Let* $\Omega = \mathbb{N} = \{1, 2, \ldots\}$*, let* $\mathcal{F} = 2^{\Omega}$*, and set* $\mathbb{P}(\{i\}) = 2^{-i}$ *for* $i \in \mathbb{N}$*.*

1. *Verify that the singleton probabilities sum to one.*
2. *Compute the probability that the outcome is even.*
3. *For* $m \in \mathbb{N}$*, compute* $\mathbb{P}(\{m, m+1, \ldots\})$ *and show that it tends to zero as* m *tends to infinity.*

Chapter 2

Random Variables

Course guide

Scheduled lecture(s): Lectures 2–3 (and review in Lecture 10)
Required for those lectures: Lecture 2: Sections 2.1–2.5; Lecture 3: Sections 2.6–2.7 and the core exercises in Section 2.8
Prerequisites: Chapter 1, especially Sections 1.2–1.3; elementary calculus, and elementary vectors for Section 2.6
Optional or advanced: Sections 2.6.4 and 2.6.6 and the exercises marked optional are optional.

A random variable assigns a numerical value to each outcome of an experiment. The probability model describes which outcomes can occur and their probabilities. The random variable specifies which numerical feature of the outcome we record.

2.1 Basic Definitions

Example 2.1 (Counting Heads in two tosses)**.** *Toss a fair coin twice, with independent tosses. The finite probability model is*

$$\Omega = \{\texttt{HH}, \texttt{HT}, \texttt{TH}, \texttt{TT}\}, \qquad \mathcal{F} = 2^{\Omega}, \qquad \mathbb{P}(\{\omega\}) = \frac{1}{4}$$

for every $\omega \in \Omega$. Define $X : \Omega \to \mathbb{R}$ to be the number of Heads:

outcome ω	HH	HT	TH	TT
value $X(\omega)$	2	1	1	0

For example, HT *is an outcome,* $\{\texttt{HT}\}$ *is a singleton event, and* $X(\texttt{HT}) = 1$ *is a numerical value. The event that X equals 1 is*

$$\{X = 1\} := \{\omega \in \Omega : X(\omega) = 1\} = \{\texttt{HT}, \texttt{TH}\},$$

and hence $\mathbb{P}(X = 1) = 1/2$.

Definition 2.2 (Random variable)**.** *Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability triple. A* **random variable (RV)** *is a function*

$$X : \Omega \to \mathbb{R}$$

such that, for every $x \in \mathbb{R}$,

$$X^{[-1]}((-\infty, x]) := \{\omega \in \Omega : X(\omega) \le x\} \in \mathcal{F}.$$

Thus the probability of the numerical statement $X \le x$ is defined by

$$\mathbb{P}(X \le x) := \mathbb{P}\big(\{\omega \in \Omega : X(\omega) \le x\}\big). \tag{2.1}$$

When Ω is finite or countable and $\mathcal{F} = 2^{\Omega}$, every function $X : \Omega \to \mathbb{R}$ is a random variable. Indeed, every inverse image is then a subset of Ω, and every subset is an event.

The condition in Definition 2.2 also makes other familiar numerical statements events. For example,

$$\begin{aligned} \{X < x\} &= \bigcup_{n=1}^{\infty} \{X \le x - 1/n\}, \\ \{X = x\} &= \{X \le x\} \setminus \{X < x\}, \\ \{a < X \le b\} &= \{X \le b\} \setminus \{X \le a\}. \end{aligned}$$

The closure properties of $\mathcal{F}$ therefore ensure that each set on the left is an event.

Definition 2.3 (Distribution function)**.** *Let X be a random variable. Its* **distribution function (DF)***, also called its* **cumulative distribution function (CDF)***, is the function $F_X : \mathbb{R} \to [0, 1]$ defined by*

$$F_X(x) := \mathbb{P}(X \le x), \qquad x \in \mathbb{R}. \tag{2.2}$$

When no confusion is possible, we write F instead of F_X. The notation $X \sim F$ means that X has distribution function F.

Proposition 2.4 (Elementary properties of a distribution function)**.** *Let X be a random variable with distribution function F_X. Then F_X is nondecreasing and right-continuous, and*

$$\lim_{x \to -\infty} F_X(x) = 0, \qquad \lim_{x \to \infty} F_X(x) = 1.$$

Proof. If $x \le y$, then $\{X \le x\} \subseteq \{X \le y\}$, so monotonicity of probability gives $F_X(x) \le F_X(y)$.

We use one consequence of countable additivity. If events A_n decrease to $A = \bigcap_{n=1}^{\infty} A_n$, then $\mathbb{P}(A_n) \to \mathbb{P}(A)$. To see this, the events $A_n \setminus A_{n+1}$ are pairwise disjoint and

$$A_1 = A \cup \bigcup_{n=1}^{\infty} (A_n \setminus A_{n+1}).$$

Countable additivity shows that the probability of the tail beginning at n tends to zero. Apply this fact to $A_n = \{X \le x + 1/n\}$. Their intersection is $\{X \le x\}$, so $F_X(x + 1/n) \to F_X(x)$. Monotonicity then gives right-continuity.

Finally, $\{X \le n\}$ increases to Ω, while $\{X \le -n\}$ decreases to $\emptyset$. Countable additivity gives the two limits. □

Proposition 2.5. *Let X be a random variable with distribution function F_X. If $a < b$, then*

$$\mathbb{P}(a < X \le b) = F_X(b) - F_X(a). \tag{2.3}$$

Proof. The events $\{X \le a\}$ and $\{a < X \le b\}$ are disjoint, and their union is $\{X \le b\}$. Finite additivity therefore gives

$$F_X(b) = F_X(a) + \mathbb{P}(a < X \le b),$$

which proves the result. $\square$

Definition 2.6 (Indicator function). *Let $A \in \mathcal{F}$ be an event. The* **indicator function** *of A is the function $\mathbb{1}_A : \Omega \to \{0,1\}$ defined by*

$$\mathbb{1}_A(\omega) := \begin{cases} 1, & \omega \in A, \\ 0, & \omega \notin A. \end{cases}$$

Lemma 2.7. *The indicator $\mathbb{1}_A$ in Definition 2.6 is a random variable.*

Proof. For $x \in \mathbb{R}$,

$$\{\mathbb{1}_A \le x\} = \begin{cases} \emptyset, & x < 0, \\ A^c, & 0 \le x < 1, \\ \Omega, & x \ge 1. \end{cases}$$

Each of these sets belongs to $\mathcal{F}$, so Definition 2.2 applies. $\square$

Model 2.8 (Indicator of an event as a Bernoulli RV). *Let $A \in \mathcal{F}$, set $\theta := \mathbb{P}(A)$, and let $X = \mathbb{1}_A$. Then*

$$\mathbb{P}(X = 1) = \theta, \qquad \mathbb{P}(X = 0) = 1 - \theta.$$

We say that X is Bernoulli(θ) *distributed.*

Lemma 2.9. *Let $A, B \in \mathcal{F}$. The following identities hold pointwise on Ω:*

1. $\mathbb{1}_A = 1 - \mathbb{1}_{A^c}$;
2. $\mathbb{1}_{A\cap B} = \mathbb{1}_A \mathbb{1}_B$;
3. $\mathbb{1}_{A\cup B} = \mathbb{1}_A + \mathbb{1}_B - \mathbb{1}_A\mathbb{1}_B$.

Proof. Fix $\omega \in \Omega$. Exactly one of $\omega \in A$ and $\omega \in A^c$ holds, which proves the first identity. The product $\mathbb{1}_A(\omega)\mathbb{1}_B(\omega)$ equals 1 exactly when $\omega \in A \cap B$, which proves the second. For the last identity, if neither event occurs both sides are 0; if exactly one occurs both sides are 1; and if both occur the right-hand side is $1 + 1 - 1 = 1$. $\square$

2.2 Discrete Random Variables

The random variable in Definition 2.1 takes only the three values $0, 1, 2$. This is the simplest type of discrete random variable.

Definition 2.10 (Discrete random variable). *A real-valued random variable X is* **discrete** *if there is a finite or countably infinite set*

$$\mathbb{X} = \{x_1, x_2, \ldots\} \subseteq \mathbb{R}$$

such that $\mathbb{P}(X \in \mathbb{X}) = 1$. The set may be finite, in which case the list ends after some x_k. Thus changing X on an event of probability zero does not change whether it is discrete.

Definition 2.11 (Probability mass function)**.** *Let X be a discrete random variable supported on $\mathbb{X} = \{x_1, x_2, \ldots\}$. Its* **probability mass function (PMF)** *is $f_X : \mathbb{R} \to [0,1]$, where*

$$f_X(x) := \mathbb{P}(X = x) = \begin{cases} \theta_i, & x = x_i \text{ for some } i, \\ 0, & x \notin \mathbb{X}, \end{cases} \qquad \theta_i := \mathbb{P}(X = x_i). \tag{2.4}$$

Example 2.12. *For the number of Heads X in Definition 2.1,*

$$f_X(0) = \frac{1}{4}, \qquad f_X(1) = \frac{1}{2}, \qquad f_X(2) = \frac{1}{4},$$

and $f_X(x) = 0$ for every other x.

Theorem 2.13 (PMF and distribution function)**.** *Let X be as in Definition 2.11. Then, for every $x \in \mathbb{R}$,*

$$F_X(x) = \sum_{\{i : x_i \le x\}} f_X(x_i) = \sum_{\{i : x_i \le x\}} \theta_i. \tag{2.5}$$

If $a < b$, then

$$F_X(b) - F_X(a) = \sum_{\{i : a < x_i \le b\}} \theta_i. \tag{2.6}$$

Finally,

$$\sum_i \theta_i = 1. \tag{2.7}$$

Proof. Set $A_i = \{X = x_i\}$. These events are pairwise disjoint, and their union has probability one. Moreover, $\{X \le x\}$ and

$$\bigcup_{\{i : x_i \le x\}} A_i$$

differ by at most an event of probability zero. Countable additivity gives (2.5) and (2.7). Applying the same argument to $\{a < X \le b\}$, or using Definition 2.5, gives (2.6). □

Remark 2.14 (Finite probability vectors)**.** *If X has the finite support $\{x_1, \ldots, x_k\}$, then its PMF is specified by a point in the unit $(k-1)$-simplex*

$$\Delta^{k-1} := \left\{ (\theta_1, \ldots, \theta_k) \in \mathbb{R}^k : \theta_i \ge 0 \text{ for every } i, \ \sum_{i=1}^{k} \theta_i = 1 \right\}. \tag{2.8}$$

For $k = 2$, this becomes

$$\Delta^1 = \{(\theta, 1 - \theta) : 0 \le \theta \le 1\}. \tag{2.9}$$

Model 2.15 (Discrete uniform)**.** *Let $k \in \mathbb{N}$ with $k \ge 1$, and let $x_1 < \cdots < x_k$. A random variable X is uniformly distributed on $\mathbb{X} = \{x_1, \ldots, x_k\}$ if*

$$f_X(x) = \begin{cases} 1/k, & x \in \mathbb{X}, \\ 0, & x \notin \mathbb{X}. \end{cases} \tag{2.10}$$

Its distribution function is

$$F_X(x) = \frac{\#\{i \in \{1, \ldots, k\} : x_i \le x\}}{k}. \tag{2.11}$$

In particular, the uniform distribution on $\{1, \ldots, k\}$ models a fair k-sided die.

Model 2.16 (Point mass)**.** *Given $\theta \in \mathbb{R}$, a random variable X has a point mass at θ, written $X \sim \text{Point Mass}(\theta)$, if $\mathbb{P}(X = \theta) = 1$. Its distribution function is*

$$F_X(x;\theta) = \begin{cases} 0, & x < \theta, \\ 1, & x \geq \theta, \end{cases} \tag{2.12}$$

and its PMF equals 1 at θ and 0 elsewhere. Thus $X = \theta$ with probability one, although values on an event of probability zero are irrelevant to the distribution. The constant version $X(\omega) = \theta$ for every ω is discrete in the sense of Definition 2.10 and has this distribution.

2.3 Continuous Random Variables

Consider the standard uniform probability model on $[0, 1]$: the event collection is the standard collection containing all intervals, and

$$\mathbb{P}((a, b]) = b - a, \qquad 0 \leq a \leq b \leq 1.$$

For the identity random variable $X(\omega) = \omega$,

$$F_X(x) = \begin{cases} 0, & x < 0, \\ x, & 0 \leq x \leq 1, \\ 1, & x > 1. \end{cases}$$

The slope of this distribution function is 1 on $(0, 1)$ and 0 outside $[0, 1]$. This slope is the density.

Definition 2.17 (Continuous random variable)**.** *Let X be a real-valued random variable with distribution function F_X. In this course, we call X* **continuous** *if there is a nonnegative function $f_X : \mathbb{R} \to [0, \infty)$ that is piecewise continuous away from isolated points, has finite one-sided improper integrals at those points, and satisfies*

$$F_X(x) = \mathbb{P}(X \leq x) = \int_{-\infty}^{x} f_X(v)\, dv, \qquad x \in \mathbb{R}. \tag{2.13}$$

The function f_X is a **probability density function (PDF)** *of X.*

This definition is more precisely called an absolutely continuous distribution. Not every nondiscrete random variable has a density; a concrete mixed example appears in optional Section 2.6.6.

Theorem 2.18. *Let X be continuous with density f_X. Then:*

1. *For every $x \in \mathbb{R}$,*
$$\mathbb{P}(X = x) = 0.$$

2. *At every point where f_X is continuous,*
$$F'_X(x) = f_X(x). \tag{2.14}$$

3. *If $a < b$, then all choices of endpoints give the same probability, and*

$$\begin{aligned}\mathbb{P}(a < X < b) = \mathbb{P}(a < X \leq b) = \mathbb{P}(a \leq X < b) = \mathbb{P}(a \leq X \leq b) \\ = F_X(b) - F_X(a) = \int_a^b f_X(v)\,dv. \qquad (2.15)\end{aligned}$$

4. *The density is normalized:*

$$\int_{-\infty}^{\infty} f_X(x)\,dx = 1.$$

Proof. An indefinite integral of a piecewise-continuous function with only integrable singularities is continuous. Hence F_X is continuous. For $h > 0$,

$$0 \leq \mathbb{P}(X = x) \leq \mathbb{P}(x - h < X \leq x) = F_X(x) - F_X(x - h).$$

Letting h decrease to zero proves 1. The fundamental theorem of calculus proves 2. By Definition 2.5 and (2.13),

$$\mathbb{P}(a < X \leq b) = F_X(b) - F_X(a) = \int_a^b f_X(v)\,dv.$$

Singletons have probability zero, so adding or removing either endpoint does not change this value. Finally, let $x \to \infty$ in (2.13). Since $F_X(x) \to 1$, the improper integral of f_X over $\mathbb{R}$ equals 1. □

Model 2.19 (Standard normal or Gaussian RV)**.** *A continuous random variable Z is* **standard normal***, or standard Gaussian, if its density is*

$$\phi(z) = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{z^2}{2}\right), \qquad z \in \mathbb{R}. \qquad (2.16)$$

The constant $1/\sqrt{2\pi}$ normalizes the density. We write $Z \sim \mathrm{Normal}(0, 1)$.

The density ϕ is symmetric about zero. Since

$$\phi'(z) = -z\phi(z), \qquad \phi''(z) = (z^2 - 1)\phi(z),$$

it has its maximum at zero and has the familiar bell shape.

2.4 Transformations of random variables

Suppose that X is a random variable and that we record $Y = g(X)$ instead. In Definition 2.1, for example, let $g(x) = \mathbb{1}_{[1,\infty)}(x)$. Then Y records whether at least one Head occurred, and

$$\mathbb{P}(Y = 1) = \frac{3}{4}, \qquad \mathbb{P}(Y = 0) = \frac{1}{4}.$$

For a function $g : \mathbb{X} \to \mathbb{Y}$ and a set $A \subseteq \mathbb{Y}$, the **inverse image** of A is

$$g^{[-1]}(A) := \{x \in \mathbb{X} : g(x) \in A\}.$$

This notation denotes a set even when g has no inverse function.

2.4.1 Transformations of discrete random variables

Let X be discrete with support $\{x_1, x_2, \ldots\}$, let $g : \mathbb{R} \to \mathbb{R}$ be a function for which $Y = g(X)$ is a random variable. Then Y is discrete, with support contained in the countable set $\{g(x_1), g(x_2), \ldots\}$. More generally, for any set A such that $\{Y \in A\}$ is an event,

$$\mathbb{P}(Y \in A) = \mathbb{P}(g(X) \in A) = \mathbb{P}\big(X \in g^{[-1]}(A)\big). \tag{2.17}$$

Consequently, the PMF of Y is

$$f_Y(y) = \mathbb{P}(Y = y) = \sum_{\{i : g(x_i) = y\}} f_X(x_i), \tag{2.18}$$

where an empty sum is zero. Thus a many-to-one transformation adds the probability masses of all support points with the same image.

2.4.2 Transformations of continuous random variables

In a continuous model, the standard event collection on $\mathbb{R}$ contains intervals and all open and closed sets.

Definition 2.20 (Valid transformation of a random variable)**.** *Let X be a random variable and let $g : \mathbb{R} \to \mathbb{R}$. We call g a valid transformation of X if, for every $y \in \mathbb{R}$,*

$$\{\omega \in \Omega : g(X(\omega)) \le y\} \in \mathcal{F}.$$

Equivalently, $g(X)$ is a random variable.

For the models used here, continuous, piecewise-continuous, and monotone functions are valid. The definition records the only event condition needed below.

Direct method

If g is a valid transformation of X and $Y = g(X)$, then its distribution function can be found directly from

$$\begin{aligned} F_Y(y) &= \mathbb{P}(Y \le y) = \mathbb{P}(g(X) \le y) \\ &= \mathbb{P}\big(X \in g^{[-1]}((-\infty, y])\big) = \mathbb{P}\big(X \in \{x : g(x) \le y\}\big). \end{aligned} \tag{2.19}$$

Example 2.21 (Squaring a continuous random variable)**.** *Let X be continuous with distribution function F_X and a continuous density f_X, and set $Y = X^2$. If $y < 0$, then $F_Y(y) = 0$. If $y \ge 0$, then*

$$\begin{aligned} F_Y(y) &= \mathbb{P}(-\sqrt{y} \le X \le \sqrt{y}) \\ &= F_X(\sqrt{y}) - F_X(-\sqrt{y}). \end{aligned}$$

The last equality uses $\mathbb{P}(X = -\sqrt{y}) = 0$. Therefore,

$$F_Y(y) = \begin{cases} 0, & y < 0, \\ F_X(\sqrt{y}) - F_X(-\sqrt{y}), & y \ge 0. \end{cases} \tag{2.20}$$

Differentiating at $y > 0$ gives a density of Y:

$$f_Y(y) = \begin{cases} 0, & y \le 0, \\ \dfrac{f_X(\sqrt{y}) + f_X(-\sqrt{y})}{2\sqrt{y}}, & y > 0. \end{cases} \tag{2.21}$$

The density may be unbounded near zero. It still has a finite improper integral there because f_X is bounded near zero and $\int_0^\epsilon y^{-1/2}\,dy < \infty$. The value assigned to a density at the single point $y = 0$ does not affect any probability.

One-to-one transformations

Proposition 2.22 (Change-of-variable formula)**.** *Let X be continuous, suppose $X(\Omega) \subseteq I$ for an open interval I, and let f_X be a density that is continuous on I. Let $g : I \to J$ be a continuously differentiable, strictly monotone bijection onto an open interval J, and suppose $g'(x) \neq 0$ for every $x \in I$. Then $Y = g(X)$ has density*

$$f_Y(y) = f_X\big(g^{-1}(y)\big) \left| \frac{d}{dy} g^{-1}(y) \right|, \qquad y \in J, \tag{2.22}$$

and $f_Y(y) = 0$ for $y \notin J$.

Proof. If g is increasing and $y \in J$, then

$$F_Y(y) = \mathbb{P}(g(X) \le y) = \mathbb{P}(X \le g^{-1}(y)) = F_X(g^{-1}(y)).$$

Differentiation gives the formula without the absolute value, and the derivative of g^{-1} is positive. If g is decreasing, then

$$F_Y(y) = \mathbb{P}(X \ge g^{-1}(y)) = 1 - F_X(g^{-1}(y)).$$

The second equality uses the fact that a continuous random variable has no point masses. Differentiation introduces a minus sign, while $(g^{-1})'(y) < 0$, which again gives the absolute-value formula. □

Example 2.23 (Affine transformation of a standard Gaussian)**.** *Let $Z \sim \mathrm{Normal}(0,1)$, let $\mu \in \mathbb{R}$ and $\sigma > 0$, and set $Y = \sigma Z + \mu$. Here*

$$g^{-1}(y) = \frac{y-\mu}{\sigma}, \qquad \frac{d}{dy} g^{-1}(y) = \frac{1}{\sigma}.$$

By (2.16) and (2.22),

$$f_Y(y) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left[-\frac{1}{2} \left(\frac{y-\mu}{\sigma} \right)^2 \right].$$

Model 2.24 ($\mathrm{Normal}(\mu,\sigma^2)$ RV)**.** *Given $\mu \in \mathbb{R}$ and $\sigma > 0$, a random variable X is* $\mathrm{Normal}(\mu,\sigma^2)$ *distributed if it has density*

$$f_X(x;\mu,\sigma^2) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left[-\frac{1}{2} \left(\frac{x-\mu}{\sigma} \right)^2 \right]. \tag{2.23}$$

The parameter μ is its mean, σ is its standard deviation, and σ^2 is its variance; these terms are defined in the next section.

The distribution function is

$$F_X(x;\mu,\sigma^2) = \frac{1}{\sigma\sqrt{2\pi}}\int_{-\infty}^{x} \exp\left[-\frac{1}{2}\left(\frac{v-\mu}{\sigma}\right)^2\right] dv. \tag{2.24}$$

Proposition 2.25 (Standardization)**.** *Let $\mu \in \mathbb{R}$ and $\sigma > 0$. Suppose that $X \sim \mathrm{Normal}(\mu,\sigma^2)$, and let Φ be the distribution function of $Z \sim \mathrm{Normal}(0,1)$. Then*

$$F_X(x;\mu,\sigma^2) = \Phi\left(\frac{x-\mu}{\sigma}\right).$$

Proof. In (2.24), make the substitution $z = (v-\mu)/\sigma$. Since $\sigma > 0$, the upper limit becomes $(x-\mu)/\sigma$, and the resulting integral is the defining integral for $\Phi((x-\mu)/\sigma)$. □

2.5 Expectations

For the number of Heads X in Definition 2.1, the probability weighted average of the possible values is

$$0\cdot\frac{1}{4} + 1\cdot\frac{1}{2} + 2\cdot\frac{1}{4} = 1.$$

This calculation motivates the expectation of a discrete random variable. For a continuous random variable, sums are replaced by ordinary improper integrals against its density.

Definition 2.26 (Expectation)**.** *Let X be discrete or continuous. In the discrete case, write its support as $\{x_1, x_2, \ldots\}$. In either case, denote its PMF or PDF by f_X, as appropriate. Its* **expectation***, expected value, or mean is*

$$\mathbb{E}[X] := \begin{cases} \displaystyle\sum_i x_i f_X(x_i), & X \text{ discrete,} \\ \displaystyle\int_{-\infty}^{\infty} x f_X(x)\,dx, & X \text{ continuous,} \end{cases} \tag{2.25}$$

provided the corresponding absolute sum or integral is finite:

$$\mathbb{E}[|X|] := \begin{cases} \displaystyle\sum_i |x_i| f_X(x_i), & X \text{ discrete,} \\ \displaystyle\int_{-\infty}^{\infty} |x| f_X(x)\,dx, & X \text{ continuous} \end{cases} < \infty. \tag{2.26}$$

A random variable satisfying (2.26) is called **integrable***.*

Absolute convergence prevents an expression such as $\infty - \infty$ from being assigned an artificial value. Expectation is a property of the distribution, not a value that every realization must be close to.

Definition 2.27 (Expectation of a function of a RV)**.** *Let X be a discrete or continuous random variable, and let $g : \mathbb{R} \to \mathbb{R}$ be a function for which $g(X)$ is a random variable. Assume, in the discrete and continuous cases respectively, that*

$$\sum_i |g(x_i)| f_X(x_i) < \infty \qquad \text{or} \qquad \int_{-\infty}^{\infty} |g(x)| f_X(x)\,dx < \infty.$$

Then

$$\mathbb{E}[g(X)] := \begin{cases} \displaystyle\sum_i g(x_i) f_X(x_i), & X \text{ discrete,} \\ \displaystyle\int_{-\infty}^{\infty} g(x) f_X(x)\,dx, & X \text{ continuous.} \end{cases}$$

This formula is sometimes called the law of the unconscious statistician. It computes the expectation of $g(X)$ directly from the distribution of X, without first finding the distribution of $g(X)$.

For a standard numerical event $A \subseteq \mathbb{R}$, we also write $\mathbb{1}_A : \mathbb{R} \to \{0, 1\}$ for the numerical-set indicator

$$\mathbb{1}_A(x) = \begin{cases} 1, & x \in A, \\ 0, & x \notin A. \end{cases}$$

This has a different domain from the event indicator in Definition 2.6, and the two are related by

$$\mathbb{1}_A(X) = \mathbb{1}_{\{X \in A\}}.$$

Example 2.28. *Let $A \subseteq \mathbb{R}$ be a standard event. Since $\mathbb{1}_A(X)$ is 1 exactly on the event $\{X \in A\}$,*

$$\mathbb{E}\left[\mathbb{1}_A(X)\right] = \mathbb{P}(X \in A).$$

Thus probabilities are expectations of indicator random variables.

Definition 2.29 (Variance). *Let X satisfy $\mathbb{E}\left[X^2\right] < \infty$. Its* **variance** *is*

$$\mathbb{V}(X) := \mathbb{E}\big[(X - \mathbb{E}\left[X\right])^2\big],$$

and its **standard deviation** *is*

$$\mathrm{sd}(X) := \sqrt{\mathbb{V}(X)}.$$

The condition $\mathbb{E}\left[X^2\right] < \infty$ also implies $\mathbb{E}\left[|X|\right] < \infty$, since $|x| \leq 1 + x^2$.

Expanding the square and using linearity of sums or integrals gives

$$\mathbb{V}(X) = \mathbb{E}\left[X^2\right] - (\mathbb{E}\left[X\right])^2.$$

Example 2.30 (Mean and variance of a Gaussian). *Let $Z \sim \mathrm{Normal}(0, 1)$. The function $z\phi(z)$ is odd, so $\mathbb{E}\left[Z\right] = 0$. Since $\phi'(z) = -z\phi(z)$, integration by parts gives*

$$\mathbb{E}\left[Z^2\right] = -\int_{-\infty}^{\infty} z\phi'(z)\,dz = \int_{-\infty}^{\infty} \phi(z)\,dz = 1.$$

The boundary term vanishes because $z\phi(z) \to 0$ as $|z| \to \infty$. Therefore, if $X = \sigma Z + \mu$ with $\sigma > 0$, then

$$\mathbb{E}\left[X\right] = \mu, \qquad \mathbb{V}(X) = \sigma^2.$$

This verifies the interpretation of the parameters in (2.23).

Definition 2.31 (Moment). *Let $k \in \mathbb{N}$ with $k \geq 1$. If $\mathbb{E}\left[|X|^k\right] < \infty$, then*

$$\mathbb{E}\left[X^k\right]$$

is the k-th moment of X. This hypothesis implies that X is integrable because $|x| \leq 1 + |x|^k$. The quantity

$$\mathbb{E}\big[(X - \mathbb{E}\left[X\right])^k\big]$$

is its k-th central moment.

2.6 Multivariate Random Variables

Return to two independent tosses of a fair coin. Let X_1 and X_2 be the indicators of Heads on the first and second toss, respectively. The random vector $Z = (X_1, X_2)$ has the four values

$$(1, 1), (1, 0), (0, 1), (0, 0),$$

each with probability 1/4. One outcome of the experiment is `HT`, while the corresponding value of the random vector is $Z(\texttt{HT}) = (1, 0)$.

Definition 2.32 (Random vector). *Let $m \in \mathbb{N}$ with $m \geq 1$. An $\mathbb{R}^m$-valued random variable, or* **random vector***, is a tuple*

$$Z = (Z_1, \ldots, Z_m) : \Omega \to \mathbb{R}^m$$

whose components $Z_i : \Omega \to \mathbb{R}$ are real-valued random variables. Consequently, for every $z = (z_1, \ldots, z_m) \in \mathbb{R}^m$,

$$\{Z \leq z\} := \bigcap_{i=1}^{m} \{Z_i \leq z_i\} \in \mathcal{F}.$$

Vector inequalities in this chapter are interpreted componentwise.

Definition 2.33 (Joint distribution function). *Let $Z = (Z_1, \ldots, Z_m)$ be a random vector. Its* **joint distribution function (JDF)** *is $F_Z : \mathbb{R}^m \to [0, 1]$, where*

$$\begin{aligned} F_Z(z_1, \ldots, z_m) &:= \mathbb{P}(Z_1 \leq z_1, \ldots, Z_m \leq z_m) \\ &= \mathbb{P}\big(\{\omega : Z_1(\omega) \leq z_1, \ldots, Z_m(\omega) \leq z_m\}\big). \end{aligned}$$

The distribution function $F_{Z_i}(t) = \mathbb{P}(Z_i \leq t)$ of a component is called a **marginal distribution**. It is defined directly; there is no need to treat $+\infty$ as a real input to the joint distribution function.

Definition 2.34 (Independence of two RVs). *Real-valued random variables X and Y are* **independent** *if, for every $(x, y) \in \mathbb{R}^2$,*

$$F_{X,Y}(x, y) = F_X(x)F_Y(y).$$

In the discrete and density models of this chapter, the PMF and PDF criteria below give the equivalent identity $\mathbb{P}(X \in A, Y \in B) = \mathbb{P}(X \in A)\,\mathbb{P}(Y \in B)$ for standard events A, B.

More generally, a family $(X_i)_{i \in I}$ is mutually independent if, for every nonempty finite set of distinct indices $J \subseteq I$ and every choice of thresholds $x_j \in \mathbb{R}$,

$$\mathbb{P}\left(\bigcap_{j \in J} \{X_j \leq x_j\}\right) = \prod_{j \in J} F_{X_j}(x_j).$$

Pairwise independence requires this identity only when $|J| = 2$, and is weaker than mutual independence. The family is independent and identically distributed (i.i.d.) if it is mutually independent and all its members have the same distribution function.

2.6.1 Discrete random vectors

Definition 2.35 (Joint probability mass function). *A random vector Z is discrete if there is a finite or countable set $\mathbb{Z} = \{z_1, z_2, \ldots\} \subseteq \mathbb{R}^m$ such that $\mathbb{P}(Z \in \mathbb{Z}) = 1$. Its joint PMF is*

$$f_Z(z) := \mathbb{P}(Z = z), \qquad z \in \mathbb{R}^m.$$

Hence $f_Z(z) \geq 0$ and $\sum_{z \in \mathbb{Z}} f_Z(z) = 1$.

For a discrete pair (X, Y), the marginal PMFs are obtained by summing out the other coordinate:

$$f_X(x) = \sum_y f_{X,Y}(x, y), \qquad f_Y(y) = \sum_x f_{X,Y}(x, y).$$

If X and Y are independent, then

$$f_{X,Y}(x, y) = f_X(x) f_Y(y)$$

for all x, y. Conversely, this factorization implies independence. The forward implication follows by taking successive jumps of the joint distribution function at (x, y); the converse follows by summing the product PMF over $(-\infty, x] \times (-\infty, y]$.

Example 2.36. *For the two fair coin indicators at the start of this section,*

$$f_{X_1,X_2}(x_1, x_2) = \frac{1}{4}, \qquad (x_1, x_2) \in \{0, 1\}^2.$$

Each marginal mass is $1/2$, so the joint PMF is the product of the marginal PMFs.

2.6.2 Continuous random vectors

Definition 2.37 (Joint probability density function). *A random vector $Z = (Z_1, \ldots, Z_m)$ is continuous if there is a nonnegative, piecewise-continuous function $f_Z : \mathbb{R}^m \to [0, \infty)$ such that*

$$F_Z(z) = \int_{-\infty}^{z_1} \cdots \int_{-\infty}^{z_m} f_Z(v_1, \ldots, v_m)\, dv_m \cdots dv_1.$$

The function f_Z is a joint PDF. As part of this course-level definition, we assume that the displayed improper iterated integrals exist and that their order may be changed. This regularity holds for the standard piecewise-smooth densities used in these notes; it replaces any need for a general integration theorem here.

For a continuous pair (X, Y),

$$F_{X,Y}(x, y) = \int_{-\infty}^{x} \int_{-\infty}^{y} f_{X,Y}(u, v)\, dv\, du.$$

The marginal densities are

$$f_X(x) = \int_{-\infty}^{\infty} f_{X,Y}(x, y)\, dy, \qquad f_Y(y) = \int_{-\infty}^{\infty} f_{X,Y}(x, y)\, dx.$$

Theorem 2.38 (Density factorization). *Let (X, Y) be a continuous random vector with marginal densities f_X, f_Y. Then X and Y are independent if and only if*

$$(x, y) \longmapsto f_X(x) f_Y(y)$$

is a joint density of (X, Y).

Proof. If the product is a joint density, then

$$\begin{aligned} F_{X,Y}(x,y) &= \int_{-\infty}^{x}\int_{-\infty}^{y} f_X(u)f_Y(v)\,dv\,du \\ &= F_X(x)F_Y(y), \end{aligned}$$

so X and Y are independent. Conversely, independence gives $F_{X,Y}(x,y) = F_X(x)F_Y(y)$, and the same calculation shows that the product integrates to the joint distribution function. A density may be changed at isolated points without changing any probability, so this formulation does not require equality between arbitrary density versions at every point. □

For every $r \geq 1$, the standard events in $\mathbb{R}^r$ include open and closed sets, rectangles, and half-spaces. This event collection is part of the given probability model; no construction of it is needed in this course.

Definition 2.39 (Valid vector transformation)**.** *Let $m, k \in \mathbb{N}$ with $m, k \geq 1$, let Z be an $\mathbb{R}^m$-valued random variable, and let $g = (g_1, \ldots, g_k) : \mathbb{R}^m \to \mathbb{R}^k$. We call g a valid transformation of Z if, for every $j \in \{1, \ldots, k\}$ and $y \in \mathbb{R}$,*

$$\{\omega \in \Omega : g_j(Z(\omega)) \leq y\} \in \mathcal{F}.$$

Equivalently, every component $g_j(Z)$ is a random variable.

If g is valid for Z, then $g(Z)$ is an $\mathbb{R}^k$-valued random variable by Definition 2.32. Continuous functions between Euclidean spaces are valid for the standard models in these notes.

2.6.3 Properties of expectations

The expectation rule in Definition 2.27 extends directly to a function of a random vector. If Z is discrete with support $\{z_1, z_2, \ldots\}$, then

$$\mathbb{E}[h(Z)] = \sum_i h(z_i) f_Z(z_i).$$

If Z has a joint density, then

$$\mathbb{E}[h(Z)] = \int_{\mathbb{R}^m} h(z) f_Z(z)\,dz,$$

where the right-hand side is an ordinary iterated improper integral. In both cases the corresponding absolute sum or integral must be finite. The results below concern random variables that are functions of a common discrete or continuous random vector. This covers the finite, countable, and density models developed in this chapter without invoking an abstract integral.

An event occurs **almost surely**, abbreviated a.s., if it has probability one. Thus $X \leq Y$ a.s. means $\mathbb{P}(\{\omega : X(\omega) \leq Y(\omega)\}) = 1$.

Theorem 2.40 (Properties of expectation)**.** *Let the random variables in each statement be functions of a common discrete or continuous random vector, as described above. Then:*

1. *If X is integrable and $\alpha \in \mathbb{R}$, then*

$$\mathbb{E}[\alpha X] = \alpha\,\mathbb{E}[X].$$

2. *If X and Y are integrable, then $X+Y$ is integrable and*

$$\mathbb{E}[X+Y] = \mathbb{E}[X] + \mathbb{E}[Y].$$

3. *If (X,Y) is itself a discrete or continuous random vector, and if X and Y are independent and integrable, then XY is integrable and*

$$\mathbb{E}[XY] = \mathbb{E}[X]\,\mathbb{E}[Y].$$

4. *If X and Y are integrable and $X \le Y$ a.s., then*

$$\mathbb{E}[X] \le \mathbb{E}[Y].$$

5. *If $A \subseteq \mathbb{R}$ is a standard event, then*

$$\mathbb{E}[\mathbb{1}_A(X)] = \mathbb{P}(X \in A).$$

Proof. Write $X = h(Z)$ and $Y = k(Z)$ for the common random vector Z. Homogeneity and additivity follow by applying the corresponding algebraic operation to the absolutely convergent sum or integral defining expectation. The bound $|h+k| \le |h| + |k|$ first gives the required integrability. For example,

$$\begin{aligned}\mathbb{E}[X+Y] &= \int_{\mathbb{R}^m} \big(h(z)+k(z)\big) f_Z(z)\,dz \\ &= \mathbb{E}[X] + \mathbb{E}[Y]\end{aligned}$$

in the density case, and the discrete calculation is the same with a sum.

For the product assertion, first suppose that (X,Y) is discrete. Independence gives

$$\begin{aligned}\mathbb{E}[|XY|] &= \sum_{x,y} |xy| f_X(x) f_Y(y) = \mathbb{E}[|X|]\,\mathbb{E}[|Y|] < \infty, \\ \mathbb{E}[XY] &= \sum_{x,y} xy f_X(x) f_Y(y) = \mathbb{E}[X]\,\mathbb{E}[Y].\end{aligned}$$

For a continuous pair, use Definition 2.38 and replace the sums by integrals. The regularity in the definition of a joint density permits their order to be changed.

If $X \le Y$ a.s., then $Y - X \ge 0$ except on an event of probability zero. The sum or integral of $k - h$ is nonnegative, and linearity gives $\mathbb{E}[Y] - \mathbb{E}[X] \ge 0$. Finally, $\mathbb{1}_A(X)$ equals 1 on $\{X \in A\}$ and 0 elsewhere, which proves the last identity. □

If X and Y are covered by the preceding expectation rule and have finite second moments, their covariance is

$$\operatorname{Cov}(X,Y) := \mathbb{E}\big[(X - \mathbb{E}[X])(Y - \mathbb{E}[Y])\big] = \mathbb{E}[XY] - \mathbb{E}[X]\,\mathbb{E}[Y].$$

The expectation of the product is finite because $2|XY| \le X^2 + Y^2$. In particular, independent random variables with finite second moments have zero covariance. The converse need not hold.

Definition 2.41 (Convex function). *A function $\phi : \mathbb{R} \to \mathbb{R}$ is convex if, for all $x_1, x_2 \in \mathbb{R}$ and $t \in [0, 1]$,*

$$\phi(tx_1 + (1-t)x_2) \le t\phi(x_1) + (1-t)\phi(x_2).$$

Lemma 2.42 (Jensen's inequality). *Let $I \subseteq \mathbb{R}$ be an open interval. Suppose that $\mathbb{P}(X \in I) = 1$, that X and $\phi(X)$ are integrable, and that $\phi : I \to \mathbb{R}$ is differentiable and convex. Then*

$$\phi(\mathbb{E}[X]) \le \mathbb{E}[\phi(X)].$$

If ϕ is concave, the inequality is reversed.

Proof. A differentiable convex function lies above each of its tangent lines. At $m = \mathbb{E}[X] \in I$,

$$\phi(x) \ge \phi(m) + \phi'(m)(x - m), \qquad x \in I.$$

The mean lies in I because I is open and X lies strictly between its finite endpoints, when present, with probability one. Substitute $x = X$ and take expectations. Since $\mathbb{E}[X - m] = 0$, the result follows. Apply this result to $-\phi$ for the concave case. □

2.6.4 Optional note on moment spaces

This subsection packages moment conditions into the language of normed spaces. It is not needed for the required route through this chapter. Fix one discrete or continuous random vector for this subsection. Every random variable below is a function of that vector. Thus all expectations are the concrete sums or iterated integrals defined above.

Definition 2.43 (The space $L^p(\mathbb{P})$). *Let $1 \le p < \infty$. The space $L^p(\mathbb{P})$ consists of random variables X satisfying*

$$\mathbb{E}[|X|^p] < \infty.$$

Random variables that are equal almost surely are identified as the same element of $L^p(\mathbb{P})$. Pointwise addition and scalar multiplication give the usual vector-space operations.

Definition 2.44 (The L^p norm). *For $X \in L^p(\mathbb{P})$, define*

$$\|X\|_{L^p(\mathbb{P})} := \mathbb{E}[|X|^p]^{1/p}.$$

Theorem 2.45 (Hölder's inequality). *Let $1 < p, q < \infty$ satisfy $1/p + 1/q = 1$. If $X \in L^p(\mathbb{P})$ and $Y \in L^q(\mathbb{P})$, then*

$$\mathbb{E}[|XY|] \le \|X\|_{L^p(\mathbb{P})} \|Y\|_{L^q(\mathbb{P})}.$$

Proof. Young's inequality states that

$$ab \le \frac{a^p}{p} + \frac{b^q}{q}, \qquad a, b \ge 0.$$

It follows from elementary calculus by maximizing $ab - a^p/p$ over $a \ge 0$. If either norm in the desired inequality is zero, then $XY = 0$ almost surely and the result is immediate. Otherwise, apply Young's inequality to

$$a = \frac{|X|}{\|X\|_{L^p(\mathbb{P})}}, \qquad b = \frac{|Y|}{\|Y\|_{L^q(\mathbb{P})}}$$

and take expectations. The right-hand side becomes $1/p + 1/q = 1$. □

Theorem 2.46 (Minkowski's inequality)**.** *For $1 \le p < \infty$ and $X, Y \in L^p(\mathbb{P})$,*

$$\|X+Y\|_{L^p(\mathbb{P})} \le \|X\|_{L^p(\mathbb{P})} + \|Y\|_{L^p(\mathbb{P})}.$$

Consequently, $L^p(\mathbb{P})$, with random variables equal almost surely identified, is a normed vector space.

Proof. The elementary bound

$$|X+Y|^p \le 2^{p-1}(|X|^p + |Y|^p)$$

first shows that $X + Y \in L^p(\mathbb{P})$. For $p = 1$, the result follows from $|X+Y| \le |X| + |Y|$. Let $p > 1$ and $q = p/(p-1)$. By Hölder's inequality,

$$\begin{aligned} \mathbb{E}\left[|X+Y|^p\right] &\le \mathbb{E}\left[|X||X+Y|^{p-1}\right] + \mathbb{E}\left[|Y||X+Y|^{p-1}\right] \\ &\le \left(\|X\|_{L^p(\mathbb{P})} + \|Y\|_{L^p(\mathbb{P})}\right)\|X+Y\|_{L^p(\mathbb{P})}^{p-1}. \end{aligned}$$

If $\|X+Y\|_{L^p(\mathbb{P})} = 0$, the conclusion is immediate. Otherwise, divide by its $(p-1)$-st power. The remaining norm axioms follow from the corresponding properties of absolute value; definiteness uses the identification of random variables equal almost surely. □

Theorem 2.47 (Consequences of Hölder's inequality)**.** *The following statements hold.*

1. *If $1 < p, q < \infty$, $1/p + 1/q = 1$, $X \in L^p(\mathbb{P})$, and $Y \in L^q(\mathbb{P})$, then $XY \in L^1(\mathbb{P})$.*
2. *If $1 \le s \le r < \infty$, then $L^r(\mathbb{P}) \subseteq L^s(\mathbb{P})$ and*

$$\|X\|_{L^s(\mathbb{P})} \le \|X\|_{L^r(\mathbb{P})}.$$

3. *Let $1 \le r < p, q < \infty$ satisfy $1/p + 1/q = 1/r$. If $X \in L^p(\mathbb{P})$ and $Y \in L^q(\mathbb{P})$, then $XY \in L^r(\mathbb{P})$.*

Proof. The first statement is Definition 2.45. For the second, the case $s = r$ is immediate. If $s < r$, apply Hölder's inequality to $|X|^s \cdot 1$ with exponents r/s and $r/(r-s)$. Since $\mathbb{P}(\Omega) = 1$,

$$\mathbb{E}\left[|X|^s\right] \le \mathbb{E}\left[|X|^r\right]^{s/r}.$$

Taking the s-th root gives the claim. For the third statement, apply Hölder's inequality to $|X|^r$ and $|Y|^r$ with conjugate exponents p/r and q/r. □

2.6.5 Conditional Random Variables

Let X and Y be independent Bernoulli(1/2) random variables, representing two fair coin tosses, and set $S = X + Y$. If $X = 0$, then $S = Y$, so

$$\mathbb{P}(S = 0 \mid X = 0) = \frac{1}{2}, \qquad \mathbb{P}(S = 1 \mid X = 0) = \frac{1}{2}.$$

If $X = 1$, then

$$\mathbb{P}(S = 1 \mid X = 1) = \frac{1}{2}, \qquad \mathbb{P}(S = 2 \mid X = 1) = \frac{1}{2}.$$

These are ordinary conditional probabilities on events of positive probability.

Definition 2.48 (Conditional distribution given an event)**.** *Let (X, Y) be a real-valued random vector, and let $A \subseteq \mathbb{R}$ be a standard event satisfying $\mathbb{P}(Y \in A) > 0$. The conditional distribution function of X given $Y \in A$ is*

$$F_{X|Y}(x \mid A) := \mathbb{P}(X \leq x \mid Y \in A) = \frac{\mathbb{P}(X \leq x, Y \in A)}{\mathbb{P}(Y \in A)}.$$

For example, if $F_Y(y) > 0$, then

$$F_{X|Y}(x \mid (-\infty, y])F_Y(y) = F_{X,Y}(x, y).$$

The positivity assumption is needed because the conditional probability contains $F_Y(y)$ in its denominator.

Definition 2.49 (Conditional PMF or PDF)**.** *Suppose first that (X, Y) is discrete. For y with $f_Y(y) > 0$, define*

$$f_{X|Y}(x \mid y) := \mathbb{P}(X = x \mid Y = y) = \frac{f_{X,Y}(x, y)}{f_Y(y)}.$$

If instead (X, Y) has a joint density, then for y with $f_Y(y) > 0$, define the conditional density

$$f_{X|Y}(x \mid y) := \frac{f_{X,Y}(x, y)}{f_Y(y)}.$$

The conditional PMF or PDF is normalized. Indeed, summing or integrating the numerator over x gives $f_Y(y)$, and division by the positive denominator gives one.

For a continuous Y, the expression $Y = y$ does not denote an event on which we can condition using Definition 1.12, because $\mathbb{P}(Y = y) = 0$. The density ratio is a separate definition. Its pointwise value depends on the chosen versions of the densities, so we use the continuous versions in the standard models below. When the joint and marginal densities are continuous near y, the ratio is motivated by conditioning on shrinking intervals around y.

Lemma 2.50. *In either the discrete or continuous setting above, if $f_Y(y) > 0$, then*

$$f_{X|Y}(x \mid y)f_Y(y) = f_{X,Y}(x, y).$$

Proof. Multiply the defining ratio by its positive denominator. □

Example 2.51. *For the pair (S, X) in the opening example, each of the four underlying coin outcomes has probability $1/4$. Hence*

$$f_{S,X}(s, x) = f_{S|X}(s \mid x)f_X(x),$$

where $f_X(0) = f_X(1) = 1/2$. The four nonzero joint masses are

$$f_{S,X}(0, 0) = f_{S,X}(1, 0) = f_{S,X}(1, 1) = f_{S,X}(2, 1) = \frac{1}{4}.$$

For a discrete pair and a value y with $f_Y(y) > 0$, define

$$\mathbb{E}[X \mid Y = y] := \sum_x x f_{X|Y}(x \mid y).$$

For a pair with a joint density and a value y with $f_Y(y) > 0$, define

$$\mathbb{E}[X \mid Y = y] := \int_{-\infty}^{\infty} x f_{X|Y}(x \mid y)\, dx.$$

These definitions require the corresponding absolute sum or integral to be finite. Write $g(y) = \mathbb{E}[X \mid Y = y]$ wherever it is defined, and set $g(y) = 0$ elsewhere. We use the following notation only when $g(Y)$ is a random variable. The conditional expectation given the random variable Y is

$$\mathbb{E}[X \mid Y] := g(Y).$$

More generally, if $H = h(X, Y)$, replace x in the preceding sum or integral by $h(x, y)$. This defines $\mathbb{E}[H \mid Y = y]$, and hence $\mathbb{E}[H \mid Y]$, provided the resulting $g(Y)$ is a random variable. The same definitions apply when X or Y is a random vector, with the corresponding multiple sums or iterated integrals.

Theorem 2.52 (Tower property)**.** *Let (X, Y) be a discrete random vector or a random vector with a joint density. Suppose $H = h(X, Y)$ is integrable and that the conditional absolute sums or integrals above are finite whenever $f_Y(y) > 0$. Assume that the resulting function $g(y) = \mathbb{E}[H \mid Y = y]$, extended by zero, makes $g(Y)$ a random variable. In the density case, also assume that the iterated improper integrals of $|h(x, y)| f_{X,Y}(x, y)$ may be evaluated in either order. Then*

$$\mathbb{E}\big[\mathbb{E}[H \mid Y]\big] = \mathbb{E}[H].$$

In particular, taking $h(x, y) = x$ gives $\mathbb{E}[\mathbb{E}[X \mid Y]] = \mathbb{E}[X]$.

Proof. We give the joint-density calculation. If $f_Y(y) > 0$, then Definition 2.50 gives

$$g(y) f_Y(y) = \int_{-\infty}^{\infty} h(x, y) f_{X,Y}(x, y)\, dx.$$

If $f_Y(y) = 0$, the nonnegative slice $f_{X,Y}(\,\cdot\,, y)$ has integral zero. Its product with $h(\,\cdot\,, y)$ therefore also has integral zero under the stated ordinary-integral regularity. Since $g(y) = 0$ in this case, the preceding identity holds for every y. Consequently,

$$\begin{aligned}
\mathbb{E}\big[\mathbb{E}[H \mid Y]\big] &= \int_{-\infty}^{\infty} g(y) f_Y(y)\, dy \\
&= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} h(x, y) f_{X,Y}(x, y)\, dx\, dy = \mathbb{E}[H].
\end{aligned}$$

In the discrete case, replace the integrals by absolutely convergent sums. The vector-valued conditioning case is the same calculation with additional sums or integrals over the components of Y. □

Filtering i.i.d. data

A filter keeps an observation when it belongs to a fixed set A. Filtering changes both the distribution of the observations and the number of observations that remain.

For example, roll a fair die independently until the result belongs to $A = \{5, 6\}$. The probability of acceptance on each roll is $p = 1/3$. The first retained value is equally likely to be 5 or 6, and

$$\mathbb{P}(T = t) = \left(\frac{2}{3}\right)^{t-1} \frac{1}{3}, \qquad t = 1, 2, \ldots,$$

where T is the index of the first retained roll. If instead the die is rolled a fixed number n of times, then the number of retained values is not fixed in advance.

For random vectors, we use the following extension of the i.i.d. definition. Let $m \in \mathbb{N}$ with $m \geq 1$. The random vectors $X_1, X_2, \ldots$ in $\mathbb{R}^m$ are i.i.d. if they have the same distribution and, for every positive integer r, every choice of distinct indices $i_1, \ldots, i_r$, and all standard events $B_1, \ldots, B_r \subseteq \mathbb{R}^m$,

$$\mathbb{P}(X_{i_1} \in B_1, \ldots, X_{i_r} \in B_r) = \prod_{j=1}^{r} \mathbb{P}(X_{i_j} \in B_j).$$

Definition 2.53 (Stopping time for a data sequence). *Let $m \in \mathbb{N}$ with $m \geq 1$, and let $X_1, X_2, \ldots$ be random vectors in $\mathbb{R}^m$. A map $T : \Omega \to \{1, 2, \ldots\} \cup \{\infty\}$ is a stopping time for this sequence if, for every integer $t \geq 1$, there is a standard event $C_t \subseteq \mathbb{R}^{mt}$ such that*

$$\{T \leq t\} = \{(X_1, \ldots, X_t) \in C_t\}.$$

Thus, after observing $X_1, \ldots, X_t$, one can determine whether stopping has occurred by time t. More advanced probability courses express the same condition using a filtration.

Theorem 2.54 (First accepted observation). *Let $m \in \mathbb{N}$ with $m \geq 1$, let $X_1, X_2, \ldots$ be i.i.d. random vectors in $\mathbb{R}^m$, and let $A \subseteq \mathbb{R}^m$ be a fixed standard event. Set*

$$p := \mathbb{P}(X_1 \in A),$$

and assume that $p > 0$. The first acceptance index

$$T := \inf\{i \geq 1 : X_i \in A\},$$

where the infimum of the empty set is ∞, is a stopping time. It is finite almost surely and satisfies

$$\mathbb{P}(T = t) = (1 - p)^{t-1} p, \qquad t = 1, 2, \ldots.$$

Define the real-valued null modification

$$\widetilde{T} := \begin{cases} T, & T < \infty, \\ 1, & T = \infty. \end{cases} \qquad \text{Then} \qquad \mathbb{E}\,[\widetilde{T}] = \frac{1}{p}.$$

Define $Z = X_T$ when $T < \infty$, and give Z any fixed value on the probability-zero event $\{T = \infty\}$. Then, for every standard event $B \subseteq \mathbb{R}^m$,

$$\mathbb{P}(Z \in B) = \mathbb{P}(X_1 \in B \mid X_1 \in A) = \frac{\mathbb{P}(X_1 \in A \cap B)}{p}.$$

Proof. For every integer $t \geq 1$, let C_t be the set of $(x_1, \ldots, x_t)$ for which $x_i \in A$ for at least one $i \leq t$. Then

$$\{T \leq t\} = \{(X_1, \ldots, X_t) \in C_t\}.$$

The set C_t is a finite union of rectangles with A in one coordinate and $\mathbb{R}^m$ in the others, so it is standard. Hence T is a stopping time. Moreover,

$$\{T = t\} = \{X_1 \notin A, \dots, X_{t-1} \notin A, X_t \in A\}.$$

Independence gives

$$\mathbb{P}(T = t) = (1-p)^{t-1}p.$$

Since

$$\sum_{t=1}^{\infty}(1-p)^{t-1}p = 1,$$

the index T is finite almost surely. Differentiating the geometric series gives

$$\mathbb{E}[\widetilde{T}] = p\sum_{t=1}^{\infty} t(1-p)^{t-1} = \frac{1}{p}.$$

Let $B \subseteq \mathbb{R}^m$ be a standard event. For $t \geq 1$, independence also gives

$$\begin{aligned}\mathbb{P}(Z \in B, T = t) &= \mathbb{P}(X_1 \notin A, \dots, X_{t-1} \notin A, X_t \in A \cap B)\\ &= (1-p)^{t-1}\,\mathbb{P}(X_1 \in A \cap B).\end{aligned}$$

The events $\{T = t\}$ are disjoint and cover an event of probability one. Therefore,

$$\begin{aligned}\mathbb{P}(Z \in B) &= \sum_{t=1}^{\infty}\mathbb{P}(Z \in B, T = t)\\ &= \frac{\mathbb{P}(X_1 \in A \cap B)}{p}.\end{aligned}$$

This is the conditional probability of $X_1 \in B$ given $X_1 \in A$. □

Corollary 2.55 (Successive accepted observations)**.** *Under the assumptions of Definition 2.54, set $T_0 = 0$ and, for $j \geq 1$, define*

$$T_j := \inf\{i > T_{j-1} : X_i \in A\}.$$

Use the same convention for the infimum of the empty set as in Definition 2.54. Choose any fixed $z_0 \in \mathbb{R}^m$. On $\{T_j < \infty\}$, set $D_j := T_j - T_{j-1}$ and $Z_j := X_{T_j}$. On $\{T_j = \infty\}$, set $D_j := 1$ and $Z_j := z_0$. Then every T_j is a stopping time and is finite almost surely. The waiting times $D_1, D_2, \dots$ are i.i.d. with

$$\mathbb{P}(D_j = t) = (1-p)^{t-1}p, \qquad t = 1, 2, \dots,$$

and the retained observations $Z_1, Z_2, \dots$ are i.i.d. with the distribution of X_1 conditional on $X_1 \in A$.

Proof. For positive integers j and t,

$$\{T_j \leq t\} = \left\{\sum_{i=1}^{t}\mathbb{1}_{\{X_i \in A\}} \geq j\right\}.$$

This event is determined by $X_1, \dots, X_t$, so T_j is a stopping time.

Fix a positive integer r, positive integers $t_1, \dots, t_r$, and standard events $B_1, \dots, B_r \subseteq \mathbb{R}^m$. Put $s_j = t_1 + \cdots + t_j$. On $\{T_r < \infty\}$, the conditions

$$\{D_j = t_j,\ Z_j \in B_j,\ j = 1, \dots, r\}$$

require rejection at the first $t_j - 1$ positions of each block and an observation in $A \cap B_j$ at position s_j. These conditions concern distinct observations, so independence gives

$$\mathbb{P}(T_r < \infty,\ D_j = t_j,\ Z_j \in B_j,\ j = 1, \dots, r) = \prod_{j=1}^{r} (1-p)^{t_j - 1}\, \mathbb{P}(X_1 \in A \cap B_j).$$

Taking every $B_j = \mathbb{R}^m$ and summing over all $t_1, \dots, t_r \geq 1$ shows that $\mathbb{P}(T_r < \infty) = 1$. We may therefore omit the condition $T_r < \infty$ from the preceding identity. Taking every $B_j = \mathbb{R}^m$ now proves the joint product formula for the waiting times. Summing over $t_1, \dots, t_r$ gives

$$\mathbb{P}(Z_1 \in B_1, \dots, Z_r \in B_r) = \prod_{j=1}^{r} \mathbb{P}(X_1 \in B_j \mid X_1 \in A).$$

$\square$

Theorem 2.56 (Filtering a fixed i.i.d. sample)**.** *Let $m, n \in \mathbb{N}$ with $m, n \geq 1$, let $X_1, \dots, X_n$ be i.i.d. random vectors in $\mathbb{R}^m$, and let $A \subseteq \mathbb{R}^m$ be a fixed standard event. Set*

$$p := \mathbb{P}(X_1 \in A), \qquad I_i := \mathbb{1}_{\{X_i \in A\}}, \qquad N_A := \sum_{i=1}^{n} I_i.$$

If $0 < p < 1$, then

$$\mathbb{P}(N_A = k) = \binom{n}{k} p^k (1-p)^{n-k}, \qquad k = 0, \dots, n.$$

If $p = 0$, then $N_A = 0$ almost surely, while if $p = 1$, then $N_A = n$ almost surely. Together, these cases define the binomial distribution with parameters n and p.

Choose a fixed $z_0 \in \mathbb{R}^m$. For $j = 1, \dots, n$, let Z_j be the j-th retained observation in the original order when $N_A \geq j$, and set $Z_j = z_0$ when $N_A < j$.

Assume now that $p > 0$, and let $k \in \{1, \dots, n\}$ satisfy $\mathbb{P}(N_A = k) > 0$. Then, for every choice of standard events $B_1, \dots, B_k \subseteq \mathbb{R}^m$,

$$\mathbb{P}(Z_1 \in B_1, \dots, Z_k \in B_k \mid N_A = k) = \prod_{j=1}^{k} \frac{\mathbb{P}(X_1 \in A \cap B_j)}{p}.$$

Thus, conditional on retaining k observations, the retained values are i.i.d. with the distribution of X_1 conditional on $X_1 \in A$.

Proof. Fix $k \in \{0, \dots, n\}$. For each set $S \subseteq \{1, \dots, n\}$ with $|S| = k$, let

$$E_S := \left(\bigcap_{i \in S} \{X_i \in A\} \right) \cap \left(\bigcap_{i \notin S} \{X_i \notin A\} \right).$$

The events E_S are disjoint, and their union is $\{N_A = k\}$. If $0 < p < 1$, independence gives

$$\mathbb{P}(E_S) = p^k(1-p)^{n-k}.$$

There are $\binom{n}{k}$ possible sets S, which proves the binomial formula. The cases $p = 0$ and $p = 1$ follow directly from the definition of N_A.

Suppose that $0 < p < 1$, that $k \geq 1$, and that $S = \{i_1 < \cdots < i_k\}$. On E_S, the retained values satisfy $Z_j = X_{i_j}$. Hence independence gives

$$\mathbb{P}(E_S, Z_1 \in B_1, \ldots, Z_k \in B_k) = (1-p)^{n-k} \prod_{j=1}^{k} \mathbb{P}(X_1 \in A \cap B_j).$$

Summing over the $\binom{n}{k}$ possible sets S and dividing by

$$\mathbb{P}(N_A = k) = \binom{n}{k} p^k (1-p)^{n-k}$$

gives the stated conditional probability. If $p = 1$, the only possible value is $k = n$, the conditioning event has probability one, and the same product identity follows directly from independence. $\square$

Remark 2.57 (The sample size after filtering)**.** *The original sample size n is fixed, but the retained sample size N_A is random unless p is zero or one. Conditional on $N_A = k$, a calculation or concentration bound for the retained observations uses k, not n. If $N_A = 0$, there is no retained observation, and a sample mean or ratio based on the filtered data is undefined unless a separate convention is specified.*

The set A is fixed before the observations are inspected. If the filter is selected using the same data, the retained observations need not have the conditional i.i.d. distribution in Definition 2.56. A filter learned from separate training data is fixed after conditioning on that training data, so the theorem applies to an independent test sample.

2.6.6 Optional: Mixed random variables

A random variable may have both point masses and a continuous part. Let $B \sim \text{Bernoulli}(1/2)$, let U be uniform on $[0, 1]$, assume that B and U are independent, and set $X = BU$. If $B = 0$, then $X = 0$; if $B = 1$, then $X = U$. Therefore,

$$\mathbb{P}(X = 0) = \mathbb{P}(B = 0) + \mathbb{P}(B = 1, U = 0) = \frac{1}{2},$$

where $\mathbb{P}(U = 0) = 0$ by Definition 2.18, and

$$F_X(x) = \begin{cases} 0, & x < 0, \\ \dfrac{1}{2} + \dfrac{x}{2}, & 0 \leq x \leq 1, \\ 1, & x > 1. \end{cases}$$

This distribution is not discrete. Indeed, if $C \subseteq \mathbb{R}$ is countable, then countable additivity and Definition 2.18 give $\mathbb{P}(U \in C) = 0$. Hence $\mathbb{P}(X \in C) \leq 1/2$, even when $0 \in C$, so no countable set supports X with probability one. The distribution is also not continuous in the sense of Definition 2.17 because its distribution function jumps by $1/2$ at zero. The jump records the point mass, while the slope $1/2$ on $(0, 1)$ records the continuous part.

2.7 Examples of modeling

A probability model distinguishes an outcome from the numerical information recorded from that outcome. The examples below specify these objects before introducing a prediction question. In both examples the probability law is unknown; the model assumes that the law exists and later chapters explain how data can be used to estimate quantities defined from it.

2.7.1 Email spam filtering

Fix a finite character alphabet $\mathcal{A}$. Record the first 1000 characters of the next incoming email, padding shorter messages with a special blank character. A concrete finite sample space is then

$$\Omega = \mathcal{A}^{1000}, \qquad \mathcal{F} = 2^{\Omega}.$$

An outcome $\omega \in \Omega$ is one recorded message. Let $\mathbb{P}$ be its unknown probability law. The law need not be uniform: common messages and character patterns may have much larger probabilities than unusual ones.

Suppose that every message has a correct spam label. Define

$$X(\omega) = \begin{cases} 1, & \omega \text{ is spam}, \\ 0, & \omega \text{ is not spam}. \end{cases}$$

The value $X(\omega)$ is a label, whereas $\{X = 1\}$ is the event consisting of all spam messages. Since the model is finite, X is automatically a random variable.

As a simple feature, let

$$D := \{\omega \in \Omega : \omega \text{ contains the word "donate"}\}, \qquad Z := \mathbb{1}_D.$$

Assume that $\mathbb{P}(Z = 1) > 0$. The comparison

$$\mathbb{P}(X = 1 \mid Z = 1) > \mathbb{P}(X = 1) \tag{2.27}$$

says that spam is more common among messages containing this word than among all messages. It describes an association under $\mathbb{P}$; it does not by itself give a complete classifier or show that the word causes spam.

If $\mathbb{P}(X = 1 \mid Z = 1) = 1$, then the rule that flags messages with $Z = 1$ has no false positives under the model. It may still miss messages for which $X = 1$ and $Z = 0$, so it need not be a perfect classifier. If instead

$$\mathbb{P}(X = 1 \mid Z = 1) = \mathbb{P}(X = 1),$$

then the events $\{X = 1\}$ and $\{Z = 1\}$ are independent. Because X and Z are binary, this is equivalent to independence of the two random variables. Practical bag-of-words methods use many such features. Their prediction rules and their performance criteria are developed in later chapters.

2.7.2 Number of website requests during a day

Consider the experiment of recording all requests received by a website on a randomly selected day. We take a probability triple $(\Omega, \mathcal{F}, \mathbb{P})$ as part of the model. There is no need to construct the event collection from every possible log file. Instead, we explicitly assume that each recorded quantity below is a random variable.

Let

$$X : \Omega \to \{0, 1, 2, \ldots\}$$

be the number of requests. Its distribution function is

$$F_X(x) = \mathbb{P}(X \le x).$$

If extra capacity is needed when more than ten requests arrive, then the probability of needing that capacity is

$$\mathbb{P}(X > 10) = 1 - F_X(10).$$

Information known before the requests arrive can refine the prediction. Let $Z : \Omega \to \{1, \ldots, 7\}$ record the day of the week. For any $z \in \{1, \ldots, 7\}$ satisfying $\mathbb{P}(Z = z) > 0$, define

$$F_{X|Z}(x \mid z) := \mathbb{P}(X \le x \mid Z = z).$$

The capacity probability for that day is

$$\mathbb{P}(X > 10 \mid Z = z) = 1 - F_{X|Z}(10 \mid z). \tag{2.28}$$

The positivity assumption is required because the conditional probability has $\mathbb{P}(Z = z)$ in its denominator.

One may include another binary variable H that records whether the day is a holiday. If $\mathbb{P}(Z = z, H = h) > 0$, then

$$\mathbb{P}(X > 10 \mid Z = z, H = h) = 1 - F_{X|Z,H}(10 \mid z, h).$$

Whether this additional information improves a prediction is a question about the joint distribution of (X, Z, H), not a consequence of the names of the variables.

2.7.3 A modeling checklist

For the finite and standard continuous models in this course, the following steps keep the mathematical objects separate.

1. Specify the experiment and an outcome ω. Use $\mathcal{F} = 2^\Omega$ when Ω is finite or countable. For a continuous sample space, treat the standard event collection as part of the given model.

2. State the probability law when it is known. If it is unknown, say explicitly which probabilities or distributions are the targets of estimation.

3. Define each observed feature or label as a random variable and state its range.

4. State sampling assumptions. Repeating an experiment creates a product experiment, but independence and identical distribution are additional modeling assumptions.

5. Check that every conditioning event has positive probability.

For example, let X_i be the indicator that the i-th sampled image contains a cat. If $X_1, \ldots, X_n$ are assumed to be i.i.d., then their average is a candidate estimator of $\mathbb{P}(X_1 = 1)$. This conclusion does not require the analyst to list every possible image. Its mathematical justification comes from the law of large numbers in Definition 3.26, which also makes the i.i.d. assumption explicit.

2.8 Exercises

Exercise 2.58 (A scalar transformation)**.** *Spell out the event condition in Definition 2.39 when $k = 1$. Compare it with the definition of a valid transformation in Definition 2.20.*

Exercise 2.59 (A discrete transformation)**.** *Let X be uniform on $\{1, 2, 3, 4, 5, 6\}$ and set $Y = (X - 3)^2$.*

1. *List the support of Y and compute its PMF using (2.18).*
2. *Compute the distribution function F_Y at every point where it jumps.*
3. *Compute $\mathbb{E}[Y]$ in two ways: from the PMF of Y and directly as $\mathbb{E}[(X - 3)^2]$.*

Exercise 2.60 (Squaring a uniform random variable)**.** *Let X be uniform on $[-1, 1]$ and set $Y = X^2$.*

1. *Use the direct method to derive $F_Y(y)$ for every $y \in \mathbb{R}$.*
2. *Differentiate where appropriate to find a density of Y. Explain why an unbounded density near zero can still define a probability model.*
3. *Compute $\mathbb{E}[Y]$ directly from X and again from the density of Y.*

Exercise 2.61 (Variance under an affine transformation)**.** *Suppose X has finite second moment and let $a, b \in \mathbb{R}$. Prove from the definitions that*

$$\mathbb{E}[aX + b] = a\,\mathbb{E}[X] + b, \qquad \mathbb{V}(aX + b) = a^2\,\mathbb{V}(X).$$

State where each moment assumption is used. Include the case $a = 0$.

Exercise 2.62 (A conditional probability table)**.** *The joint PMF of binary random variables X and Z is*

	$Z = 0$	$Z = 1$
$X = 0$	0.54	0.06
$X = 1$	0.16	0.24

Find both marginal PMFs, $f_{X|Z}(x \mid z)$ for each z, and $\mathbb{E}[X \mid Z]$. Verify the tower property by computing $\mathbb{E}[\mathbb{E}[X \mid Z]]$ and $\mathbb{E}[X]$ separately. Are X and Z independent?

Exercise 2.63 (A complete finite model)**.** *Refine the spam example by recording only strings of length three over the alphabet $\{a, b\}$. Give the eight strings the probabilities proportional to $1, 2, \ldots, 8$ in lexicographic order. Let X indicate that a string contains at least two occurrences of b, and let Z indicate that its first character is b.*

1. *Write down $(\Omega, \mathcal{F}, \mathbb{P})$ and verify that $\mathbb{P}$ is normalized.*
2. *Compute $\mathbb{P}(X = 1)$, $\mathbb{P}(Z = 1)$, and $\mathbb{P}(X = 1 \mid Z = 1)$.*
3. *Decide whether (2.27) holds and whether X and Z are independent.*

Exercise 2.64 (Computational distribution transformation)**.** *Use Python with `numpy` and `matplotlib`. Set the seed with `numpy.random.seed(2026)`, draw 20 000 independent standard normal values X_i with `numpy.random.normal`, and set $Y_i = X_i^2$.*

1. *Plot a density-normalized histogram of the* Y_i.

2. *On the same axes, plot the density in* (2.21) *on a grid that begins at a small positive number rather than at zero.*

3. *For* $y \in \{0.25, 1, 4\}$, *compare the empirical proportion* $n^{-1} \sum_{i=1}^{n} \mathbb{1}_{\{Y_i \le y\}}$ *with the value obtained from* (2.20). *State which observations in your output are numerical approximations and which identities follow from the probability model.*

Exercise 2.65 (Optional: filtering a fixed sample)**.** *Roll a fair die independently* $n = 60$ *times and retain only values in* $A = \{5, 6\}$. *Use* **`numpy`** *to repeat the full experiment* 5000 *times. Plot a histogram of the retained sample size* N_A *and compare its empirical mean with* $np = 20$. *Then condition your simulation output on* $N_A = 20$ *and compare the observed proportions of retained* 5*s and* 6*s. Explain which conclusions are predicted by Definition 2.56 and why the retained sample size must not be silently replaced by* 60.

Chapter 3

Concentration and Limits

Course guide

Scheduled lecture(s): Lectures 4–5 (and review in Lecture 10)
Required for those lectures: Lecture 4: Section 3.1 except 3.1.1; Lecture 5: Sections 3.2–3.4; Exercises: Section 3.5
Prerequisites: Chapter 2, especially Sections 2.5–2.6
Optional or advanced: Sections 3.1.1 and 3.2.1 are advanced.

3.1 Concentration inequalities

A concentration inequality bounds the probability that a random variable is far from a specified value, usually its expectation. We begin with a bound that requires only a finite expectation.

Theorem 3.1 (Markov's inequality). *Let* $(\Omega, \mathcal{F}, \mathbb{P})$ *be a probability triple and let* X *be a nonnegative real-valued RV with* $\mathbb{E}[X] < \infty$. *Then, for every* $\epsilon > 0$,

$$\mathbb{P}(X \geq \epsilon) \leq \frac{\mathbb{E}[X]}{\epsilon}. \tag{3.1}$$

Proof. For every outcome $\omega \in \Omega$,

$$X(\omega) \geq \epsilon \mathbb{1}_{\{X \geq \epsilon\}}(\omega).$$

Taking expectations and using the expectation of an indicator gives

$$\mathbb{E}[X] \geq \epsilon\, \mathbb{E}[\mathbb{1}_{\{X \geq \epsilon\}}] = \epsilon\, \mathbb{P}(X \geq \epsilon).$$

Division by ϵ proves the result. □

Applying Markov's inequality to transformations of a random variable gives the following consequences.

Proposition 3.2 (Chebyshev's inequality). *Let* X *be a real-valued RV and let* $\epsilon > 0$. *If* $\mathbb{E}[|X|] < \infty$, *then*

$$\mathbb{P}(|X| \geq \epsilon) \leq \frac{\mathbb{E}[|X|]}{\epsilon}.$$

If $\mathbb{E}[X^2] < \infty$, *then also*

$$\mathbb{P}(|X| \geq \epsilon) = \mathbb{P}(X^2 \geq \epsilon^2) \leq \frac{\mathbb{E}[X^2]}{\epsilon^2},$$
$$\mathbb{P}(|X - \mathbb{E}[X]| \geq \epsilon) = \mathbb{P}((X - \mathbb{E}[X])^2 \geq \epsilon^2) \leq \frac{\mathbb{E}[(X - \mathbb{E}[X])^2]}{\epsilon^2} = \frac{\mathbb{V}(X)}{\epsilon^2}.$$

Proof. Apply Definition 3.1 to $|X|$, X^2, and $(X - \mathbb{E}[X])^2$, respectively. □

Recall that a family is independent and identically distributed (i.i.d.) if it is mutually independent and all its members have the same distribution. For real-valued RVs with common distribution function F, we write $X_1, \ldots, X_n \stackrel{\text{IID}}{\sim} F$.

For example, let X_i be the indicator that the i-th toss of a fair coin is Heads. Then $\overline{X}_n$ is the observed fraction of Heads. A concentration bound makes precise how unlikely it is for $\overline{X}_n$ to be far from $1/2$ at a fixed sample size. This is a probability statement, not a claim that every infinite sequence of tosses has limiting frequency $1/2$. The latter distinction is addressed by the laws of large numbers later in the chapter.

Lemma 3.3 (Hoeffding's lemma). *Let X be a real-valued RV. Suppose that $a < b$, $\mathbb{P}(X \in [a, b]) = 1$, and $\mathbb{E}[X] = 0$. Then, for every $s \in \mathbb{R}$,*

$$\mathbb{E}[e^{sX}] \leq \exp\left(\frac{s^2(b-a)^2}{8}\right).$$

Proof. Since $x \mapsto e^{sx}$ is convex,

$$e^{sx} \leq \frac{b-x}{b-a}e^{sa} + \frac{x-a}{b-a}e^{sb}$$

for $a \leq x \leq b$. Taking expectations gives

$$\mathbb{E}[e^{sX}] \leq \frac{b}{b-a}e^{sa} - \frac{a}{b-a}e^{sb}.$$

Set

$$h = s(b-a), \qquad p = \frac{-a}{b-a}, \qquad L(h) = -hp + \ln(1 - p + pe^h).$$

The assumptions imply $a \leq 0 \leq b$, so $p \in [0, 1]$. Moreover,

$$\frac{b}{b-a}e^{sa} - \frac{a}{b-a}e^{sb} = e^{L(h)}. \tag{3.2}$$

Direct differentiation gives $L(0) = L'(0) = 0$ and

$$L''(h) = q(h)(1 - q(h)), \qquad q(h) := \frac{pe^h}{1 - p + pe^h}.$$

Since $q(h) \in [0, 1]$, we have $L''(h) \leq 1/4$. Taylor's theorem therefore gives

$$L(h) \leq \frac{h^2}{8} = \frac{s^2(b-a)^2}{8}.$$

Combine this bound with (3.2). □

Theorem 3.4 (Hoeffding's inequality)**.** *Let $n \geq 1$, and let $X_1, \dots, X_n$ be independent real-valued RVs with finite expectations. Suppose that, for constants $a_i < b_i$,*

$$\mathbb{P}(X_i - \mathbb{E}[X_i] \in [a_i, b_i]) = 1, \qquad i = 1, \dots, n,$$

Then, for every $\epsilon > 0$,

$$\mathbb{P}(\overline{X}_n - \mathbb{E}[\overline{X}_n] \geq \epsilon) \leq \exp\left(-\frac{2n^2\epsilon^2}{\sum_{i=1}^n (b_i - a_i)^2}\right), \qquad \overline{X}_n = \frac{1}{n}\sum_{i=1}^n X_i.$$

Proof. Let $S_n = \sum_{i=1}^n X_i$ and let $s, t > 0$. Markov's inequality and independence give

$$\begin{aligned}
\mathbb{P}(S_n - \mathbb{E}[S_n] \geq t) &= \mathbb{P}(e^{s(S_n - \mathbb{E}[S_n])} \geq e^{st}) && (3.3)\\
&\leq e^{-st}\,\mathbb{E}(e^{s(S_n - \mathbb{E}[S_n])}) && (3.4)\\
&= e^{-st}\prod_{i=1}^n \mathbb{E}(e^{s(X_i - \mathbb{E}[X_i])}) && (3.5)
\end{aligned}$$

Applying Definition 3.3 to each centered variable gives

$$e^{-st}\prod_{i=1}^n \mathbb{E}(e^{s(X_i - \mathbb{E}[X_i])}) \leq \exp\left(-st + \frac{s^2}{8}\sum_{i=1}^n (b_i - a_i)^2\right). \tag{3.6}$$

The exponent is minimized at

$$s^* = \frac{4t}{\sum_{i=1}^n (b_i - a_i)^2}.$$

Therefore, (3.3) and (3.6) yield

$$\mathbb{P}(S_n - \mathbb{E}[S_n] \geq t) \leq \exp\left(-\frac{2t^2}{\sum_{i=1}^n (b_i - a_i)^2}\right).$$

Set $t = n\epsilon$ to complete the proof. □

Corollary 3.5. *Let $n \geq 1$, and let $X_1, \dots, X_n$ be independent real-valued RVs such that $\mathbb{P}(X_i \in [a, b]) = 1$ for every i, where $a < b$. Then, for every $\epsilon > 0$,*

$$\mathbb{P}(\overline{X}_n - \mathbb{E}[\overline{X}_n] \leq -\epsilon) \leq e^{-\frac{2n\epsilon^2}{(b-a)^2}},$$

$$\mathbb{P}(|\overline{X}_n - \mathbb{E}[\overline{X}_n]| \geq \epsilon) \leq 2e^{-\frac{2n\epsilon^2}{(b-a)^2}}.$$

Proof. Since $X_i - \mathbb{E}[X_i] \in [a - \mathbb{E}[X_i], b - \mathbb{E}[X_i]]$ with probability one, Definition 3.4 gives the upper-tail bound with denominator $n(b-a)^2$. Apply Definition 3.4 to $-X_1, \dots, -X_n$ for the lower tail. The two-sided bound follows from the union bound. □

Hoeffding's inequality gives a confidence interval for a Bernoulli success probability without using an asymptotic approximation.

Lemma 3.6 (Estimating p in Bernoulli)**.** *Let $n \geq 1$, $p \in [0, 1]$, and $X_1, \dots, X_n \overset{\text{IID}}{\sim}$ Bernoulli(p). For $\alpha \in (0, 1)$, set*

$$\delta_{n,\alpha} := \sqrt{\frac{\ln(2/\alpha)}{2n}}.$$

Then

$$\mathbb{P}\left(\overline{X}_n - \delta_{n,\alpha} \leq p \leq \overline{X}_n + \delta_{n,\alpha}\right) \geq 1 - \alpha.$$

Remark 3.7. *For $\alpha = 0.05$, the half-width is $\delta_{n,0.05} \approx 1.36/\sqrt{n}$. Intersecting the interval with $[0,1]$ cannot reduce its coverage because $p \in [0,1]$.*

Proof. Since $\mathbb{E}[\overline{X}_n] = p$, Definition 3.5 with $a = 0$ and $b = 1$ gives

$$\mathbb{P}\left(|\overline{X}_n - p| \geq \delta_{n,\alpha}\right) \leq 2e^{-2n\delta_{n,\alpha}^2} = \alpha.$$

The event in the statement contains the complement of the event on the left, which proves the claimed coverage. □

Corollary 3.8 (Confidence interval after filtering)**.** *Let $m, n \in \mathbb{N}$ with $m, n \geq 1$, let $X_1, \dots, X_n$ be i.i.d. random vectors in $\mathbb{R}^m$, and let $A, C \subseteq \mathbb{R}^m$ be fixed standard events. Assume that*

$$p := \mathbb{P}(X_1 \in A) > 0, \qquad q := \mathbb{P}(X_1 \in C \mid X_1 \in A).$$

Define

$$N_A := \sum_{i=1}^{n} \mathbb{1}_{\{X_i \in A\}}, \qquad S_{A,C} := \sum_{i=1}^{n} \mathbb{1}_{\{X_i \in A \cap C\}}.$$

When $N_A \geq 1$, set

$$\widehat{q}_A := \frac{S_{A,C}}{N_A}, \qquad \delta(k,\alpha) := \sqrt{\frac{\log(2/\alpha)}{2k}}, \qquad \alpha \in (0,1).$$

The random interval

$$I_A(\alpha) := \begin{cases} [0,1], & N_A = 0, \\ \left[\max\{0, \widehat{q}_A - \delta(N_A,\alpha)\}, \min\{1, \widehat{q}_A + \delta(N_A,\alpha)\}\right], & N_A \geq 1, \end{cases}$$

satisfies

$$\mathbb{P}\big(q \in I_A(\alpha)\big) \geq 1 - \alpha.$$

Proof. Conditional on $N_A = k \geq 1$, Definition 2.56 shows that the retained observations are i.i.d. with the distribution of X_1 given $X_1 \in A$. Their indicators of C are independent Bernoulli random variables with success probability q. Their sample mean is $\widehat{q}_A$.

Since $q \in [0,1]$, clipping an interval to $[0,1]$ cannot remove q from it. By Definition 3.6,

$$\mathbb{P}\big(q \notin I_A(\alpha) \mid N_A = k\big) \leq \alpha$$

for every k with $\mathbb{P}(N_A = k) > 0$. When $N_A = 0$, failure is impossible because $I_A(\alpha) = [0,1]$. Therefore, the law of total probability gives

$$\mathbb{P}\big(q \notin I_A(\alpha)\big) \leq \alpha\,\mathbb{P}(N_A \geq 1) \leq \alpha.$$

□

Remark 3.9 (Simultaneous intervals)**.** *The union bound also controls several intervals at once; independence between the intervals is not required. For $i = 1, \dots, m$, let $Z_i = (X_{i1}, \dots, X_{in})$. Assume that the variables within each row are i.i.d. and integrable. The* *rows need not be independent. Define $\mu_i = \mathbb{E}[X_{ij}]$, which does not depend on j, and suppose that*

$$\mathbb{P}\left(\left|\frac{1}{n}\sum_{j=1}^{n} X_{ij} - \mu_i\right| \geq \epsilon\right) \leq C_i$$

for every i and some $C_i \geq 0$. Then

$$\mathbb{P}\left(\left|\frac{1}{n}\sum_{j=1}^{n} X_{ij} - \mu_i\right| \geq \epsilon \text{ for some } i\right) \leq \sum_{i=1}^{m} C_i,$$

and hence

$$\mathbb{P}\left(\left|\frac{1}{n}\sum_{j=1}^{n} X_{ij} - \mu_i\right| < \epsilon \text{ for all } i\right) \geq 1 - \sum_{i=1}^{m} C_i.$$

For example, if $X_{ij} \sim \mathrm{Bernoulli}(p_i)$ and $\alpha \in (0,1)$, then applying Definition 3.6 with error probability α/m in each row gives

$$\mathbb{P}\left(\frac{1}{n}\sum_{j=1}^{n} X_{ij} - \delta \leq p_i \leq \frac{1}{n}\sum_{j=1}^{n} X_{ij} + \delta \text{ for every } i\right) \geq 1 - \alpha,$$

where

$$\delta = \sqrt{\frac{\ln(2m/\alpha)}{2n}}.$$

Thus the number m of simultaneous intervals affects the half-width only through a logarithm. This use of the union bound is also called a **Bonferroni correction***.*

The proof of Definition 3.4 only uses a bound on the moment generating function of each centered variable. This motivates the following definitions.

Definition 3.10. *Let $\lambda > 0$. An integrable real-valued RV X is* **sub-Gaussian** *with parameter λ if*

$$\mathbb{E}\left[e^{s(X-\mathbb{E}[X])}\right] \leq e^{s^2\lambda^2/2}, \qquad s \in \mathbb{R}.$$

Definition 3.11. *Let $\lambda > 0$. An integrable real-valued RV X is* **sub-exponential** *with parameter λ if*

$$\mathbb{E}\left[e^{s(X-\mathbb{E}[X])}\right] \leq e^{s^2\lambda^2/2}, \qquad |s| \leq \frac{1}{\lambda}.$$

Theorem 3.12. *Let $n \geq 1$, and let $X_1, \ldots, X_n$ be independent sub-Gaussian RVs with parameter $\sigma > 0$. Then, for every $\epsilon > 0$,*

$$\mathbb{P}(\overline{X}_n - \mathbb{E}[\overline{X}_n] \geq \epsilon) \leq e^{-n\epsilon^2/(2\sigma^2)},$$
$$\mathbb{P}(|\overline{X}_n - \mathbb{E}[\overline{X}_n]| \geq \epsilon) \leq 2e^{-n\epsilon^2/(2\sigma^2)},$$

where $\overline{X}_n = n^{-1}\sum_{i=1}^{n} X_i$.

Proof. Let $S_n = \sum_{i=1}^n X_i$. For $s, t > 0$, Markov's inequality and independence give

$$\begin{aligned} \mathbb{P}(S_n - \mathbb{E}[S_n] \ge t) &\le e^{-st} \prod_{i=1}^n \mathbb{E}\left[e^{s(X_i - \mathbb{E}[X_i])}\right] \\ &\le \exp\left(-st + \frac{ns^2\sigma^2}{2}\right). \end{aligned}$$

The exponent is minimized at $s = t/(n\sigma^2)$. Setting $t = n\epsilon$ proves the upper-tail estimate. Apply the same estimate to $-X_i$ and use the union bound for the two-sided estimate. □

For sub-exponential variables, the moment generating function estimate is only available near the origin. This gives a Gaussian bound for small deviations and an exponential bound for large deviations.

Theorem 3.13. *Let $n \ge 1$, and let $X_1, \dots, X_n$ be independent sub-exponential RVs with parameter $\lambda > 0$. Then, for every $\epsilon > 0$,*

$$\begin{aligned} \mathbb{P}(\overline{X}_n - \mathbb{E}[\overline{X}_n] \ge \epsilon) &\le \exp\left(-\frac{n}{2}\min\left\{\frac{\epsilon^2}{\lambda^2}, \frac{\epsilon}{\lambda}\right\}\right), \\ \mathbb{P}(|\overline{X}_n - \mathbb{E}[\overline{X}_n]| \ge \epsilon) &\le 2\exp\left(-\frac{n}{2}\min\left\{\frac{\epsilon^2}{\lambda^2}, \frac{\epsilon}{\lambda}\right\}\right), \end{aligned}$$

where $\overline{X}_n = n^{-1}\sum_{i=1}^n X_i$.

Proof. Let $S_n = \sum_{i=1}^n X_i$. For $0 < s \le 1/\lambda$, Markov's inequality and independence give

$$\mathbb{P}(S_n - \mathbb{E}[S_n] \ge t) \le \exp\left(-st + \frac{ns^2\lambda^2}{2}\right). \tag{3.7}$$

The unconstrained minimizer is $s^* = t/(n\lambda^2)$. If $t \le n\lambda$, then $s^* \le 1/\lambda$ and

$$\mathbb{P}(S_n - \mathbb{E}[S_n] \ge t) \le e^{-t^2/(2n\lambda^2)}.$$

If $t > n\lambda$, choose $s = 1/\lambda$. Since

$$\frac{n}{2} - \frac{t}{\lambda} \le -\frac{t}{2\lambda},$$

(3.7) yields

$$\mathbb{P}(S_n - \mathbb{E}[S_n] \ge t) \le e^{-t/(2\lambda)}.$$

Set $t = n\epsilon$. Applying the same argument to $-X_i$ and using the union bound gives the two-sided estimate. □

Lemma 3.14. *The following properties hold.*

1. *If X is sub-Gaussian with parameter λ and $\alpha \ne 0$, then αX is sub-Gaussian with parameter $|\alpha|\lambda$.*

2. *If X is sub-exponential with parameter λ and $\alpha \ne 0$, then αX is sub-exponential with parameter $|\alpha|\lambda$.*

3. *A sub-Gaussian RV with parameter λ is sub-exponential with parameter λ.*

4. *If $\mathbb{P}(X \in [a,b]) = 1$, then X is sub-Gaussian with parameter $(b-a)/2$. In particular, a Bernoulli RV is sub-Gaussian with parameter $1/2$.*

5. *If X is sub-Gaussian with parameter λ, then $(X - \mathbb{E}[X])^2$ is sub-exponential with parameter $8\sqrt{2}\lambda^2$.*

6. *If X and Y are independent and sub-Gaussian with parameters σ_1 and σ_2, then $X+Y$ is sub-Gaussian with parameter $\sqrt{\sigma_1^2+\sigma_2^2}$.*

If $\alpha = 0$, then αX is sub-Gaussian and sub-exponential with every positive parameter.

Proof. The first and third assertions follow from Definition 3.10, while the second follows from Definition 3.11. The fourth assertion is Definition 3.3, and independence gives

$$\mathbb{E}\left[e^{s((X+Y)-\mathbb{E}[X+Y])}\right] \le e^{s^2\sigma_1^2/2} e^{s^2\sigma_2^2/2},$$

which proves the sixth assertion.

It remains to prove the fifth assertion. Set $W = X - \mathbb{E}[X]$ and $Z = W^2 - \mathbb{E}[W^2]$. The sub-Gaussian inequalities at s and $-s$ give

$$\mathbb{E}[\cosh(sW)] = \frac{1}{2}\left(\mathbb{E}[e^{sW}] + \mathbb{E}[e^{-sW}]\right) \le e^{s^2\lambda^2/2}.$$

For every integer $k \ge 1$ and $s > 0$, the power series for cosh gives the pointwise inequality

$$\cosh(sW) \ge \frac{s^{2k}|W|^{2k}}{(2k)!}.$$

Choose $s = \sqrt{2k}/\lambda$. Since $(2k)!/k! \le (2k)^k$, it follows that

$$\mathbb{E}[|W|^{2k}] \le (e\lambda^2)^k k!.$$

For $k \ge 2$, convexity gives

$$\mathbb{E}[|Z|^k] \le 2^{k-1}\left(\mathbb{E}[|W|^{2k}] + \mathbb{E}[W^2]^k\right) \le 2^k\, \mathbb{E}[|W|^{2k}] \le (2e\lambda^2)^k k!.$$

Let $|s| \le 1/(8\sqrt{2}\lambda^2)$ and set $r = 2e\lambda^2|s|$. Since $e < 2\sqrt{2}$, we have $r < 1/2$. The preceding moment bound makes the exponential series absolutely integrable, and hence

$$\mathbb{E}[e^{sZ}] \le 1 + \sum_{k=2}^{\infty} r^k = 1 + \frac{r^2}{1-r} \le 1 + 2r^2 \le e^{8e^2s^2\lambda^4} \le e^{64s^2\lambda^4}.$$

This is Definition 3.11 with parameter $8\sqrt{2}\lambda^2$. $\square$

Some standard examples are summarized in Fig. 3.1. Whether a distribution belongs to a class does not depend on its location, while the parameter in the bound depends on its scale.

Distribution	sub-exponential	sub-Gaussian
Gaussian	Yes	Yes
Bernoulli	Yes	Yes
Uniform	Yes	Yes
Bounded	Yes	Yes
Exponential	Yes	No
χ^2	Yes	No
Weibull ($1 \leq k < 2$)	Yes	No
Weibull ($k \geq 2$)	Yes	Yes
Laplace	Yes	No
Pareto	No	No
Lognormal	No	No

Figure 3.1: Examples of distributions that are sub-exponential and sub-Gaussian

3.1.1 Finite moments and polynomial tails*

The sub-Gaussian and sub-exponential definitions require the moment generating function to be finite on a two-sided interval around the origin. If X has density f_X, this means that

$$\int_{-\infty}^{\infty} e^{sx} f_X(x)\, dx < \infty$$

for both positive and negative s of sufficiently small magnitude. This is a weighted integrability condition; it does not require X to have a density or the density to satisfy a pointwise exponential bound. For comparison,

$$\int_{-\infty}^{\infty} e^{sx} e^{-|x|/\lambda}\, dx < \infty \quad \iff \quad |s| < \frac{1}{\lambda}.$$

Two-sided exponential integrability implies $\mathbb{E}\left[|X|^p\right] < \infty$ for every $1 \leq p < \infty$. With only finitely many moments, one instead obtains a polynomial tail bound.

Theorem 3.15 (Finite-moment tail bound). *Let $X_1, \dots, X_n$ be independent real-valued RVs and let $q \geq 2$ be an even integer. Suppose that $\mathbb{E}\left[|X_i|^q\right] < \infty$ and, for some $\sigma > 0$,*

$$\mathbb{V}(X_i) \leq \sigma^2,$$
$$|\,\mathbb{E}\left[(X_i - \mathbb{E}\left[X_i\right])^r\right]| \leq \sigma^2 r!, \qquad r = 3, \dots, q,$$

for every $i = 1, \dots, n$. If $q \leq n\sigma^2/2$, then, for every $\epsilon > 0$,

$$\mathbb{P}(|\overline{X}_n - \mathbb{E}\left[\overline{X}_n\right]| \geq \epsilon) \leq \left(\frac{2q\sigma^2}{n\epsilon^2}\right)^{q/2}.$$

If in addition $q \geq n\epsilon^2/(4\sigma^2)$, then

$$\mathbb{P}(|\overline{X}_n - \mathbb{E}\left[\overline{X}_n\right]| \geq \epsilon) \leq 3e^{-n\epsilon^2/(12\sigma^2)}.$$

Proof. Set $Y_i = X_i - \mathbb{E}[X_i]$ and $S_n = \sum_{i=1}^n Y_i$. The multinomial expansion and independence give, for every even $r \le q$,

$$\mathbb{E}[S_n^r] \le \frac{r!}{(r/2)!} 2^{r/2} (n\sigma^2)^{r/2} \le (2rn\sigma^2)^{r/2}. \tag{3.8}$$

The combinatorial estimate is proved in [BlHo, Theorem 12.5]. Applying Definition 3.1 to S_n^q and setting $a = n\epsilon$ gives

$$\mathbb{P}(|S_n| \ge a) \le \left(\frac{2qn\sigma^2}{a^2}\right)^{q/2} = \left(\frac{2q\sigma^2}{n\epsilon^2}\right)^{q/2}.$$

For the exponential estimate, assume that $q \ge a^2/(4n\sigma^2)$. If $a^2 \le 12n\sigma^2$, then the claimed upper bound is at least $3/e > 1$ and is immediate. Otherwise, let r be the largest even integer satisfying $r \le a^2/(6n\sigma^2)$. Then $2 \le r \le q$, and (3.8) and Markov's inequality yield

$$\mathbb{P}(|S_n| \ge a) \le \left(\frac{2rn\sigma^2}{a^2}\right)^{r/2} \le 3^{-r/2} \le 3e^{-a^2/(12n\sigma^2)}.$$

Substituting $a = n\epsilon$ completes the proof. □

3.2 Convergence of Random Variables

We study the limiting behavior of a sequence of real-valued RVs $X_1, X_2, \ldots$. Different notions of convergence retain different information. Convergence in distribution depends only on the laws of the variables, so the variables need not be defined on a common probability space. Convergence in probability and almost-sure convergence compare the variables on one common probability space. The optional subsection adds a moment-based mode of convergence that also requires a common space.

In statistical applications, the index n usually measures the amount of data and tends to infinity. The candidate limit may be a constant RV or a non-degenerate RV; the mode of convergence specifies the meaning of this assertion.

Definition 3.16 (**Convergence in Distribution** (or Weakly, or in Law))**.** *Let $X_1, X_2, \ldots$ and X be real-valued RVs, possibly defined on different probability spaces. Let F_n denote the DF of X_n and let F denote the DF of X. We say that X_n converges to X in distribution, and write*

$$X_n \rightsquigarrow X$$

if, for every continuity point t of F,

$$\lim_{n\to\infty} F_n(t) = F(t).$$

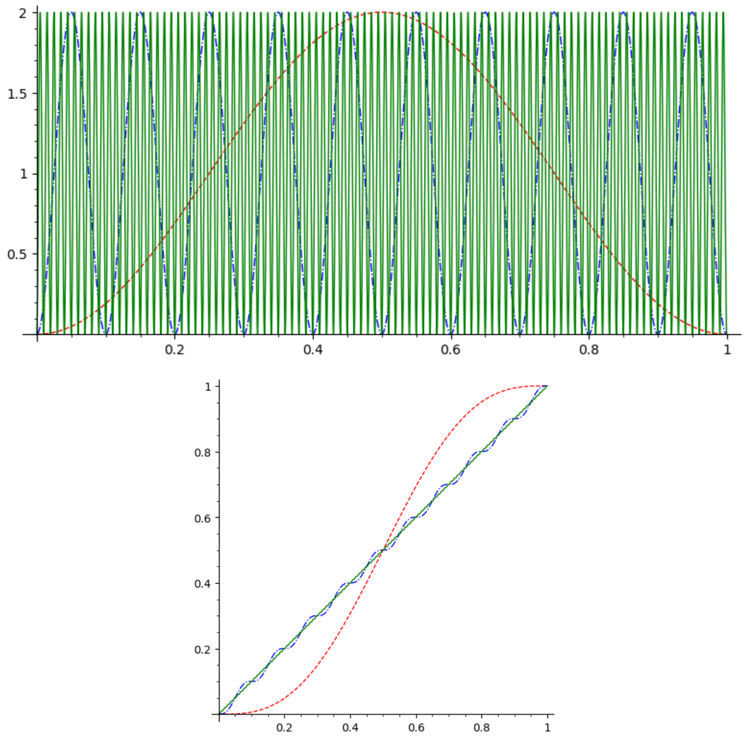


Figure 3.2: The densities $f_n(x) = \mathbb{1}_{(0,1)}(x)(1 - \cos(2\pi n x))$ (left) and their distribution functions (right) for $n = 1, 10, 100$. The distribution functions converge to that of the Uniform$(0, 1)$ distribution, while the densities do not converge pointwise on $(0, 1)$.

Convergence in distribution does not imply pointwise convergence of densities. For the example in Fig. 3.2,

$$F_n(x) = \begin{cases} 0, & x \le 0, \\ x - \dfrac{\sin(2\pi n x)}{2\pi n}, & 0 < x < 1, \\ 1, & x \ge 1. \end{cases}$$

Thus $F_n(x)$ converges to the DF of the $\mathrm{Uniform}(0,1)$ distribution at every x. However, $f_n(1/2)$ alternates between 0 and 2, and

$$\int_{\mathbb{R}} |f_n(x) - \mathbb{1}_{(0,1)}(x)|\, dx = \frac{2}{\pi}.$$

The next elementary criterion is enough for the discrete approximation used below. It avoids any need for abstract measures or convergence theorems for integrals.

Lemma 3.17 (Convergence of integer-valued probabilities)**.** *Suppose that $X_1, X_2, \ldots$ and X take values in $\mathbb{Z}_+ := \{0, 1, 2, \ldots\}$. If, for every fixed $k \in \mathbb{Z}_+$,*

$$\mathbb{P}(X_n = k) \longrightarrow \mathbb{P}(X = k),$$

then $X_n \rightsquigarrow X$.

Proof. Fix $t \in \mathbb{R}$. If $t < 0$, then $\mathbb{P}(X_n \le t) = \mathbb{P}(X \le t) = 0$. If $t \ge 0$, set $m = \lfloor t \rfloor$. The sum below has only finitely many terms, so

$$\mathbb{P}(X_n \le t) = \sum_{k=0}^{m} \mathbb{P}(X_n = k) \longrightarrow \sum_{k=0}^{m} \mathbb{P}(X = k) = \mathbb{P}(X \le t).$$

Thus the distribution functions converge at every t, which proves the result by Definition 3.16. $\square$

We apply this criterion to the Poisson approximation of a binomial random variable.

Theorem 3.18. *Fix $\lambda \ge 0$. For every integer n satisfying*

$$n \ge \max\{1, \lceil \lambda \rceil\},$$

let $X_n \sim \mathrm{Binomial}(n, \lambda/n)$, and let $Y \sim \mathrm{Poisson}(\lambda)$. Then

$$X_n \rightsquigarrow Y.$$

Proof. If $\lambda = 0$, then $X_n = Y = 0$ almost surely, so there is nothing to prove. Assume henceforth that $\lambda > 0$. Fix $k \in \mathbb{Z}_+$. For all sufficiently large n,

$$\begin{aligned} \mathbb{P}(X_n = k) &= \binom{n}{k} \left(\frac{\lambda}{n}\right)^k \left(1 - \frac{\lambda}{n}\right)^{n-k} \\ &= \frac{\lambda^k}{k!} \prod_{j=0}^{k-1} \left(1 - \frac{j}{n}\right) \left(1 - \frac{\lambda}{n}\right)^{n-k}. \end{aligned}$$

The empty product is interpreted as 1. The finite product tends to 1, while the last factor tends to $e^{-\lambda}$. Therefore,

$$\mathbb{P}(X_n = k) \to e^{-\lambda} \frac{\lambda^k}{k!} = \mathbb{P}(Y = k).$$

The conclusion follows from Definition 3.17. $\square$

The second notion of convergence of RVs is convergence in probability.

Definition 3.19 (**Convergence in Probability**). *Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability triple, and let $X_1, X_2, \ldots$ and X be real-valued RVs on this space. We say that X_n converges to X in probability, and write*

$$X_n \xrightarrow{\mathbb{P}} X$$

if, for every $\epsilon > 0$,

$$\mathbb{P}(|X_n - X| > \epsilon) \longrightarrow 0.$$

Here $\{|X_n - X| > \epsilon\}$ denotes the event

$$\{\omega \in \Omega : |X_n(\omega) - X(\omega)| > \epsilon\}.$$

Definition 3.20 (**Convergence Almost Surely** (or with Probability 1)). *Let $X_1, X_2, \ldots$ and X be real-valued RVs on the same probability space $(\Omega, \mathcal{F}, \mathbb{P})$. We say that X_n converges to X almost surely if*

$$\mathbb{P}(\{\omega \in \Omega : X_n(\omega) \to X(\omega)\}) = 1.$$

We write

$$X_n \stackrel{a.s.}{\to} X.$$

Equivalently,

$$\mathbb{P}(\{\omega \in \Omega : X_n(\omega) \not\to X(\omega)\}) = 0.$$

Proposition 3.21 (Almost-sure convergence implies convergence in probability). *If $X_n \stackrel{a.s.}{\to} X$, then $X_n \stackrel{\mathbb{P}}{\to} X$.*

Proof. Fix $\epsilon > 0$ and define

$$B_m := \bigcup_{n=m}^{\infty} \{|X_n - X| > \epsilon\}.$$

The events B_m decrease as m increases. Their intersection is the event that $|X_n - X| > \epsilon$ for infinitely many n, which is contained in the event that X_n does not converge to X. Hence

$$\mathbb{P}\left(\bigcap_{m=1}^{\infty} B_m\right) = 0.$$

To see the required continuity directly, set $C_m = B_m \setminus B_{m+1}$. The events C_m are pairwise disjoint and

$$B_m = \left(\bigcap_{j=1}^{\infty} B_j\right) \cup \bigcup_{j=m}^{\infty} C_j.$$

Countable additivity shows that $\mathbb{P}(B_m)$ is the probability of the intersection plus the tail of a convergent series. Hence $\mathbb{P}(B_m) \to 0$. Since $\{|X_m - X| > \epsilon\} \subseteq B_m$, the definition of convergence in probability now gives the result. $\square$

3.2.1 Optional: Moment convergence and further properties

Mean-square convergence is the case $p = 2$ of the following definition.

Definition 3.22 (**Convergence in L^p**). *Fix $1 \le p < \infty$. Let $X_1, X_2, \ldots$ and X be RVs in $L^p(\mathbb{P})$ on the same probability space. We say that X_n converges to X in $L^p(\mathbb{P})$ if*

$$\|X_n - X\|_{L^p(\mathbb{P})} \to 0.$$

By the definition of the $L^p(\mathbb{P})$ norm in Section 2.6.4, this is equivalent to

$$\mathbb{E}\left[|X_n - X|^p\right] \to 0.$$

We record further relations among the notions of convergence introduced above. This subsection uses the optional L^p material from Section 2.6.4.

Theorem 3.23 (Portmanteau theorem). *Let $X_1, X_2, \ldots$ and X be real-valued RVs, possibly defined on different probability spaces. The following are equivalent:*

- $\mathbb{E}\left[f(X_n)\right] \to \mathbb{E}\left[f(X)\right]$ *for every bounded continuous function $f : \mathbb{R} \to \mathbb{R}$.*
- $\mathbb{E}\left[f(X_n)\right] \to \mathbb{E}\left[f(X)\right]$ *for every bounded Lipschitz function $f : \mathbb{R} \to \mathbb{R}$.*
- $X_n \rightsquigarrow X$.
- $\mathbb{P}(X_n \in A) \to \mathbb{P}(X \in A)$ *for every standard event $A \subseteq \mathbb{R}$ satisfying $\mathbb{P}(X \in \partial A) = 0$.*

Remark 3.24. *For $A = (-\infty, t]$, the boundary is $\partial A = \{t\}$. Thus $\mathbb{P}(X \in \partial A) = 0$ exactly when the DF of X is continuous at t, which recovers Definition 3.16.*

- The converse to almost-sure convergence implying convergence in probability fails. For a concrete example, take $\Omega = [0, 1)$ with the uniform probability law. In block m, partition $[0, 1)$ into 2^m intervals and let the corresponding 2^m random variables be their indicators. Enumerate these indicators block by block. Their probabilities of equalling 1 tend to zero, so the sequence converges to zero in probability. Every outcome belongs to one interval in every block, so the sequence does not converge pointwise to zero.

- Convergence in probability implies convergence in distribution:

$$\boxed{X_n \xrightarrow{\mathbb{P}} X \implies X_n \rightsquigarrow X.}$$

- Suppose that $X_1, X_2, \ldots$ are defined on a common probability space, and let X satisfy $\mathbb{P}(X = \theta) = 1$. If X_n converges in distribution to X, then X_n converges in probability to the constant θ:

$$X_n \rightsquigarrow X \implies X_n \xrightarrow{\mathbb{P}} \theta.$$

- Convergence in L^p implies convergence in L^q whenever $1 \le q \le p < \infty$, by Definition 2.45.

- Convergence in L^p, for $1 \le p < \infty$, implies convergence in probability by Definition 3.1.

- In general, convergence in distribution does not imply convergence in probability. For example, if X is a Rademacher RV and $X_n = (-1)^n X$, then $X_n \rightsquigarrow X$ but X_n does not converge to X in probability.

3.3 Law of Large Numbers

Theorem 3.25. *Let $X_1, X_2, \ldots$ be pairwise independent real-valued RVs with finite second moments and a common mean $\mathbb{E}[X_i] = \mu$. Suppose that, for some finite σ^2,*

$$\mathbb{V}(X_i) \le \sigma^2 \qquad \text{for every } i \ge 1.$$

Then

$$\overline{X}_n \xrightarrow{\mathbb{P}} \mu.$$

Proof. Fix $\epsilon > 0$. By Chebyshev's inequality,

$$\mathbb{P}\big(|\overline{X}_n - \mu| \ge \epsilon\big) \le \frac{\mathbb{V}(\overline{X}_n)}{\epsilon^2}.$$

Pairwise independence gives

$$\mathbb{V}(\overline{X}_n) = \frac{1}{n^2}\sum_{i=1}^{n} \mathbb{V}(X_i) \le \frac{\sigma^2}{n}.$$

Consequently,

$$\mathbb{P}\big(|\overline{X}_n - \mu| \ge \epsilon\big) \le \frac{\sigma^2}{n\epsilon^2} \longrightarrow 0,$$

which is the claimed convergence in probability. □

In the i.i.d. setting, a finite absolute first moment is enough for the corresponding almost-sure statement.

Theorem 3.26 (Strong law of large numbers)**.** *Let $X_1, X_2, \ldots$ be i.i.d. real-valued RVs with $\mathbb{E}[|X_1|] < \infty$, and set $\mu = \mathbb{E}[X_1]$. Then*

$$\overline{X}_n \xrightarrow{a.s.} \mu.$$

The proof of the strong law is omitted.

3.4 Central Limit Theorem

The law of large numbers describes the first-order behavior of the sample mean. When the variables have common mean μ, the next question is whether the centered fluctuations

$$\frac{1}{\sqrt{n}}\sum_{i=1}^{n}(X_i - \mu) = \sqrt{n}(\overline{X}_n - \mu)$$

have a nondegenerate limiting distribution.

Theorem 3.27. *Let $X_1, X_2, \ldots$ be i.i.d. RVs with*

$$\mu := \mathbb{E}[X_1] \qquad \text{and} \qquad 0 < \sigma^2 := \mathbb{V}(X_1) < \infty.$$

Then

$$Z_n := \frac{\sqrt{n}(\overline{X}_n - \mu)}{\sigma} = \frac{1}{\sigma\sqrt{n}}\sum_{i=1}^{n}(X_i - \mu) \rightsquigarrow Z, \qquad Z \sim \text{Normal}(0, 1).$$

Independence gives

$$\mathbb{V}(\overline{X}_n) = \frac{\sigma^2}{n} \qquad \text{and} \qquad \mathbb{V}(Z_n) = 1.$$

Thus the normalization keeps the variance equal to 1 while the variance of the unscaled sample mean tends to zero. The conclusion of the theorem is only convergence in distribution. It does not assert convergence in probability or almost surely, and the limiting normal RV need not be defined on the same probability space as the sequence (Z_n).

The proof of the central limit theorem is omitted.

3.5 Exercises

Exercise 3.28 (Markov and Chebyshev bounds)**.** *Let X have probability mass function*

$$\mathbb{P}(X=0) = \frac{1}{2}, \qquad \mathbb{P}(X=2) = \frac{1}{3}, \qquad \mathbb{P}(X=8) = \frac{1}{6}.$$

1. *Compute $\mathbb{E}[X]$, $\mathbb{V}(X)$, and $\mathbb{P}(X \geq 2)$.*

2. *Compare the exact probability with the Markov bound for $\mathbb{P}(X \geq 2)$.*

3. *Compute $\mathbb{P}(|X - \mathbb{E}[X]| \geq 2)$ and compare it with the Chebyshev bound. A probability bound may exceed 1; explain why replacing such a bound by 1 is always valid.*

Exercise 3.29 (Hoeffding confidence intervals)**.** *Let $X_1, \ldots, X_n \overset{\text{IID}}{\sim} \text{Bernoulli}(p)$ with $p = 0.3$.*

1. *Find the Hoeffding 95% confidence interval from Definition 3.6 when $n = 100$ and the sample contains 34 successes. Clip the interval to $[0, 1]$.*

2. *Use Python and `numpy.random.binomial` to generate 5000 independent samples for each $n \in \{25, 100, 400\}$. Estimate the coverage probability and average width of the clipped interval.*

3. *Use `matplotlib` to plot average interval width against n on logarithmic axes. Add a reference curve proportional to $n^{-1/2}$ and explain the comparison.*

Exercise 3.30 (The Poisson tail class)**.** *Let $X \sim \text{Poisson}(\nu)$, where $\nu > 0$.*

1. *Use*
$$\mathbb{E}[e^{sX}] = \exp(\nu(e^s - 1))$$
to find $\mathbb{E}[e^{s(X-\nu)}]$.

2. *Show that $e^s - 1 - s \leq s^2$ when $|s| \leq 1/2$. Deduce that X is sub-exponential with parameter $\max\{2, \sqrt{2\nu}\}$ according to Definition 3.11.*

3. *Show that no finite parameter can make X sub-Gaussian. Hint: compare the growth of e^s and s^2 as $s \to \infty$.*

Exercise 3.31 (Modes of convergence)**.** *Let $U \sim \text{Uniform}([0, 1])$ and define*

$$X_n := \mathbb{1}_{\{U \leq 1/n\}}.$$

1. *Prove directly from the definitions that* $X_n \to 0$ *almost surely and in probability.*

2. *(Optional) For each* $p \geq 1$, *compute* $\mathbb{E}[|X_n|^p]$ *and determine whether* $X_n \to 0$ *in* L^p.

3. *Explain why these conclusions do not contradict the sequence* $Z_n = n\mathbb{1}_{\{X_1 \leq 1/n\}}$ *in Definition 5.12.*

Exercise 3.32 (A central-limit simulation). *Let* $X_1, X_2, \ldots$ *be i.i.d. exponential RVs with mean* 1.

1. *Identify* μ *and* σ *in Definition 3.27.*

2. *With Python, generate* 10000 *independent samples of size* $n \in \{2, 10, 50\}$ *using* `numpy.random.exponential(scale=1, size=(10000,n))`. *For every sample compute* $Z_n = \sqrt{n}(\overline{X}_n - \mu)/\sigma$.

3. *For each* n, *use* `matplotlib` *to draw a density-normalized histogram of* Z_n. *Overlay the standard normal density* $(2\pi)^{-1/2}e^{-z^2/2}$, *computed with* `numpy`. *Describe what changes with* n *and what the CLT does not claim for a fixed* n.

Chapter 4

Risk

Course guide

Scheduled lecture(s): Lectures 5 and 7 (revisited in Lecture 15)
Required for those lectures: Section 4.1 in Lecture 5; Sections 4.1–4.2 in Lecture 7; Exercises: Section 4.3
Prerequisites: Chapter 2, Section 2.6, and Chapter 3, Section 3.1
Optional or advanced: General optimization theory is outside scope; regression is introduced in Sections 4.1.2 and 4.2.1 and revisited in Chapter 9.

Suppose that W is one toss of a coin whose probability of Heads is an unknown number $\theta \in [0, 1]$. With `H` and `T` denoting Heads and Tails, respectively, the possible laws are

$$\mathbb{P}_\theta(W = \mathtt{H}) = \theta, \qquad \mathbb{P}_\theta(W = \mathtt{T}) = 1 - \theta, \qquad \theta \in [0, 1].$$

This family is a simple statistical model. It specifies which distributions are possible without specifying which value of θ generated the data.

Definition 4.1 (Statistical model)**.** *A* **statistical model** *is an indexed family*

$$\mathcal{S} = \{F_\theta : \theta \in \Theta\}$$

of probability distributions on the same data space. The set Θ is the parameter space. A model is **parametric** *if its chosen parameter θ belongs to a subset of $\mathbb{R}^d$ for some fixed finite d. A model that is not described by a fixed finite list of real-valued parameters is called* **nonparametric***.*

The notation F_θ may denote a distribution function. When all distributions in the model have densities or probability mass functions, we may instead specify the family by f_θ.

Example 4.2 (A parametric model)**.** *The normal location-scale model consists of the densities*

$$f_{\mu,\sigma}(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right), \qquad (\mu, \sigma) \in \mathbb{R} \times (0, \infty).$$

It is parametric with a two-dimensional parameter (μ, σ).

Example 4.3 (A nonparametric model)**.** *Let*

$$\mathcal{E} = \{F : F \text{ is a distribution function on } \mathbb{R}\}.$$

In this course, the family of all distribution functions is treated as a nonparametric model: the unknown object is the entire function F, not a fixed finite list of its values.

Statistical inference uses observations to draw conclusions about the unknown distribution in a statistical model. Examples include estimating a probability, a distribution function, a regression function, or a classification rule. A statistical model records the assumptions on the data-generating distribution. It need not contain every distribution, and conclusions drawn under the model depend on those assumptions.

4.1 The supervised learning problem

We begin with a finite example. Let $X, Y \in \{0,1\}$ and suppose that

$$\begin{aligned} \mathbb{P}(X=0) = \mathbb{P}(X=1) &= \frac{1}{2}, \\ \mathbb{P}(Y=1 \mid X=0) &= \frac{1}{4}, \\ \mathbb{P}(Y=1 \mid X=1) &= \frac{3}{4}. \end{aligned}$$

The input is X and the response is Y. Consider the prediction rule $g(x) = x$ and the 0–1 loss, which is 1 for a wrong prediction and 0 for a correct prediction. The probability of an error is

$$\begin{aligned} \mathbb{P}(Y \neq g(X)) &= \mathbb{P}(X=0, Y=1) + \mathbb{P}(X=1, Y=0) \\ &= \frac{1}{2}\frac{1}{4} + \frac{1}{2}\frac{1}{4} = \frac{1}{4}. \end{aligned}$$

Thus the quality of g can be expressed as an expected loss.

In the general supervised learning setup, the input X takes values in a data space $\mathbb{X} \subseteq \mathbb{R}^m$ and the response Y takes values in $\mathbb{Y} \subseteq \mathbb{R}^q$. The pair $Z = (X, Y)$ has an unknown distribution $F_{X,Y}$ belonging to a specified statistical model. In a discrete model, or in a model with a joint density, this distribution may be written as

$$f_{X,Y}(x,y) = f_{Y|X}(y \mid x) f_X(x)$$

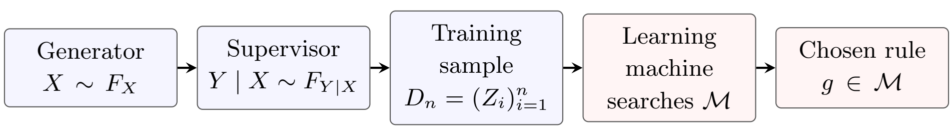
at values of x for which the conditional PMF or PDF is defined.

The terminology used in machine learning divides this setup into three parts. The generator produces an input X. The supervisor produces a response Y according to the conditional law given X. The learning machine uses observed input-response pairs to choose a prediction rule. Repeated observations are modelled explicitly as i.i.d. pairs

$$Z_1 = (X_1, Y_1), \ldots, Z_n = (X_n, Y_n) \quad \text{with common distribution } F_{X,Y}.$$

The training sample is denoted by $D_n = (Z_1, \ldots, Z_n)$. Its realized values are $(x_1, y_1), \ldots, (x_n, y_n)$.

Let $\mathcal{M}$ be a class of functions $g : \mathbb{X} \to \mathbb{Y}$ for which $g(X)$ is a random variable. This is the model space, or prediction class, searched by the learning machine. The data-generating and learning steps are summarized below.



Let

$$\ell : \mathbb{Y} \times \mathbb{Y} \to [0, \infty)$$

be a loss function, where $\ell(y, u)$ is the loss incurred when the response is y and the prediction is u. For $g \in \mathcal{M}$, assume that $\ell(Y, g(X))$ is a random variable with finite expectation. The **risk** of g is

$$R(g) := \mathbb{E}\left[\ell(Y, g(X))\right].$$

The population learning problem is to find a member of $\mathcal{M}$ with small risk. The distribution $F_{X,Y}$, and hence $R(g)$, is usually unknown. Later chapters replace it by a quantity computed from the training data.

4.1.1 Learning an unknown function

The deterministic supervisor is the special case

$$Y = f(X)$$

for an unknown function $f : \mathbb{X} \to \mathbb{Y}$. For each fixed x, the conditional distribution of Y is a point mass at $f(x)$. Its conditional distribution function, when Y is real-valued, is

$$F_{Y|X}(y \mid x) = \mathbb{1}_{\{f(x) \le y\}}.$$

This formula describes a distribution function, not a density.

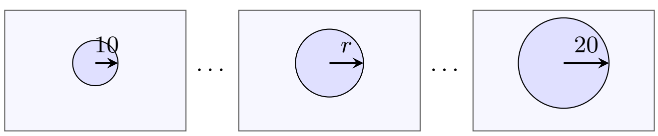
Example 4.4 (Measuring a reference mark)**.** *Let* $R \sim \mathrm{Uniform}([10, 20])$*. For each* $r \in [10, 20]$*, let* $x_r \in [0, 1]^m$ *be a digital image of a circular reference mark with radius* r *pixels. Set*

$$X = x_R, \qquad Y = R.$$

Thus $Y = f(X)$*, where* $f(x_r) = r$*. The inputs range over an uncountable collection of images and the response is continuous. For* $10 \le a \le b \le 20$*, the model gives*

$$\mathbb{P}(Y \in [a, b]) = \frac{b - a}{10}.$$

A learning machine is given images and their measured radii and seeks a rule that predicts the radius in a new image.



$x_{10} \mapsto Y = 10$ $\quad x_r \mapsto Y = r$ $\quad x_{20} \mapsto Y = 20$

$R \sim \mathrm{Uniform}([10, 20])$

More generally, fix a distribution F_X for X and a collection $\mathcal{H}$ of possible deterministic supervisor functions. The statistical model is

$$\mathcal{S} = \{\text{the distribution of } (X, f(X)) : X \sim F_X,\ f \in \mathcal{H}\}.$$

This describes the possible data-generating laws. Separately, let

$$\mathcal{M} = \{g_\lambda : \lambda \in \Lambda\}$$

be the class of prediction rules available to the learning machine. The statistical model $\mathcal{S}$ and the model space $\mathcal{M}$ need not be the same.

For a loss ℓ, define

$$Q((x, y), \lambda) := \ell(y, g_\lambda(x)).$$

If the true supervisor is f, then

$$R(\lambda) := R(g_\lambda) = \mathbb{E}\left[\ell(f(X), g_\lambda(X))\right].$$

If X is discrete, this expectation is a probability-weighted sum. If X has a density f_X, it is the ordinary integral

$$R(\lambda) = \int_{\mathbb{R}^m} \ell(f(x), g_\lambda(x)) f_X(x)\, dx.$$

If the minimum is attained, we may choose

$$\lambda^* \in \operatorname*{argmin}_{\lambda \in \Lambda} R(\lambda) \qquad \text{and set} \qquad g^* = g_{\lambda^*}.$$

Existence or uniqueness of a minimizer requires additional assumptions and is not automatic.

4.1.2 Finding the regression function

Consider another finite example. Let X be Bernoulli(1/2) and let $Y \in \{0, 2\}$ have conditional probabilities

$$\mathbb{P}(Y = 2 \mid X = 0) = \frac{1}{4}, \qquad \mathbb{P}(Y = 2 \mid X = 1) = \frac{3}{4}.$$

The remaining conditional probability is assigned to $Y = 0$. The conditional means are

$$r(0) = \mathbb{E}\left[Y \mid X = 0\right] = \frac{1}{2}, \qquad r(1) = \mathbb{E}\left[Y \mid X = 1\right] = \frac{3}{2}.$$

Under squared loss, the constant predictor $g(x) = 1$ has risk 1, while the predictor r has risk 3/4. The difference is

$$\mathbb{E}\left[(r(X) - g(X))^2\right] = \frac{1}{4}.$$

The result below shows that this decomposition holds in general.

We work with the discrete and regular joint-density models of Section 2.6.5, for which the conditional expectations and the tower property in Definition 2.52 apply. The **regression function** is

$$r(x) := \mathbb{E}\left[Y \mid X = x\right]$$

at values of x where the conditional PMF or PDF is defined. We may set $r(x) = 0$ elsewhere, since those values do not affect $r(X)$ or its expectations.

Theorem 4.5 (Squared-risk decomposition)**.** *Let Y be real-valued and suppose that $\mathbb{E}\left[Y^2\right] < \infty$. Let g be a prediction rule such that $g(X)$ is a random variable and $\mathbb{E}\left[g(X)^2\right] < \infty$. Then*

$$\mathbb{E}\left[(Y - g(X))^2\right] = \mathbb{E}\left[(Y - r(X))^2\right] + \mathbb{E}\left[(r(X) - g(X))^2\right]. \tag{4.1}$$

Proof. Jensen's inequality from Definition 2.42, applied to each conditional distribution, gives at the values needed when evaluating at X,

$$r(x)^2 \le \mathbb{E}\,[Y^2 \mid X = x].$$

The tower property therefore gives $\mathbb{E}\,[r(X)^2] \le \mathbb{E}\,[Y^2] < \infty$. All terms below are integrable because $2|UV| \le U^2 + V^2$.

Add and subtract $r(X)$ and expand:

$$\begin{aligned}\mathbb{E}\,[(Y - g(X))^2] = \mathbb{E}\,[(Y - r(X))^2] + \mathbb{E}\,[(r(X) - g(X))^2] \\ + 2\,\mathbb{E}\,[(Y - r(X))(r(X) - g(X))].\end{aligned}$$

For every x at which the conditional expectation is defined,

$$\begin{aligned}&\mathbb{E}\,[(Y - r(X))(r(X) - g(X)) \mid X = x] \\ &\quad = (r(x) - g(x))\big(\mathbb{E}\,[Y \mid X = x] - r(x)\big) = 0.\end{aligned}$$

Applying the tower property to the cross term proves (4.1). □

The first term on the right-hand side of (4.1) does not depend on g. Hence

$$\inf_{g \in \mathcal{M}} R(g) = \mathbb{E}\,[(Y - r(X))^2] + \inf_{g \in \mathcal{M}} \mathbb{E}\,[(r(X) - g(X))^2]$$

for every prediction class whose members satisfy the theorem's moment condition. If the second infimum is attained, the risk minimizers are exactly the members of $\mathcal{M}$ closest to r in mean square. If $r \in \mathcal{M}$, then r is a risk minimizer. Any other minimizer equals r with probability one when evaluated at X.

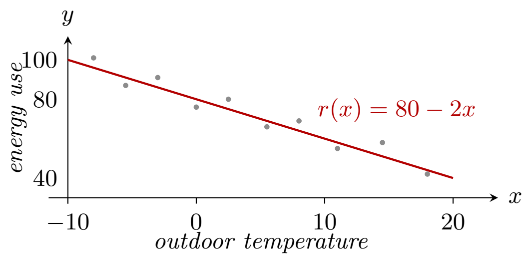
Example 4.6 (Forecasting energy use)**.** *Let X be the outdoor temperature in degrees Celsius, uniformly distributed on $[-10, 20]$. Let Y be the energy used by a building in one day, measured in kilowatt-hours. Suppose that the conditional model is*

$$Y \mid X = x \sim \mathrm{Normal}(80 - 2x, 25), \qquad -10 \le x \le 20.$$

Here 25 is the conditional variance. The regression function is

$$r(x) = \mathbb{E}\,[Y \mid X = x] = 80 - 2x.$$

Energy use remains random after the temperature is known. Under squared loss, the prediction $r(x)$ is the conditional mean. The grey points below show possible observations; the red line is the regression function.



The website-request model in Section 2.7.2 is another regression example. Its regression function is the conditional mean number of requests.

4.1.3 The pattern recognition problem (classification)

Let $k \geq 2$ be an integer. Suppose that Y takes values in the finite class set $\{0, \ldots, k-1\}$. A **decision rule**, or classifier, is a function

$$g : \mathbb{X} \to \{0, \ldots, k-1\}$$

for which $g(X)$ is a random variable. The 0–1 loss is

$$\ell(y, u) := \mathbb{1}_{\{y \neq u\}}.$$

Consequently, the risk of g is its probability of misclassification:

$$R(g) = \mathbb{E}\left[\mathbb{1}_{\{Y \neq g(X)\}}\right] = \mathbb{P}(Y \neq g(X)).$$

The finite model at the start of this section has $k = 2$, and its rule $g(x) = x$ has risk $1/4$.

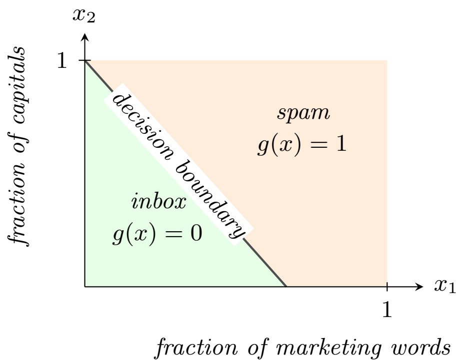
Example 4.7 (Filtering email)**.** *Represent an email by* $X = (X_1, X_2) \in [0,1]^2$*, where* X_1 *is the fraction of words that are marketing terms and* X_2 *is the fraction of letters written in capitals. In a simplified model,* X *is uniformly distributed on the unit square: the probability of a rectangle is its area. Let* $Y = 1$ *mean that the email is spam. Given* $X = x$*, suppose that* Y *is* Bernoulli$(\eta(x))$*, where*

$$\eta(x) = \frac{1}{1 + \exp(4 - 6x_1 - 4x_2)}.$$

Thus the input varies continuously even though the class label is binary. The rule that selects the more likely class predicts spam when

$$-4 + 6x_1 + 4x_2 > 0.$$

The Bayes-rule subsection below proves that this rule minimizes the misclassification risk.



Bayes rule

For binary classification, suppose the model supplies the conditional class probability

$$\eta(x) := \mathbb{P}(Y = 1 \mid X = x).$$

We assume that $\eta(X)$ is a random variable. For a finite input space this is the ordinary conditional probability from Definition 1.12. For a general input space, the conditional class probabilities are treated as part of the given model; their construction is outside the scope of this course. We assume the usual total-expectation formula

$$R(g) = \mathbb{E}\left[\eta(X)\mathbb{1}_{\{g(X)=0\}} + (1 - \eta(X))\mathbb{1}_{\{g(X)=1\}}\right] \tag{4.2}$$

for every binary decision rule g under consideration.

Definition 4.8 (Bayes classification rule)**.** *The* **Bayes classification rule** h^* *is*

$$h^*(x) := \begin{cases} 1, & \eta(x) > 1/2, \\ 0, & \eta(x) \leq 1/2. \end{cases}$$

Thus ties are assigned to class 0.

Theorem 4.9 (Optimality of the Bayes rule)**.** *Let g be any binary decision rule. Assume that* (4.2) *holds for both g and h^*. Then*

$$R(g) - R(h^*) = \mathbb{E}\left[|2\eta(X) - 1|\mathbb{1}_{\{g(X)\neq h^*(X)\}}\right] \geq 0. \tag{4.3}$$

In particular, $R(h^*) \leq R(g)$.

Proof. Fix x. If $\eta(x) > 1/2$, then $h^*(x) = 1$. Choosing class 0 instead increases the conditional error probability from $1-\eta(x)$ to $\eta(x)$, a difference of $2\eta(x)-1$. If $\eta(x) \leq 1/2$, then $h^*(x) = 0$. Choosing class 1 instead increases the conditional error probability from $\eta(x)$ to $1-\eta(x)$, a difference of $1-2\eta(x)$. In both cases the difference is

$$|2\eta(x) - 1|\mathbb{1}_{\{g(x)\neq h^*(x)\}}.$$

Taking expectations in (4.2) proves (4.3). □

The factor $|2\eta(x)-1|$ records the cost of disagreeing with the Bayes rule at x. The cost is zero when $\eta(x) = 1/2$, because both classes have the same conditional error probability. It increases as one class becomes more likely than the other.

4.2 Maximum Likelihood Estimation

First consider a Bernoulli model. If $Z \sim \text{Bernoulli}(q^*)$ and a candidate parameter is $q \in (0,1)$, the log-loss risk is

$$R(q) = -q^*\ln q - (1-q^*)\ln(1-q).$$

For observed values $z_1, \dots, z_n \in \{0,1\}$, the corresponding empirical risk is

$$\widehat{R}_n(q) = -\frac{1}{n}\sum_{i=1}^{n}\big(z_i\ln q + (1-z_i)\ln(1-q)\big).$$

The product of the probabilities assigned to the observations is the likelihood. Taking its negative logarithm gives $n\widehat{R}_n(q)$.

Now let

$$\mathcal{S} = \{p_\alpha : \alpha \in \Theta\}$$

be a family of probability densities on $\mathbb{R}^m$, and suppose that the true density is p_{α^*}. The same calculation applies to probability mass functions, with sums replacing integrals. Define the log loss and its risk by

$$L(z,\alpha) := -\ln p_\alpha(z), \qquad R(\alpha) := \mathbb{E}_{\alpha^*}\left[L(Z,\alpha)\right] = -\int_{\mathbb{R}^m} \ln p_\alpha(z)p_{\alpha^*}(z)\,dz.$$

We use the convention $-\ln 0 = +\infty$. When comparing two finite risks, we assume that the corresponding absolute log integrals are finite. More precisely, for every $\alpha \in \Theta$, assume that $R(\alpha) = +\infty$ or that

$$\int_{\mathbb{R}^m} |\ln p_\alpha(z)| p_{\alpha^*}(z)\, dz < \infty.$$

Unlike a negative log-PMF, a negative log-density can be negative because a density value may exceed 1. Its expectation is still called a log-loss risk, and only comparisons between candidate distributions will be used.

Theorem 4.10 (Log loss is minimized by the true distribution)**.** *Suppose $R(\alpha^*) < \infty$. Then*

$$R(\alpha^*) \le R(\alpha)$$

for every $\alpha \in \Theta$. If both risks are finite, equality holds only if p_α and p_{α^*} determine the same distribution. Therefore, if the model is identifiable, meaning that

$$p_\alpha \text{ and } p_\beta \text{ determine the same distribution} \quad \Longrightarrow \quad \alpha = \beta,$$

then α^* is the unique minimizer.

Proof. If $R(\alpha) = +\infty$, the inequality is immediate. We may therefore assume that both risks are finite. In particular, $p_\alpha(Z) > 0$ with probability one under the true distribution. Let

$$A := \{z : p_{\alpha^*}(z) > 0\}.$$

The elementary inequality $\ln t \le t - 1$ for $t > 0$ gives

$$\begin{aligned} R(\alpha^*) - R(\alpha) &= \int_A \ln\left(\frac{p_\alpha(z)}{p_{\alpha^*}(z)}\right) p_{\alpha^*}(z)\, dz \\ &\le \int_A \left(\frac{p_\alpha(z)}{p_{\alpha^*}(z)} - 1\right) p_{\alpha^*}(z)\, dz \\ &= \int_A p_\alpha(z)\, dz - 1 \le 0. \end{aligned}$$

Equality in $\ln t \le t - 1$ occurs only at $t = 1$. Equality of the risks therefore requires the two densities to agree on A except on a set whose density integral is zero. It also requires $\int_A p_\alpha(z)\, dz = 1$, so p_α assigns no probability outside A. Thus the two distributions are equal. Identifiability then gives $\alpha = \alpha^*$. □

For i.i.d. data $Z_1, \dots, Z_n$ from the true distribution, define the empirical log-loss

$$\widehat{R}_n(\alpha) := -\frac{1}{n} \sum_{i=1}^n \ln p_\alpha(Z_i).$$

For observed data $z = (z_1, \dots, z_n)$, the likelihood under parameter α is the function

$$\mathcal{L}_n(\alpha; z) := \prod_{i=1}^n p_\alpha(z_i).$$

Whenever the likelihood is positive and finite,

$$-\ln \mathcal{L}_n(\alpha; z) = n\widehat{R}_n(\alpha; z),$$

where $\widehat{R}_n(\alpha; z)$ is the observed value of the empirical risk. Hence minimizing empirical log-loss and maximizing likelihood give the same parameter values, when their optima are attained.

For example, the normal location-scale family is

$$p_{\mu,\sigma}(z) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(z-\mu)^2}{2\sigma^2}\right), \qquad (\mu, \sigma) \in \mathbb{R} \times (0, \infty).$$

Every density is positive on $\mathbb{R}$, and distinct parameter pairs give distinct distributions. Thus the model is identifiable.

4.2.1 Maximum likelihood and regression

Let (X, Y) have a joint density or joint PMF of the form

$$p_\theta(x, y) = q_\theta(y \mid x) p_X(x), \qquad \theta \in \Theta,$$

where only the conditional density or PMF q_θ depends on θ. For an i.i.d. observed sample $(x_1, y_1), \ldots, (x_n, y_n)$, suppose that the density or mass values in the following logarithms are positive and finite. Then

$$-\sum_{i=1}^{n} \ln p_\theta(x_i, y_i) = -\sum_{i=1}^{n} \ln q_\theta(y_i \mid x_i) - \sum_{i=1}^{n} \ln p_X(x_i).$$

The second sum does not depend on θ. Therefore, maximizing the joint likelihood over θ is equivalent to maximizing the conditional likelihood.

Example 1: Linear regression

Suppose X and Y are real-valued and the proposed conditional density is normal with mean $ax + b$ and variance σ^2:

$$q_{a,b,\sigma}(y \mid x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(y-(ax+b))^2}{2\sigma^2}\right), \qquad \sigma > 0.$$

The negative conditional log-likelihood is

$$n \ln \sigma + \frac{n}{2} \ln(2\pi) + \frac{1}{2\sigma^2} \sum_{i=1}^{n} (y_i - (ax_i + b))^2. \tag{4.4}$$

For each fixed $\sigma > 0$, minimizing (4.4) over (a, b) is exactly the least-squares problem. If the least-squares minimum is positive and is attained at $(\widehat{a}, \widehat{b})$, then joint minimization also gives

$$\widehat{\sigma}^2 = \frac{1}{n} \sum_{i=1}^{n} (y_i - (\widehat{a}x_i + \widehat{b}))^2.$$

If the minimum sum of squares is zero, the likelihood increases without bound as σ decreases to zero, so there is no maximizer with $\sigma > 0$. For fixed σ, the population log-loss risk also differs from the squared risk in Section 4.1.2 only by a positive scaling and a constant.

Example 2: Logistic regression

Suppose X is real-valued and $Y \in \{0,1\}$. For $\beta = (\beta_0, \beta_1) \in \mathbb{R}^2$, define

$$G(t) := \frac{1}{1+e^{-t}}, \qquad p_\beta(x) := G(\beta_0 + \beta_1 x).$$

The proposed conditional PMF is

$$q_\beta(y \mid x) = p_\beta(x)^y (1 - p_\beta(x))^{1-y}, \qquad y \in \{0,1\}.$$

For observed data (x_i, y_i), its negative conditional log-likelihood is

$$-\sum_{i=1}^{n} \ln q_\beta(y_i \mid x_i) = -\sum_{i=1}^{n} \big(y_i \ln p_\beta(x_i) + (1-y_i)\ln(1 - p_\beta(x_i))\big).$$

If $z_i := 2y_i - 1 \in \{-1, 1\}$, then the same expression is

$$\sum_{i=1}^{n} \ln\left(1 + e^{-z_i(\beta_0 + \beta_1 x_i)}\right).$$

The logistic function is used because the conditional log-odds are linear:

$$\begin{aligned}\ln\left(\frac{\mathbb{P}_\beta(Y=1 \mid X=x)}{\mathbb{P}_\beta(Y=0 \mid X=x)}\right) &= \ln\left(\frac{p_\beta(x)}{1-p_\beta(x)}\right) \\ &= \beta_0 + \beta_1 x.\end{aligned}$$

As with the general risk-minimization problem, a finite optimizer is not automatic. For example, perfectly separated binary data can make the logistic likelihood approach its supremum while $\|\beta\|$ tends to infinity. Software returning fitted coefficients does not replace checking whether the stated model has an attained optimum.

4.3 Exercises

Exercise 4.11 (A finite classification model)**.** *Let $k \geq 2$ be an integer, let X take values in a finite set $\mathbb{X}$, and let $Y \in \{0, \dots, k-1\}$. Specify a statistical model for classification by giving a probability mass function for X and conditional class probabilities for Y given each $x \in \mathbb{X}$. State the conditions these probabilities must satisfy. Write the resulting joint probability mass function of (X, Y) and verify that it sums to 1.*

Exercise 4.12 (Squared loss in a finite model)**.** *Let $X \sim \mathrm{Bernoulli}(1/2)$ and suppose that, conditionally on $X = x$,*

$$Y = x + \epsilon, \qquad \mathbb{P}(\epsilon = -1 \mid X = x) = \mathbb{P}(\epsilon = 1 \mid X = x) = \frac{1}{2}.$$

1. *Find the regression function $r(x) = \mathbb{E}[Y \mid X = x]$.*

2. *Compute the squared risks of r, $g_0(x) = 0$, and $g_1(x) = 1$.*

3. *Verify (4.1) separately for g_0 and g_1.*

Exercise 4.13 (Bayes classification risk). *Suppose X takes the values a, b, c with respective probabilities $1/4$, $1/2$, and $1/4$, and*

$$\eta(a) = 0.1, \qquad \eta(b) = 0.5, \qquad \eta(c) = 0.8.$$

1. *Find the Bayes rule with the tie convention in Definition 4.8 and compute its risk.*
2. *Compute the risk of the rule that always predicts class 1. Verify the excess-risk identity in (4.3).*
3. *Describe all risk-minimizing rules. Explain the role of the tie at b.*

Exercise 4.14 (Linear regression with Python). *Use `numpy.random.uniform` and `numpy.random.normal` to generate $n = 100$ independent pairs from*

$$X \sim \text{Uniform}([0,1]), \qquad Y = 2 - 3X + \epsilon, \qquad \epsilon \sim \text{Normal}(0, 0.5^2),$$

with X and ϵ independent.

1. *Fit `sklearn.linear_model.LinearRegression` and report the fitted intercept and slope.*
2. *Use `matplotlib` to plot the data, the true regression function, and the fitted line.*
3. *Repeat the experiment 500 times. Plot histograms of the fitted intercepts and slopes and explain why the fitted line is a random function before the training data are observed.*

Exercise 4.15 (Logistic loss and separation). *Consider the four observations*

$$(-2, 0), \quad (-1, 0), \quad (1, 1), \quad (2, 1).$$

1. *For $\beta_0 = 0$, show that the negative conditional log-likelihood decreases to 0 as $\beta_1 \to \infty$. Explain why no finite maximum-likelihood estimate exists.*
2. *Fit `sklearn.linear_model.LogisticRegression` for several positive values of its regularization parameter `C`. Record the fitted slope and predicted probabilities.*
3. *Use `matplotlib` to plot the fitted probability curves. Explain why the finite software output does not contradict part 1.*

Chapter 5

Fundamentals of Estimation

Course guide

Scheduled lecture(s): Lectures 6–7
Required for those lectures: Lecture 6: Sections 5.1–5.2; Lecture 7: Sections 5.2–5.4; Exercises: Section 5.5
Prerequisites: Chapter 3, Sections 3.2–3.4, and Chapter 4, Section 4.1
Optional or advanced: The brief remark on function-valued targets in Section 5.1 is optional. The proof of the DKW inequality is outside the scope of the course.

5.1 Point estimates from data

We begin with a finite model. Let $X_i = 1$ if the i-th toss of a coin is Heads and let $X_i = 0$ otherwise. Assume that the tosses are independent and that

$$X_i \sim \text{Bernoulli}(\theta), \qquad \theta \in [0,1].$$

The unknown parameter θ is the probability of Heads. After $n = 5$ tosses, suppose that the observed data are

$$x = (1,0,1,1,0).$$

The fraction of Heads is

$$\overline{x}_5 = \frac{1+0+1+1+0}{5} = 0.6.$$

Thus 0.6 is one point estimate of θ. Before the tosses are observed, the same rule is the random variable

$$\overline{X}_5 = \frac{1}{5}\sum_{i=1}^{5} X_i.$$

This distinction between a rule and its observed value is central to the definitions below.

Let $X_1, \ldots, X_n$ be real-valued RVs on a probability triple $(\Omega, \mathcal{F}, \mathbb{P})$. The random vector

$$X = (X_1, \ldots, X_n) : \Omega \to \mathbb{X}_n \subseteq \mathbb{R}^n$$

is the data before the experiment is performed. Its value $x = (x_1, \ldots, x_n) \in \mathbb{X}_n$ is the observed data after the experiment is performed. For finite $\mathbb{X}_n$, every subset of the data space is an event in the finite data model with event collection $2^{\mathbb{X}_n}$. Consequently, every function on $\mathbb{X}_n$ is a valid transformation of X.

Definition 5.1 (Statistic). *Let $k \in \mathbb{N}$ with $k \geq 1$, and let $X : \Omega \to \mathbb{X}_n \subseteq \mathbb{R}^n$ be the data. A* **statistic** *is a function*

$$T : \mathbb{X}_n \to \mathbb{R}^k$$

for which $T(X)$ is a random vector. Thus T is a valid transformation in the sense of Definition 2.39. A statistic may use the data and known features of the model, but it may not use an unknown parameter value.

For a finite data space, the validity condition in Definition 5.1 is automatic. For the standard continuous models in these notes, familiar continuous functions of the data, such as sums and averages, are valid transformations.

Recall from Definition 4.1 that a statistical model is a family of possible distributions. For fixed n, write the possible joint distributions of the data vector as

$$\mathcal{S}_n = \{F_{\lambda,n} : \lambda \in \mathbf{\Lambda}\}.$$

The fixed but unknown value λ^* determines the distribution that generates the data. In an i.i.d. model, $F_{\lambda,n}$ is determined by the common one-observation distribution and the independence assumption. We write $\mathbb{P}_\lambda$, $\mathbb{E}_\lambda$, and $\mathbb{V}_\lambda$ for probabilities, expectations, and variances computed under the parameter value λ.

A numerical feature of the model at λ is denoted by $\theta(\lambda)$. For example, $\theta(\lambda)$ may be a success probability or the mean of the distribution. If the indexing parameter itself is the target, then $\theta(\lambda) = \lambda$.

Definition 5.2 (Point estimator). *Let $\theta(\lambda) \in \mathbb{R}$ be a numerical target defined for every $\lambda \in \mathbf{\Lambda}$. A* **point estimator** *of $\theta(\lambda)$ based on n observations is a real-valued statistic*

$$\widehat{\Theta}_n : \mathbb{X}_n \to \mathbb{R}.$$

The random variable $\widehat{\Theta}_n(X)$ is the estimator before the data are observed. The number $\widehat{\Theta}_n(x)$ is the point estimate obtained from observed data x.

In the coin model, $\mathbf{\Lambda} = [0, 1]$, $F_{\theta,n}$ is the joint distribution of n independent Bernoulli(θ) RVs, and the target is the indexing parameter itself. The sample mean

$$\widehat{\Theta}_n(X) = \overline{X}_n := \frac{1}{n}\sum_{i=1}^{n} X_i$$

is a point estimator. It does not use the unknown value of θ.

Remark 5.3. *Later chapters estimate an entire distribution function, regression function, or classification rule. Such an estimator is function-valued. The scalar definitions in this chapter can be applied to its value at one fixed input. Comparing whole functions requires a specified loss or distance and is outside the required material here.*

5.2 Bias, standard error, and mean squared error

Fix n and λ. The distribution of $\widehat{\Theta}_n(X)$ under the parameter value λ is its **sampling distribution**. It describes how the estimator would vary over repetitions of the same statistical experiment. The next definitions summarize its centre and spread.

Definition 5.4 (Bias)**.** *Suppose that $\widehat{\Theta}_n(X)$ is integrable under the parameter value λ. Its bias at λ is*

$$\mathsf{bias}_\lambda(\widehat{\Theta}_n) := \mathbb{E}_\lambda\left[\widehat{\Theta}_n(X)\right] - \theta(\lambda). \tag{5.1}$$

If the estimator is integrable for every λ and n, the sequence $(\widehat{\Theta}_n)_{n\geq 1}$ is **unbiased** *for θ on the model if*

$$\mathsf{bias}_\lambda(\widehat{\Theta}_n) = 0$$

for every $\lambda \in \mathbf{\Lambda}$ and every $n \geq 1$. It is **asymptotically unbiased** *if, for every $\lambda \in \mathbf{\Lambda}$,*

$$\mathsf{bias}_\lambda(\widehat{\Theta}_n) \longrightarrow 0.$$

Example 5.5 (The sample mean)**.** *Fix λ. Suppose that the observations are integrable under the parameter value λ and have the common mean*

$$\theta(\lambda) := \mathbb{E}_\lambda\left[X_1\right].$$

By linearity of expectation,

$$\mathbb{E}_\lambda\left[\overline{X}_n\right] = \frac{1}{n}\sum_{i=1}^{n}\mathbb{E}_\lambda\left[X_i\right] = \theta(\lambda).$$

Therefore, the sample mean is unbiased. Independence is not needed for this calculation; common finite means are enough.

Definition 5.6 (Standard error)**.** *Suppose that $\widehat{\Theta}_n(X)$ has a finite second moment under the parameter value λ. Its* **standard error** *at λ is the standard deviation of its sampling distribution:*

$$\mathsf{se}_\lambda(\widehat{\Theta}_n) := \sqrt{\mathbb{V}_\lambda(\widehat{\Theta}_n(X))}. \tag{5.2}$$

The standard error is a property of the sampling distribution, not of one observed data set. It may depend on the unknown parameter and may therefore need to be estimated.

Definition 5.7 (Mean squared error)**.** *Suppose that $\widehat{\Theta}_n(X)$ has a finite second moment under the parameter value λ. Its* **mean squared error** *(MSE) at λ is*

$$\mathsf{MSE}_\lambda(\widehat{\Theta}_n) := \mathbb{E}_\lambda\left[\left(\widehat{\Theta}_n(X) - \theta(\lambda)\right)^2\right]. \tag{5.3}$$

Proposition 5.8 (Bias–variance decomposition)**.** *If $\widehat{\Theta}_n(X)$ has a finite second moment under the parameter value λ, then*

$$\mathsf{MSE}_\lambda(\widehat{\Theta}_n) = \mathsf{se}_\lambda(\widehat{\Theta}_n)^2 + \mathsf{bias}_\lambda(\widehat{\Theta}_n)^2. \tag{5.4}$$

Proof. Set

$$B := \mathbb{E}_\lambda\left[\widehat{\Theta}_n(X)\right] - \theta(\lambda).$$

Then

$$\widehat{\Theta}_n(X) - \theta(\lambda) = \widehat{\Theta}_n(X) - \mathbb{E}_\lambda\left[\widehat{\Theta}_n(X)\right] + B.$$

Expanding the square and taking expectations gives

$$\begin{aligned}\mathsf{MSE}_\lambda(\widehat{\Theta}_n) = {} & \mathbb{V}_\lambda(\widehat{\Theta}_n(X)) + B^2 \\ & + 2B\,\mathbb{E}_\lambda\left[\widehat{\Theta}_n(X) - \mathbb{E}_\lambda\left[\widehat{\Theta}_n(X)\right]\right].\end{aligned}$$

The last expectation is zero. By (5.1) and (5.2), the first two terms are the squared standard error and squared bias. ∎

Example 5.9 (Estimating a Bernoulli probability)**.** *Let $X_1, \dots, X_n$ be independent* Bernoulli(θ) *RVs, where $\theta \in [0, 1]$. For the sample mean,*

$$\mathbb{E}_\theta\left[\overline{X}_n\right] = \theta,$$

so its bias is zero. Independence gives

$$\mathbb{V}_\theta(\overline{X}_n) = \frac{1}{n^2}\sum_{i=1}^{n}\mathbb{V}_\theta(X_i) = \frac{\theta(1-\theta)}{n}.$$

Therefore,

$$\begin{aligned}\mathsf{se}_\theta(\overline{X}_n) &= \sqrt{\frac{\theta(1-\theta)}{n}},\\ \mathsf{MSE}_\theta(\overline{X}_n) &= \frac{\theta(1-\theta)}{n}.\end{aligned}$$

These formulas also cover the endpoints $\theta = 0$ and $\theta = 1$.

Unbiasedness is useful, but it is not the only criterion for an estimator. A biased estimator can have a smaller variance. The MSE records both effects on the same scale, so it is often the more direct measure of accuracy.

5.3 Consistency

An estimator can have nonzero error for every fixed sample size and still become accurate as the amount of data increases.

Definition 5.10 (Consistency)**.** *Fix $\lambda \in \boldsymbol{\Lambda}$. A sequence of point estimators $(\widehat{\Theta}_n)_{n\geq 1}$ is* **consistent** *for $\theta(\lambda)$ if*

$$\widehat{\Theta}_n(X_1, \dots, X_n) \xrightarrow{\mathbb{P}_\lambda} \theta(\lambda).$$

Equivalently, for every $\epsilon > 0$,

$$\mathbb{P}_\lambda\left(|\widehat{\Theta}_n(X_1, \dots, X_n) - \theta(\lambda)| > \epsilon\right) \longrightarrow 0.$$

The sequence is consistent on the statistical model if this holds for every $\lambda \in \boldsymbol{\Lambda}$.

The word "consistent" therefore refers to a sequence of estimators and a specified model. It is not a property of one observed estimate.

Proposition 5.11 (Vanishing bias and standard error imply consistency)**.** *Fix $\lambda \in \boldsymbol{\Lambda}$ and suppose that $\widehat{\Theta}_n(X_1, \dots, X_n)$ has a finite second moment for every n. If*

$$\mathsf{bias}_\lambda(\widehat{\Theta}_n) \longrightarrow 0 \qquad \text{and} \qquad \mathsf{se}_\lambda(\widehat{\Theta}_n) \longrightarrow 0,$$

then $\widehat{\Theta}_n$ is consistent for $\theta(\lambda)$.

Proof. Write

$$T_n := \widehat{\Theta}_n(X_1, \dots, X_n)$$

for the estimator as an RV under $\mathbb{P}_\lambda$. By (5.4),

$$\mathsf{MSE}_\lambda(\widehat{\Theta}_n) \longrightarrow 0.$$

Markov's inequality applied to the nonnegative RV $(T_n - \theta(\lambda))^2$ gives, for every $\epsilon > 0$,

$$\begin{aligned}\mathbb{P}_\lambda\left(|T_n - \theta(\lambda)| > \epsilon\right) &\le \mathbb{P}_\lambda\left((T_n - \theta(\lambda))^2 \ge \epsilon^2\right)\\ &\le \frac{\mathsf{MSE}_\lambda(\widehat{\Theta}_n)}{\epsilon^2} \longrightarrow 0.\end{aligned}$$

This is the definition of consistency. □

For the Bernoulli sample mean,

$$\mathsf{bias}_\theta(\overline{X}_n) = 0, \qquad \mathsf{se}_\theta(\overline{X}_n) = \sqrt{\frac{\theta(1-\theta)}{n}} \longrightarrow 0.$$

Hence Definition 5.11 shows that $\overline{X}_n$ is consistent for every $\theta \in [0, 1]$.

The sample mean also illustrates exactly which assumptions are needed for different conclusions. If $X_1, X_2, \ldots$ are i.i.d. and integrable, the strong law of large numbers in Definition 3.26 gives

$$\overline{X}_n \xrightarrow{\text{a.s.}} \mathbb{E}[X_1], \qquad \text{and hence} \qquad \overline{X}_n \xrightarrow{\mathbb{P}} \mathbb{E}[X_1].$$

Thus integrability is enough for consistency of the sample mean. If the stronger condition $\mathbb{V}(X_1) < \infty$ holds, then its standard error and MSE are also available:

$$\mathsf{se}(\overline{X}_n) = \sqrt{\frac{\mathbb{V}(X_1)}{n}}, \qquad \mathsf{MSE}(\overline{X}_n) = \frac{\mathbb{V}(X_1)}{n}.$$

Example 5.12 (Consistency need not imply asymptotic unbiasedness)**.** *Let $X_1, X_2, \ldots$ be i.i.d.* Uniform$(0, 1)$ *RVs. To estimate the constant target* 0*, consider the estimator*

$$Z_n := n\mathbb{1}_{\{X_1 \le 1/n\}}.$$

This is a statistic based on $(X_1, \ldots, X_n)$, although it ignores all but the first observation. For every $\epsilon > 0$, $\mathbb{P}(|Z_n| \ge \epsilon) = 1/n$ for all sufficiently large n, so $Z_n \xrightarrow{\mathbb{P}} 0$. However,

$$\mathbb{E}[Z_n] = n\,\mathbb{P}(X_1 \le 1/n) = 1.$$

Consequently, convergence in probability, including convergence obtained from a law of large numbers, does not by itself justify taking limits of expectations.

5.4 Estimating a distribution function

Return to the coin-toss data from Section 5.1. For the observed sample

$$x = (1, 0, 1, 1, 0),$$

the fraction of observations at or below a real number t is

$$\widehat{F}_5(t; x) = \frac{1}{5}\sum_{i=1}^{5} \mathbb{1}_{\{x_i \le t\}} = \begin{cases} 0, & t < 0, \\ 2/5, & 0 \le t < 1, \\ 1, & t \ge 1. \end{cases}$$

This function is a distribution function. If the observations are independent Bernoulli(θ) RVs, the unknown distribution function is

$$F_\theta(t) = \begin{cases} 0, & t < 0, \\ 1-\theta, & 0 \le t < 1, \\ 1, & t \ge 1. \end{cases}$$

Thus $\widehat{F}_5$ estimates the whole function F_θ, not only the single number θ.

We now allow any distribution function on $\mathbb{R}$. Let

$$\mathcal{E} := \{F : F \text{ is a distribution function on } \mathbb{R}\}.$$

This is a statistical model in the sense of Definition 4.1. It is called nonparametric because its unknown member F is an entire function rather than a fixed finite list of real parameters. Assume that $X_1, X_2, \ldots$ are i.i.d. with some fixed but unknown $F \in \mathcal{E}$. In this section, $\mathbb{P}$, $\mathbb{E}$, and $\mathbb{V}$ refer to the law determined by F and the independence assumption.

Definition 5.13 (Empirical distribution function)**.** *The* **empirical distribution function** *(empirical DF or ECDF) is*

$$\widehat{F}_n(t; X) := \frac{1}{n}\sum_{i=1}^{n} \mathbb{1}_{\{X_i \le t\}}, \qquad t \in \mathbb{R}.$$

For observed data $x = (x_1, \ldots, x_n)$, its value is the function

$$\widehat{F}_n(t; x) = \frac{1}{n}\sum_{i=1}^{n} \mathbb{1}_{\{x_i \le t\}}.$$

For each observed sample, $\widehat{F}_n(\,\cdot\,; x)$ is a distribution function: it is nondecreasing and right-continuous, tends to 0 as $t \to -\infty$, and tends to 1 as $t \to \infty$. It places mass $1/n$ at each observation, with repeated observations contributing repeated mass. For each fixed t, $\widehat{F}_n(t; X)$ is a real-valued statistic, so the definitions of bias, standard error, MSE, and consistency from Sections 5.2–5.3 apply directly.

Proposition 5.14 (Pointwise properties of the ECDF)**.** *Let $X_1, \ldots, X_n$ be i.i.d. with distribution function F. For every fixed $t \in \mathbb{R}$,*

$$\begin{aligned} \mathbb{E}\,[\widehat{F}_n(t; X)] &= F(t), \\ \mathbb{V}(\widehat{F}_n(t; X)) &= \frac{F(t)(1 - F(t))}{n}. \end{aligned}$$

Consequently, $\widehat{F}_n(t; X)$ *is unbiased for* $F(t)$ *and*

$$\mathsf{se}(\widehat{F}_n(t;X)) = \sqrt{\frac{F(t)(1-F(t))}{n}},$$
$$\mathsf{MSE}(\widehat{F}_n(t;X)) = \frac{F(t)(1-F(t))}{n}.$$

In particular, $\widehat{F}_n(t; X)$ *is consistent for* $F(t)$*.*

Proof. Fix $t \in \mathbb{R}$ and set

$$I_i(t) := \mathbb{1}_{\{X_i \le t\}}.$$

The RVs $I_1(t), \dots, I_n(t)$ are independent Bernoulli($F(t)$) RVs. Therefore,

$$\mathbb{E}\,[\widehat{F}_n(t;X)] = \frac{1}{n}\sum_{i=1}^{n} \mathbb{E}\,[I_i(t)] = F(t),$$
$$\mathbb{V}(\widehat{F}_n(t;X)) = \frac{1}{n^2}\sum_{i=1}^{n} \mathbb{V}(I_i(t)) = \frac{F(t)(1-F(t))}{n}.$$

The standard-error and MSE formulas follow from unbiasedness. Since the standard error tends to zero, consistency follows from Definition 5.11. □

The common distribution determines the success probability $F(t)$ in the proof. Independence is used when the variances are added.

No moment assumption on X_i is needed for this result. For each fixed t, the proof concerns bounded indicator RVs, even when X_i itself has no finite mean or variance.

Pointwise consistency answers a separate question for every fixed t. Because the target is the whole function F, we would prefer one error bound that holds for all t at the same time. Measure the largest vertical distance between two distribution functions by

$$\|\widehat{F}_n - F\|_\infty := \sup_{t\in\mathbb{R}} |\widehat{F}_n(t;X) - F(t)|.$$

Theorem 5.15 (Dvoretzky–Kiefer–Wolfowitz inequality)**.** *Let* $X_1, \dots, X_n$ *be i.i.d. with distribution function* F*. Then, for every* $\epsilon > 0$*,*

$$\mathbb{P}\left(\|\widehat{F}_n - F\|_\infty > \epsilon\right) \le 2e^{-2n\epsilon^2}. \tag{5.5}$$

The theorem requires no continuity or density assumption on F. Its proof is beyond the required material, but its conclusion is important here. For every $\epsilon > 0$, the right-hand side of (5.5) tends to zero. Hence

$$\|\widehat{F}_n - F\|_\infty \xrightarrow{\mathbb{P}} 0.$$

This is uniform consistency: with high probability, the entire ECDF is close to the entire unknown distribution function.

The DKW inequality also gives a simultaneous confidence band. Fix $\alpha \in (0, 1)$ and set

$$\epsilon_{n,\alpha} := \sqrt{\frac{1}{2n}\ln\left(\frac{2}{\alpha}\right)}.$$

Define

$$\underline{C}_n(t) := \max\{\widehat{F}_n(t; X) - \epsilon_{n,\alpha}, 0\},$$
$$\overline{C}_n(t) := \min\{\widehat{F}_n(t; X) + \epsilon_{n,\alpha}, 1\}.$$

Substitution in (5.5) gives

$$\mathbb{P}\left(\underline{C}_n(t) \le F(t) \le \overline{C}_n(t) \text{ for every } t \in \mathbb{R}\right) \ge 1 - \alpha.$$

The word "simultaneous" matters: this is one event on which the band covers $F(t)$ at every real t, not a separate coverage statement for each fixed t.

There is a useful connection with later chapters. For the half-line $A_t = (-\infty, t]$,

$$F(t) = \mathbb{P}(X_1 \in A_t), \qquad \widehat{F}_n(t; X) = \frac{1}{n}\sum_{i=1}^{n} \mathbb{1}_{\{X_i \in A_t\}}.$$

Thus the ECDF replaces each unknown probability by its empirical frequency, and DKW controls the replacement uniformly over all half-lines. The same idea reappears in the optional complexity chapter, where empirical risks are controlled uniformly over a class of prediction rules.

5.5 Exercises

Exercise 5.16 (Estimator and estimate)**.** *Suppose $X_1, \ldots, X_8$ are independent RVs, each with distribution* Bernoulli(θ)*. The observed sample is*

$$x = (1, 0, 0, 1, 1, 0, 1, 1).$$

1. *Define the sample-mean estimator as a function on the data space. State clearly which object is a function, which is an RV, and which is the observed numerical estimate.*

2. *Compute the estimate for the observed sample.*

3. *Explain why the rule $T(x) = \theta$ is not a statistic when θ is unknown.*

Exercise 5.17 (Bias–variance tradeoff)**.** *Let $X_1, \ldots, X_n$ be independent* Bernoulli(θ) *RVs. For a constant $c \in [0, 1]$, consider*

$$\widehat{\Theta}_{n,c} := c\overline{X}_n + \frac{1 - c}{2}.$$

1. *Compute its bias, standard error, and MSE at θ.*

2. *Compare its MSE with that of $\overline{X}_n$. Show that for some values of θ and $c < 1$, introducing bias reduces MSE.*

3. *Let (c_n) be a sequence in $[0, 1]$. Give a condition on c_n that guarantees consistency for every $\theta \in [0, 1]$.*

Exercise 5.18 (Sampling distributions in Python)**.** *Fix $\theta = 0.3$. For each $n \in \{10, 50, 200\}$, use* `numpy.random.binomial` *to simulate* 10000 *independent values of the estimator $\overline{X}_n$.*

1. *Estimate its bias, standard error, and MSE from the simulations. Compare them with the formulas in Definition 5.9.*

2. *Use* `matplotlib` *to draw the three sampling distributions as density-normalized histograms with common horizontal limits.*

3. *Plot the simulated standard errors and the theoretical curve* $\sqrt{\theta(1-\theta)/n}$ *against* n*. Explain the observed rate.*

Exercise 5.19 (Computing an ECDF)**.** *For the sample*

$$x = (-1, 2, 2, 0, 3),$$

write $\widehat{F}_5(t; x)$ *as a piecewise-defined function. Verify directly that it is nondecreasing, right-continuous, and has the required limits at* $-\infty$ *and* $+\infty$*. Compute* $\widehat{F}_5(-1)$*,* $\widehat{F}_5(1)$*, and* $\widehat{F}_5(2)$*.*

Exercise 5.20 (ECDFs and simultaneous bands)**.** *Let* F *be the distribution function of* Uniform$([0, 1])$*.*

1. *Use Python and* `numpy` *to generate one sample for each* $n \in \{20, 100, 500\}$*. Compute the ECDF on a grid of points in* $[-0.1, 1.1]$ *using comparisons and averages.*

2. *With* `matplotlib`*, plot each ECDF, the true* F*, and the* 95% *DKW band obtained by clipping at* 0 *and* 1*.*

3. *Repeat the experiment* 2000 *times for each* n*. If* $x_{(1)} \leq \cdots \leq x_{(n)}$ *are the sorted observations, use*

$$\|\widehat{F}_n - F\|_\infty = \max_{1\leq i\leq n} \left\{ \frac{i}{n} - x_{(i)}, x_{(i)} - \frac{i-1}{n} \right\}$$

for this continuous uniform model. Estimate the probability that the true DF leaves the DKW band and compare it with 0.05*.*

4. *Explain why checking the band only at one fixed* t *would answer a different question.*

Chapter 6

Random Variable Generation

Course guide

Scheduled lecture(s): Lecture 8
Required for those lectures: Sections 6.1–6.3
Prerequisites: Chapter 1, Section 1.3, and Chapter 2, Sections 2.1–2.4 and 2.6.5
Optional or advanced: The number-theoretic proof of the Hull–Dobell theorem and the polar-coordinate calculation in the Box–Muller proof are optional; cryptographic random-number generation is outside scope.

A computer program started from a fixed initial state follows deterministic instructions. It therefore does not produce mathematical randomness by itself. A pseudorandom number generator instead produces a deterministic sequence whose relevant numerical properties resemble those of random samples.

For a first example, compare a fair draw X from $\{0, 1\}$ with the deterministic sequence

$$0, 1, 0, 1, 0, 1, \ldots .$$

The probability model satisfies $\mathbb{P}(X = 0) = \mathbb{P}(X = 1) = 1/2$. The deterministic sequence has the same long-run proportions of zeros and ones. It is nevertheless completely predictable. Matching one-point frequencies is therefore only a first requirement; it does not imply independence or other properties of random samples.

Definition 6.1 (Uniformly pseudorandom sequence)**.** *Let $M \in \mathbb{N}$ with $M \geq 2$, and set*

$$\mathcal{M} := \{0, 1, \ldots, M - 1\}.$$

For a deterministic sequence $u_0, u_1, \ldots$ in $\mathcal{M}$ and $c \in \mathcal{M}$, define the relative frequency

$$N_n(c) := \frac{1}{n} \#\{0 \leq i \leq n - 1 : u_i = c\}, \qquad n \geq 1.$$

The sequence is **uniformly pseudorandom** *on $\mathcal{M}$ if*

$$\lim_{n\to\infty} N_n(c) = \frac{1}{M} \qquad \text{for every } c \in \mathcal{M}.$$

This is a deliberately modest definition for this chapter. It compares the long-run frequencies with the probability law of a uniform random variable on $\mathcal{M}$. A practical generator must also be tested for patterns among successive values. Cryptographic unpredictability is a stronger requirement and is outside the scope of the course.

6.1 Congruential Generators

Definition 6.2 (Congruential generator). *Let $M \in \mathbb{N}$ with $M \geq 2$, and let $a, b \in \mathbb{Z}$. A* **congruential generator** *with parameters (a, b, M) is the map $D : \mathcal{M} \to \mathcal{M}$ defined by*

$$D(x) := (ax + b) \bmod M,$$

where the right-hand side is the remainder in $\mathcal{M}$. Given a seed $u_0 \in \mathcal{M}$, the generator produces

$$u_{i+1} = D(u_i), \qquad i = 0, 1, \ldots.$$

Example 6.3. *Consider the generator $(3, 0, 16)$ on $\{0, 1, \ldots, 15\}$. Starting from $u_0 = 1$ gives*

$$u_0 = 1, \qquad u_1 = 3, \qquad u_2 = 9, \qquad u_3 = 11, \qquad u_4 = 1.$$

The sequence is therefore

$$1, 3, 9, 11, 1, 3, 9, 11, \ldots.$$

Every sequence obtained by repeatedly applying a map to a finite state space eventually repeats. A sequence is *purely periodic with period T* if T is the smallest positive integer such that

$$u_{i+T} = u_i \qquad \text{for every } i \geq 0.$$

More generally, it is eventually periodic if this identity holds for every $i \geq \mu$ for some $\mu \geq 0$. The values before index μ form the transient part of the sequence.

Lemma 6.4 (Finite deterministic sequences). *Let $D : \mathcal{M} \to \mathcal{M}$ and $u_{i+1} = D(u_i)$. Then the sequence is eventually periodic, with eventual period at most M.*

Proof. Among $u_0, u_1, \ldots, u_M$, two values must agree. Choose indices $0 \leq r < s \leq M$ with $u_r = u_s$. Applying D repeatedly gives

$$u_{r+j} = u_{s+j} \qquad \text{for every } j \geq 0.$$

Thus the sequence repeats after $s - r \leq M$ steps from index r onward. □

For the generator $(3, 0, 16)$, the seed 0 gives a constant sequence and period 1, the seed 1 gives period 4, and the seed 2 gives period 2. This dependence on the seed is one reason to distinguish an arbitrary cycle from a full-period generator.

A congruential generator on $\mathcal{M}$ has *full period* if, from every seed, its sequence is purely periodic with period M. Equivalently, one orbit visits all M states before returning to its starting point.

Lemma 6.5. *If a congruential generator on $\mathcal{M}$ has full period, then the sequence from every seed is uniformly pseudorandom on $\mathcal{M}$.*

Proof. Fix a seed. Every block of M consecutive terms contains every element of $\mathcal{M}$ exactly once. Write $n = qM + r$, where $0 \leq r < M$. For each $c \in \mathcal{M}$, the first qM terms contain q copies of c, and the remaining r terms contain either zero or one copy. Hence

$$\left| N_n(c) - \frac{1}{M} \right| \leq \frac{1}{n}.$$

Letting $n \to \infty$ proves the claim. □

The following number-theoretic theorem characterizes the full-period mixed congruential generators.

Theorem 6.6 (Hull–Dobell theorem)**.** *Let $M \geq 2$. The congruential generator (a, b, M) has full period M if and only if*

- $\gcd(b, M) = 1$*;*
- *every prime divisor of M divides $a - 1$; and*
- *if 4 divides M, then 4 divides $a - 1$.*

The number-theoretic proof is outside the declared prerequisites, so we use the theorem without proof. See [HD, K] for that argument.

A full-period generator can be mapped to a smaller discrete set without changing its limiting proportions.

Lemma 6.7 (Rescaling to a smaller set)**.** *Let a congruential generator on $\mathcal{M}$ have full period, and let $u_0, u_1, \ldots$ be its sequence. Suppose that $K \in \mathbb{N}$, $1 \leq K \leq M$, and K divides M. Define*

$$v_i := \left\lfloor \frac{K u_i}{M} \right\rfloor, \qquad i \geq 0.$$

Then $v_0, v_1, \ldots$ is uniformly pseudorandom on $\mathcal{K}$. Moreover, $v_{i+M} = v_i$ for every $i \geq 0$, although the least period of the rescaled sequence may be smaller than M.

Proof. Fix $c \in \mathcal{K}$. Since K divides M,

$$\left\lfloor \frac{Kx}{M} \right\rfloor = c$$

holds for exactly the M/K integers

$$x = \frac{cM}{K}, \frac{cM}{K} + 1, \ldots, \frac{(c+1)M}{K} - 1.$$

Every block of M consecutive values of u_i visits each state once, so every such block of v_i contains exactly M/K copies of c. A final incomplete block contributes at most M terms. Dividing the count by n and letting $n \to \infty$ gives the limiting frequency $1/K$.

Full periodicity gives $u_{i+M} = u_i$. Applying the displayed rescaling map proves $v_{i+M} = v_i$. □

The same frequencies determine two numerical summaries. For this deterministic sequence, we define the empirical mean and variance directly; they are not expectations of random variables.

Lemma 6.8 (Empirical moments)**.** *Let $u_0, u_1, \ldots$ be uniformly pseudorandom on $\mathcal{M}$. Define*

$$\overline{u}_n := \frac{1}{n} \sum_{i=0}^{n-1} u_i, \qquad s_n^2 := \frac{1}{n} \sum_{i=0}^{n-1} u_i^2 - \overline{u}_n^2.$$

Then

$$\lim_{n\to\infty} \overline{u}_n = \frac{M-1}{2}, \qquad \lim_{n\to\infty} s_n^2 = \frac{M^2-1}{12}.$$

Proof. By Definition 6.1 and the fact that $\mathcal{M}$ is finite,

$$\lim_{n\to\infty} \overline{u}_n = \sum_{c=0}^{M-1} c \lim_{n\to\infty} N_n(c) = \frac{1}{M}\sum_{c=0}^{M-1} c = \frac{M-1}{2},$$
$$\lim_{n\to\infty} \frac{1}{n}\sum_{i=0}^{n-1} u_i^2 = \frac{1}{M}\sum_{c=0}^{M-1} c^2 = \frac{(M-1)(2M-1)}{6}.$$

Subtracting the square of the first limit gives

$$\frac{(M-1)(2M-1)}{6} - \left(\frac{M-1}{2}\right)^2 = \frac{M^2-1}{12}.$$

□

Corollary 6.9. *Under the assumptions of the preceding lemma, let $w_i = u_i/M$. Then*

$$\lim_{n\to\infty} \frac{1}{n}\sum_{i=0}^{n-1} w_i = \frac{1}{2} - \frac{1}{2M},$$

and

$$\lim_{n\to\infty} \left(\frac{1}{n}\sum_{i=0}^{n-1} w_i^2 - \left(\frac{1}{n}\sum_{i=0}^{n-1} w_i\right)^2\right) = \frac{1}{12} - \frac{1}{12M^2}.$$

The normalized values w_i lie on the grid $\{0, 1/M, \ldots, (M-1)/M\}$. Their interval frequencies are close to those of a uniform random variable on $[0,1]$.

Lemma 6.10 (Interval discrepancy)**.** *Let $v_0, v_1, \ldots$ be uniformly pseudorandom on $\mathcal{M}$, and set $w_i = v_i/M$. For $0 \le a < b \le 1$, define*

$$N_n((a,b)) := \frac{1}{n}\#\{0 \le i \le n-1 : a < w_i < b\}.$$

Then

$$\left|\lim_{n\to\infty} N_n((a,b)) - (b-a)\right| \le \frac{1}{M}.$$

Proof. The limiting frequency is the fraction of grid points in (a,b):

$$\lim_{n\to\infty} N_n((a,b)) = \frac{1}{M}\#\{j \in \mathcal{M} : Ma < j < Mb\}.$$

The number in braces is

$$q = \lceil Mb \rceil - \lfloor Ma \rfloor - 1.$$

Consequently,

$$q - M(b-a) = \big(\lceil Mb \rceil - Mb\big) + \big(Ma - \lfloor Ma \rfloor\big) - 1,$$

whose absolute value is at most 1. Division by M proves the bound. □

A finite-state generator remains deterministic and eventually periodic. The preceding results establish useful frequency properties, but they do not turn its outputs into independent uniform random variables. In the sampling theorems below, independent uniform random variables are a mathematical model for the ideal inputs to an algorithm.

6.2 Sampling

6.2.1 Inversion sampling

Definition 6.11 (Quantile function)**.** *Let F be a distribution function. Its* quantile function*, or generalized inverse, is*

$$F^{-1}(u) := \inf\{x \in \mathbb{R} : F(x) \geq u\}, \qquad 0 < u < 1.$$

The limiting properties of a distribution function ensure that the infimum is a finite real number. We may assign arbitrary values to $F^{-1}(0)$ and $F^{-1}(1)$ when a formula is needed on all of $[0, 1]$.

If F is strictly increasing and continuous, this is the usual inverse function. The generalized inverse is also defined when F has jumps or flat parts.

Theorem 6.12 (Inversion sampling theorem)**.** *Let $U \sim \mathrm{Uniform}([0, 1])$, and let F be a distribution function. Then*

$$F^{-1}(U) \sim F.$$

Proof. For $0 < u < 1$ and $x \in \mathbb{R}$, the definition and right-continuity of F give

$$F^{-1}(u) \leq x \iff u \leq F(x). \tag{6.1}$$

Indeed, the reverse implication follows because x belongs to the set defining the infimum. For the forward implication, suppose instead that $F(x) < u$. Right-continuity gives a $\delta > 0$ such that $F(x + \delta) < u$. By monotonicity, no $y \leq x + \delta$ satisfies $F(y) \geq u$, so $F^{-1}(u) \geq x + \delta$, a contradiction.

A uniform random variable has probability zero of equalling either endpoint of $[0, 1]$. Therefore, (6.1) gives

$$\mathbb{P}(F^{-1}(U) \leq x) = \mathbb{P}(U \leq F(x)) = F(x).$$

Thus $F^{-1}(U)$ has distribution function F. □

Example 6.13 (A distribution with a jump)**.** *Suppose X takes the value 0 with probability $1 - p$ and the value 1 with probability p, where $0 < p < 1$. Its quantile function is*

$$F^{-1}(u) = \begin{cases} 0, & 0 < u \leq 1 - p, \\ 1, & 1 - p < u < 1. \end{cases}$$

Hence $F^{-1}(U)$ equals 0 with probability $1 - p$ and 1 with probability p, as required.

Example 6.14 (Exponential sampling)**.** *Let $\lambda > 0$. The distribution function of the exponential distribution with rate λ is*

$$F(x) = \begin{cases} 0, & x < 0, \\ 1 - e^{-\lambda x}, & x \geq 0. \end{cases}$$

For $0 < u < 1$,

$$F^{-1}(u) = -\frac{1}{\lambda} \log(1 - u).$$

Thus, if $U \sim \mathrm{Uniform}([0, 1])$, then $-\lambda^{-1} \log(1 - U)$ has the exponential distribution with rate λ. Its value may be defined arbitrarily on the probability-zero event $\{U = 1\}$.

Inversion is convenient when the quantile function is available. When it is not available in closed form or is expensive to evaluate numerically, accept–reject sampling provides another method.

6.2.2 Accept–reject sampling

Algorithm 1 Accept–Reject Sampler

input:

(1) a target density f;

(2) a proposal density g and a constant $M \geq 1$ such that $f(x) \leq Mg(x)$ for every $x \in \mathbb{R}$.

output: the requested number of i.i.d. observations with density f.
repeat
  Independently of all previous repetitions, draw X with density g and draw $U \sim \text{Uniform}([0,1])$ independently of X.
  if $g(X) > 0$ and $U \leq f(X)/(Mg(X))$ **then**
    Append X to the output sequence.
  end if
until the requested number of observations has been output.

The domination assumption implies $f(x) = 0$ whenever $g(x) = 0$, so the algorithm never needs to divide by zero.

Lemma 6.15 (Correctness of accept–reject sampling). *Let $f, g : \mathbb{R} \to [0, \infty)$ be densities, and let $M \geq 1$ satisfy $f(x) \leq Mg(x)$ for every $x \in \mathbb{R}$. Define*

$$r(x) := \begin{cases} \dfrac{f(x)}{Mg(x)}, & g(x) > 0, \\ 0, & g(x) = 0. \end{cases}$$

Let $X_1, X_2, \ldots$ be i.i.d. with density g, let $U_1, U_2, \ldots$ be i.i.d. Uniform$([0,1])$*, and assume that the two sequences are independent. Accept X_i on the event*

$$A_i := \{g(X_i) > 0,\ U_i \leq r(X_i)\}.$$

Then $\mathbb{P}(A_i) = 1/M$, and the accepted proposals, in their original order, are i.i.d. with density f. The number of proposals examined for each accepted output has the geometric distribution with parameter $1/M$ and mean M. Among the first n proposals, the number accepted has the binomial distribution with parameters n and $1/M$.

Proof. The inequality $f \leq Mg$ gives $0 \leq r \leq 1$. It also gives $f(x) = 0$ when $g(x) = 0$, and hence

$$r(x)g(x) = \frac{f(x)}{M} \qquad \text{for every } x \in \mathbb{R}.$$

Independence of X_i and U_i gives, for every standard event $B \subseteq \mathbb{R}$,

$$\begin{aligned}\mathbb{P}(X_i \in B, A_i) &= \int_B \int_0^{r(x)} g(x)\,du\,dx \\ &= \frac{1}{M}\int_B f(x)\,dx.\end{aligned}$$

Taking $B = \mathbb{R}$ gives $\mathbb{P}(A_i) = 1/M > 0$. Therefore,

$$\mathbb{P}(X_i \in B \mid A_i) = \int_B f(x)\,dx.$$

The pairs $W_i = (X_i, U_i)$ are i.i.d., and every proposal is retained using the same fixed acceptance set

$$A_{\mathrm{AR}} := \{(x,u) \in \mathbb{R} \times [0,1] : g(x) > 0,\ u \le r(x)\}.$$

For the densities considered in this course, this is a standard event. The accepted pairs, and hence their first coordinates, are i.i.d. by Definition 2.55. The same result gives the geometric waiting times and their mean. The fixed-batch count follows from Definition 2.56. □

The acceptance probability is $1/M$. Thus a smaller valid value of M gives a more efficient sampler. The proposal density should be easy to sample from and should place enough density wherever the target density is large.

6.2.3 The Box–Muller method

The standard normal distribution function has no elementary closed-form inverse. Numerical quantile approximations are widely used, but a direct transformation of two uniform inputs is also possible.

Theorem 6.16 (Box–Muller)**.** *Let U_1 and U_2 be independent* Uniform$([0,1])$ *random variables. On the event $\{U_1 > 0\}$, define*

$$\begin{aligned}Z_0 &= \sqrt{-2\log U_1}\cos(2\pi U_2), \\ Z_1 &= \sqrt{-2\log U_1}\sin(2\pi U_2).\end{aligned}$$

Assign arbitrary values to Z_0 and Z_1 when $U_1 = 0$. Then Z_0 and Z_1 are independent, and each has the Normal$(0,1)$ *distribution.*

Optional proof using polar coordinates. Set

$$R := \sqrt{-2\log U_1}, \qquad \Theta := 2\pi U_2$$

on $\{U_1 > 0\}$. For $r \ge 0$,

$$\mathbb{P}(R \le r) = \mathbb{P}(U_1 \ge e^{-r^2/2}) = 1 - e^{-r^2/2}.$$

Hence R has density $f_R(r) = re^{-r^2/2}$ for $r > 0$. Also, Θ is uniform on $[0, 2\pi]$. Since R is a function of U_1 and Θ is a function of the independent variable U_2, they are independent. Their joint density is therefore

$$f_{R,\Theta}(r,\theta) = \frac{r}{2\pi}e^{-r^2/2}, \qquad r > 0, \quad 0 < \theta < 2\pi.$$

The polar-coordinate map

$$(r, \theta) \longmapsto (z_0, z_1) = (r\cos\theta, r\sin\theta)$$

has area factor r. Away from the origin and one choice of polar ray, the density change-of-variables formula gives

$$\begin{aligned} f_{Z_0,Z_1}(z_0, z_1) &= \frac{1}{r} f_{R,\Theta}(r, \theta) \\ &= \frac{1}{2\pi}\exp\left(-\frac{z_0^2 + z_1^2}{2}\right) \\ &= \left(\frac{1}{\sqrt{2\pi}} e^{-z_0^2/2}\right)\left(\frac{1}{\sqrt{2\pi}} e^{-z_1^2/2}\right). \end{aligned}$$

The exceptional ray and the origin have probability zero, and the value of a density there is irrelevant. The final factorization is the joint density of two independent standard normal random variables. □

6.3 Exercises

Exercise 6.17. *Let $u_0, u_1, \ldots$ be the output of a full-period congruential generator on $\mathcal{M}$, where $M \geq 2$. For $x, y \in \mathcal{M}$, define the relative frequency of the consecutive pair (x, y) by*

$$N_n(x, y) := \frac{1}{n}\#\{0 \leq i \leq n-1 : (u_i, u_{i+1}) = (x, y)\}.$$

Show that every $N_n(x, y)$ has a limit and that exactly M of the M^2 possible pairs have a positive limiting frequency. Conclude that uniform one-point frequencies do not imply the pair frequencies of independent uniform random variables.

Exercise 6.18.

- *In Python, implement a congruential generator (a, b, M). Use the Hull–Dobell conditions to choose parameters with full period.*
- *Form the normalized values $w_i = u_i/M$. For the first n values, use Matplotlib to plot the empirical distribution function $\widehat{F}_n$ and compute*

  $$D_n := \sup_{x\in\mathbb{R}} |\widehat{F}_n(x) - F(x)|.$$

  Here F is the distribution function of $\mathrm{Uniform}([0, 1])$.
- *Choose $0 < \alpha < 1$. Under the hypothetical model of i.i.d. uniform observations, Definition 5.15 gives the level-α threshold*

  $$\sqrt{\frac{\log(2/\alpha)}{2n}}.$$

  Compare D_n with this threshold. Explain why the deterministic generator does not satisfy the i.i.d. hypothesis, so this comparison is a diagnostic rather than a calibrated statistical test.

Exercise 6.19.

- *In Python, use Definition 6.16 to generate standard normal values from successive pairs of uniform inputs. Ensure that the first input in each pair lies in $(0, 1)$ so that its logarithm is defined.*

- *For each standard normal output Z, form*

$$Y = 10 + \sqrt{5}Z.$$

  Then $Y \sim \mathrm{Normal}(10, 5)$; the second parameter is the variance.

- *Repeat the empirical-distribution diagnostic and plot from Definition 6.18, now using the* Normal(10, 5) *distribution function. Identify precisely which i.i.d. assumption in Definition 5.15 is not justified for outputs obtained from a deterministic congruential generator.*

Exercise 6.20. *Consider the density*

$$f(x) = \frac{1}{2}\cos x, \qquad -\frac{\pi}{2} < x < \frac{\pi}{2},$$

and $f(x) = 0$ outside this interval.

- *Use Matplotlib to plot the distribution function F.*
- *Find the quantile function F^{-1}.*
- *Implement the inversion sampler in Python and display a histogram of a large sample together with the density f.*
- *Use the proposal density of* $\mathrm{Uniform}([-\pi/2, \pi/2])$ *in the accept–reject sampler Algorithm 1. Find the smallest valid constant M. On average, how many proposals are rejected before one proposal is accepted?*

Chapter 7

Finite Markov Chains

Course guide

Scheduled lecture(s): Lectures 8–9
Required for those lectures: Lecture 8: Sections 7.1–7.2 and corresponding exercises in Section 7.6; Lecture 9: Sections 7.3–7.5 and corresponding exercises in Section 7.6
Prerequisites: Chapter 1, Section 1.3; Chapter 2, Section 2.6; Chapter 6, Section 6.2; and elementary matrix multiplication
Optional or advanced: The proofs in Section 7.4.1 are advanced. Infinite-state and continuous-time chains are outside scope.

7.1 Introduction

Consider a weather model with two states,

$$\mathbb{X} = \{\mathsf{D}, \mathsf{W}\},$$

where D means dry and W means wet. Suppose that tomorrow is dry with probability 3/4 after a dry day and with probability 1/2 after a wet day. The remaining probabilities give a wet day. Thus the one-step rules are

	D	W
D	3/4	1/4
W	1/2	1/2

The model asserts that, once today's state is known, earlier states do not change the probabilities for tomorrow. This is the Markov property.

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability triple and let $\mathbb{X}$ be finite. A finite-state random variable is a function $X : \Omega \to \mathbb{X}$ such that $\{X = x\} \in \mathcal{F}$ for every $x \in \mathbb{X}$. Finite unions then show that $\{X \in A\}$ is an event for every $A \subseteq \mathbb{X}$.

Let $\mathbb{Z}_+ := \{0, 1, 2, \ldots\}$. A **discrete-time stochastic process** on $\mathbb{X}$ is a sequence $(X_t)_{t \in \mathbb{Z}_+}$ of finite-state random variables defined on one probability model. The set $\mathbb{X}$ is the **state space**, and X_t is the state at time t.

Definition 7.1 (Stochastic matrix)**.** *Let $\mathbb{X} = \{s_1, \ldots, s_k\}$. A matrix $P = (P(x,y))_{x,y \in \mathbb{X}}$ is a* **stochastic matrix** *if*

$$P(x,y) \geq 0 \quad \text{for all } x, y \in \mathbb{X}, \qquad \sum_{y \in \mathbb{X}} P(x,y) = 1 \quad \text{for all } x \in \mathbb{X}.$$

Each row $P(x,\cdot)$ is therefore a probability distribution on $\mathbb{X}$.

Definition 7.2 (Finite time-homogeneous Markov chain)**.** *Let $(X_t)_{t\in\mathbb{Z}_+}$ be a stochastic process on the finite state space $\mathbb{X}$, and let P be a stochastic matrix on $\mathbb{X}$. The process is a* **time-homogeneous Markov chain** *with* **transition matrix** *P if, for every $t \in \mathbb{Z}_+$ and every $x_0,\dots,x_t,y \in \mathbb{X}$ such that*

$$\mathbb{P}(X_0 = x_0,\dots,X_t = x_t) > 0,$$

we have

$$\mathbb{P}(X_{t+1} = y \mid X_0 = x_0,\dots,X_t = x_t) = P(x_t,y).$$

The positivity assumption is needed because the conditional probability in the definition must be defined. The right-hand side depends on the present state x_t, but not on the earlier states or on t. Dependence only on the present state is the Markov property. Using the same matrix at every time is time homogeneity.

For the weather model, with the state order (D,W), the transition matrix is

$$P = \begin{pmatrix} 3/4 & 1/4 \\ 1/2 & 1/2 \end{pmatrix}.$$

If $X_0 = \mathsf{D}$, then the initial distribution and the distribution after one step are the row vectors

$$\mu_0 = (1,0), \qquad \mu_1 = \left(\frac{3}{4},\frac{1}{4}\right) = \mu_0 P.$$

The law of total probability gives

$$\begin{aligned}\mathbb{P}(X_2 = \mathsf{D}) &= \mathbb{P}(X_2 = \mathsf{D} \mid X_1 = \mathsf{D})\,\mathbb{P}(X_1 = \mathsf{D}) \\ &\quad + \mathbb{P}(X_2 = \mathsf{D} \mid X_1 = \mathsf{W})\,\mathbb{P}(X_1 = \mathsf{W}) \\ &= \frac{3}{4}\frac{3}{4} + \frac{1}{2}\frac{1}{4} = \frac{11}{16}.\end{aligned}$$

This is the first coordinate of $\mu_0 P^2$.

For a general initial distribution μ_0, write

$$\mu_t(y) := \mathbb{P}(X_t = y), \qquad y \in \mathbb{X}.$$

We regard μ_t as a row vector in the chosen order of the states.

Lemma 7.3. *Let $(X_t)_{t\in\mathbb{Z}_+}$ be a finite time-homogeneous Markov chain with transition matrix P and initial distribution μ_0. Then, for every $t \in \mathbb{Z}_+$,*

$$\mu_t = \mu_0 P^t.$$

Proof. The statement holds at $t = 0$ because P^0 is the identity matrix. For $y \in \mathbb{X}$, the events $\{X_t = x\}$, $x \in \mathbb{X}$, partition the sample space. Hence the law of total probability and the Markov property give

$$\begin{aligned}\mu_{t+1}(y) &= \sum_{x\in\mathbb{X}} \mathbb{P}(X_t = x, X_{t+1} = y) \\ &= \sum_{x\in\mathbb{X}} \mu_t(x)P(x,y) = (\mu_t P)(y).\end{aligned}$$

If $\mu_t(x) = 0$, both the corresponding joint probability and the product $\mu_t(x)P(x,y)$ are zero, so no conditioning on a zero-probability event is used. Thus $\mu_{t+1} = \mu_t P$. Induction proves the result. $\square$

When a chain is initialized at a fixed state x, we write $\mathbb{P}_x$ and $\mathbb{E}_x$ for probabilities and expectations in that model. This notation does not mean that we condition an existing model on an event of probability zero. If e_x is the row vector concentrated at x, then Definition 7.3 gives

$$\mathbb{P}_x(X_t = y) = (e_x P^t)(y) = P^t(x, y).$$

Thus $P^t(x, y)$ is the probability of moving from x to y in t steps.

The same matrix also acts on functions. For $f : \mathbb{X} \to \mathbb{R}$ and $t \in \mathbb{Z}_+$, define

$$(P^t f)(x) := \sum_{y \in \mathbb{X}} P^t(x, y) f(y) = \mathbb{E}_x [f(X_t)]. \tag{7.1}$$

Lemma 7.4 (Semigroup property)**.** *Let $f, g : \mathbb{X} \to \mathbb{R}$, let $a, b \in \mathbb{R}$, and let $s, t \in \mathbb{Z}_+$. Then*

$$P^t(af + bg) = aP^t f + bP^t g,$$
$$P^{t+s} f = P^t(P^s f).$$

Proof. The first identity follows by distributing the finite sum in (7.1). Matrix multiplication gives

$$P^t(P^s f) = (P^t P^s) f = P^{t+s} f.$$

□

A **semigroup** is a set with an associative binary operation. The operators $\{P^t : t \in \mathbb{Z}_+\}$ form a semigroup under composition. It also contains the identity P^0, so it is more precisely a monoid. No continuous-time process is needed for this chapter.

The transition rule may also change with time.

Definition 7.5 (Finite time-inhomogeneous Markov chain)**.** *Let $(P_t)_{t \geq 1}$ be stochastic matrices on $\mathbb{X}$. A process $(X_t)_{t \in \mathbb{Z}_+}$ is a time-inhomogeneous Markov chain with these transition matrices if, for every $t \geq 1$ and every $x_0, \ldots, x_{t-1}, y \in \mathbb{X}$ with*

$$\mathbb{P}(X_0 = x_0, \ldots, X_{t-1} = x_{t-1}) > 0,$$

we have

$$\mathbb{P}(X_t = y \mid X_0 = x_0, \ldots, X_{t-1} = x_{t-1}) = P_t(x_{t-1}, y).$$

Here P_t describes the transition from time $t - 1$ to time t. A time-homogeneous chain is the special case $P_t = P$ for every t.

Lemma 7.6. *Let $(X_t)_{t \in \mathbb{Z}_+}$ be a finite time-inhomogeneous Markov chain with initial distribution μ_0 and transition matrices $P_1, P_2, \ldots$. For $t \in \mathbb{Z}_+$, let μ_t be the distribution of X_t. Then, for every integer $t \geq 1$,*

$$\mu_t = \mu_0 P_1 P_2 \cdots P_t. \tag{7.2}$$

Proof. The same total-probability calculation as in Definition 7.3 gives $\mu_t = \mu_{t-1} P_t$. Repeating this identity from time t back to time 1 proves the formula. □

7.2 Simulation by Random Mappings

We next turn a transition matrix into a simulation rule. Recall the inversion method from Definition 6.12: intervals whose lengths are the desired probabilities can be selected with one uniform random variable.

Definition 7.7 (Random mapping representation)**.** *Let P be a stochastic matrix on a finite state space $\mathbb{X}$. A* **random mapping representation** *(RMR) of P consists of a random variable $U \sim \mathrm{Uniform}([0,1])$ and a function*

$$\rho : \mathbb{X} \times [0,1] \to \mathbb{X} \tag{7.3}$$

such that $\rho(x,U)$ is a finite-state random variable for every $x \in \mathbb{X}$ and, for every $x, y \in \mathbb{X}$,

$$\mathbb{P}(\rho(x,U) = y) = P(x,y). \tag{7.4}$$

Theorem 7.8 (Existence of a random mapping representation)**.** *Every stochastic matrix on a finite state space has a random mapping representation.*

Proof. Enumerate $\mathbb{X} = \{s_1, \dots, s_k\}$. For $1 \le i \le k$, set

$$c_{i,0} := 0, \qquad c_{i,j} := \sum_{h=1}^{j} P(s_i, s_h), \quad 1 \le j \le k.$$

Thus $c_{i,k} = 1$. For $j < k$, define

$$I_{i,j} := [c_{i,j-1}, c_{i,j}),$$

and define $I_{i,k} := [c_{i,k-1}, 1]$. For fixed i, these intervals partition $[0,1]$. Set $\rho(s_i, u) = s_j$ when $u \in I_{i,j}$. Since the uniform probability of an interval equals its length,

$$\mathbb{P}(\rho(s_i,U) = s_j) = c_{i,j} - c_{i,j-1} = P(s_i, s_j).$$

Hence (ρ, U) is an RMR of P. □

Independent random inputs are required at different time steps. Reusing one uniform value generally does not produce the intended Markov chain.

Theorem 7.9 (Simulation by random mappings)**.** *Let X_0 have distribution μ_0. Let $U_1, U_2, \dots$ be mutually independent $\mathrm{Uniform}([0,1])$ RVs, independent of X_0. For every $t \ge 1$, let $\rho_t : \mathbb{X} \times [0,1] \to \mathbb{X}$ satisfy*

$$\mathbb{P}(\rho_t(x,U_t) = y) = P_t(x,y) \qquad \textit{for all } x, y \in \mathbb{X},$$

where P_t is a stochastic matrix, and assume that $\rho_t(x,U_t)$ is a finite-state random variable for every $x \in \mathbb{X}$. Define recursively

$$X_t := \rho_t(X_{t-1}, U_t), \qquad t \ge 1.$$

Then $(X_t)_{t \in \mathbb{Z}_+}$ is a time-inhomogeneous Markov chain with initial distribution μ_0 and transition matrix P_t from time $t-1$ to time t.

Proof. The recursion defines finite-state random variables. Indeed, for $y \in \mathbb{X}$,

$$\{X_t = y\} = \bigcup_{x \in \mathbb{X}} \big(\{X_{t-1} = x\} \cap \{\rho_t(x, U_t) = y\}\big),$$

which is an event by induction and finiteness of $\mathbb{X}$.

Fix $t \geq 1$ and a history $x_0, \ldots, x_{t-1}$ having positive probability. The variables $X_0, \ldots, X_{t-1}$ are functions of $X_0, U_1, \ldots, U_{t-1}$. Independence therefore implies that U_t is independent of this history. Consequently, for every $y \in \mathbb{X}$,

$$\begin{aligned}&\mathbb{P}(X_t = y \mid X_0 = x_0, \ldots, X_{t-1} = x_{t-1}) \\ &\qquad = \mathbb{P}(\rho_t(x_{t-1}, U_t) = y) = P_t(x_{t-1}, y).\end{aligned}$$

This is the Markov property. The distribution of X_0 is μ_0 by assumption. □

For a homogeneous chain, take $\rho_t = \rho$ and $P_t = P$ for every t. Starting from a chosen X_0, draw a fresh U_t and apply $X_t = \rho(X_{t-1}, U_t)$ at each step.

7.3 Irreducibility and Aperiodicity

Two examples show why long-run behaviour needs additional hypotheses. The matrices

$$P_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad P_2 = \begin{pmatrix} 1/2 & 1/2 \\ 1/2 & 1/2 \end{pmatrix}$$

both allow movement between the two states. The first chain alternates deterministically, while the second can return to a state after either one or two steps. Irreducibility describes movement throughout the state space; aperiodicity rules out forced cycles such as the first example.

Definition 7.10 (Accessibility and communication)**.** *Let P be the transition matrix of a finite time-homogeneous Markov chain. A state y is* **accessible** *from x, written $x \to y$, if there exists $n \in \mathbb{Z}_+$ such that $P^n(x, y) > 0$. States x and y* **communicate***, written $x \leftrightarrow y$, if $x \to y$ and $y \to x$.*

Because P^0 is the identity, every state is accessible from itself.

Definition 7.11 (Irreducible)**.** *A finite time-homogeneous Markov chain is* **irreducible** *if every pair of states communicates. Otherwise it is* **reducible***.*

Definition 7.12 (Return times and period)**.** *For a state x, let*

$$\mathbb{T}(x) := \{n \in \mathbb{N} : n \geq 1 \text{ and } P^n(x, x) > 0\}.$$

If $\mathbb{T}(x)$ is nonempty, the **period** *of x is*

$$d(x) := \gcd \mathbb{T}(x).$$

The state is **aperiodic** *if $d(x) = 1$.*

An irreducible finite chain always has a possible positive return time for each state, so its periods are defined.

Proposition 7.13 (Common period). *In an irreducible finite Markov chain, all states have the same period.*

Proof. Fix distinct states x and y; there is nothing to prove when $x = y$. By irreducibility, there are positive integers r and s such that

$$P^r(x,y) > 0, \qquad P^s(y,x) > 0.$$

Hence $r + s \in \mathbb{T}(x) \cap \mathbb{T}(y)$. If $n \in \mathbb{T}(x)$, following a path from y to x, returning from x to itself in n steps, and then following a path from x to y shows that $s + n + r \in \mathbb{T}(y)$. Therefore $d(y)$ divides both $r + s$ and $s + n + r$, so it divides n. Thus $d(y)$ divides every element of $\mathbb{T}(x)$ and hence divides $d(x)$. Interchanging x and y shows that $d(x)$ divides $d(y)$. Therefore $d(x) = d(y)$. □

Definition 7.14 (Aperiodic chain). *An irreducible finite Markov chain is* **aperiodic** *if its common period is* 1.

For example, P_1 above has return times $2, 4, 6, \ldots$ and period 2. The matrix P_2 has a positive diagonal, so 1 is a possible return time and its period is 1. More generally, an irreducible chain is aperiodic as soon as $P(x,x) > 0$ for at least one state x.

7.3.1 Hitting times

Continue with the weather chain from Section 7.1, started from D. The first time at which the chain is wet is

$$\tau_{\mathsf{W}} := \inf\{t \in \mathbb{Z}_+ : X_t = \mathsf{W}\},$$

with the convention $\inf \varnothing = \infty$. To have $\tau_{\mathsf{W}} = n$, the chain must remain dry for $n - 1$ steps and then become wet. Hence, for $n \geq 1$,

$$\mathbb{P}_{\mathsf{D}}(\tau_{\mathsf{W}} = n) = \left(\frac{3}{4}\right)^{n-1} \frac{1}{4}, \qquad \mathbb{P}_{\mathsf{D}}(\tau_{\mathsf{W}} > n) = \left(\frac{3}{4}\right)^{n}.$$

For this subsection, we extend the definition of the mean for a nonnegative integer-valued random variable T as follows. If $\mathbb{P}(T = \infty) > 0$, set $\mathbb{E}[T] = \infty$. Otherwise use the nonnegative series $\sum_{k=0}^{\infty} k\,\mathbb{P}(T = k)$, which is also allowed to diverge to infinity. The tail-sum formula is

$$\mathbb{E}[T] = \sum_{n=0}^{\infty} \mathbb{P}(T > n), \tag{7.5}$$

where both sides are allowed to be infinite. This agrees with the definition of expectation in Section 2.5 whenever T is integrable. The identity follows by writing $k = \sum_{n=0}^{k-1} 1$ for each finite value k and rearranging the resulting nonnegative sums. Therefore

$$\mathbb{E}_{\mathsf{D}}[\tau_{\mathsf{W}}] = \sum_{n=0}^{\infty} \left(\frac{3}{4}\right)^{n} = 4.$$

Definition 7.15 (Hitting time). *Let A be a nonempty subset of the state space $\mathbb{X}$. The* **hitting time** *of A is*

$$\tau_A := \inf\{t \in \mathbb{Z}_+ : X_t \in A\},$$

with the convention $\inf \varnothing = \infty$*. For* $x \in \mathbb{X}$*, define*

$$h_A(x) := \mathbb{P}_x(\tau_A < \infty), \qquad m_A(x) := \mathbb{E}_x[\tau_A].$$

Thus $h_A(x)$ *is the probability of ever hitting* A*, while* $m_A(x)$ *is the expected time required to hit it. The expectation is allowed to be infinite.*

If $x \in A$, then $\tau_A = 0$ under $\mathbb{P}_x$. To study the first return to a state x instead, one uses

$$\tau_x^+ := \inf\{t \geq 1 : X_t = x\}.$$

The restriction $t \geq 1$ distinguishes a return from the hit at time zero.

The complete distribution of τ_A can be calculated by removing the target states from the transition matrix. Fix a nonempty set $A \subseteq \mathbb{X}$. Let $B := \mathbb{X} \setminus A$, let

$$Q := (P(x,y))_{x,y \in B}, \qquad r(x) := \sum_{y \in A} P(x,y), \quad x \in B,$$

and let $\mathbf{1}$ denote the column vector of ones indexed by B. The matrix Q need not be stochastic because a transition from B into A is not included in it.

Proposition 7.16 (Distribution of a hitting time)**.** *With* A, B, Q, r *as above, let* $x \in B$*. For every integer* $n \geq 1$*,*

$$\begin{aligned} \mathbb{P}_x(\tau_A = n) &= (Q^{n-1} r)(x), \\ \mathbb{P}_x(\tau_A > n) &= (Q^n \mathbf{1})(x). \end{aligned}$$

Consequently,

$$\begin{aligned} h_A(x) &= \sum_{n=1}^{\infty} (Q^{n-1} r)(x), \\ m_A(x) &= \sum_{n=0}^{\infty} (Q^n \mathbf{1})(x), \end{aligned}$$

where the second series may diverge to infinity.

Proof. The event $\{\tau_A > n\}$ occurs exactly when $X_1, \ldots, X_n \in B$. Summing the probabilities of all such paths gives $(Q^n \mathbf{1})(x)$. Similarly, $\{\tau_A = n\}$ requires the first $n-1$ states after time zero to lie in B and the next transition to enter A. Summing these path probabilities gives $(Q^{n-1} r)(x)$. The first series follows by adding the disjoint events $\{\tau_A = n\}$, and the second follows from (7.5). □

For small state spaces, first-step equations are usually faster than computing many powers of Q.

Proposition 7.17 (First-step equations)**.** *Let* $A \subseteq \mathbb{X}$ *be nonempty. For every* $x \in \mathbb{X}$*,*

$$h_A(x) = \begin{cases} 1, & x \in A, \\ \sum_{y \in \mathbb{X}} P(x,y) h_A(y), & x \notin A. \end{cases} \tag{7.6}$$

If $h_A(x) < 1$, then $m_A(x) = \infty$. Let $C := \{x \in \mathbb{X} : h_A(x) = 1\}$. The values $m_A(x)$ are finite on C and are the unique finite solution of

$$m_A(x) = \begin{cases} 0, & x \in A, \\ 1 + \sum_{y \in C} P(x,y) m_A(y), & x \in C \setminus A. \end{cases} \tag{7.7}$$

Proof. If $x \notin A$, condition on the value of X_1. The Markov property gives

$$h_A(x) = \sum_{y \in \mathbb{X}} P(x,y) h_A(y),$$

which proves (7.6). If $h_A(x) < 1$, then $\mathbb{P}_x(\tau_A = \infty) > 0$. Hence $\mathbb{P}_x(\tau_A > n) \geq \mathbb{P}_x(\tau_A = \infty)$ for every n, so (7.5) gives $m_A(x) = \infty$.

Now let $x \in C \setminus A$. The first-step equation for h_A shows that $P(x,y) > 0$ implies $y \in C$. Since the state space is finite and every state in C can reach A, there are an integer $N \geq 1$ and a number $\delta > 0$ such that, from every state in C, the probability of hitting A within N steps is at least δ. Applying the Markov property at times $N, 2N, \ldots$ gives

$$\mathbb{P}_x(\tau_A > kN) \leq (1-\delta)^k, \qquad k \in \mathbb{Z}_+.$$

Thus the tail-sum formula shows that $m_A(x) < \infty$. Conditioning on X_1 now gives (7.7).

To see uniqueness, subtract two finite solutions and choose a state where the absolute difference is largest. The homogeneous first-step equation forces the same largest value along every positive-probability transition. Following a positive-probability path to A, where the difference is zero, gives a contradiction unless the largest difference is zero. □

The equations for h_A alone need an additional boundary value when some states cannot reach A. Set $h_A(x) = 0$ at every such state, set $h_A(x) = 1$ on A, and solve (7.6) at the remaining states. These conditions determine h_A uniquely by the same largest-difference argument used in the proof.

For the weather chain and $A = \{\mathsf{W}\}$, the expectation equations are

$$m_A(\mathsf{W}) = 0, \qquad m_A(\mathsf{D}) = 1 + \frac{3}{4} m_A(\mathsf{D}) + \frac{1}{4} m_A(\mathsf{W}).$$

Solving gives $m_A(\mathsf{D}) = 4$, in agreement with the direct calculation.

As an edge case, consider a chain on $\{a, b, c\}$ that moves from a to b or c with probability $1/2$ each, while b and c are absorbing. For $A = \{b\}$,

$$h_A(b) = 1, \qquad h_A(c) = 0, \qquad h_A(a) = \frac{1}{2}.$$

Thus $m_A(a) = \infty$, even though the chain reaches either absorbing state after one step.

7.4 Stationarity

Return to the weather chain. The probability vector

$$\pi = \left(\frac{2}{3}, \frac{1}{3}\right)$$

satisfies

$$\pi P = \left(\frac{2}{3}, \frac{1}{3}\right) \begin{pmatrix} 3/4 & 1/4 \\ 1/2 & 1/2 \end{pmatrix} = \left(\frac{2}{3}, \frac{1}{3}\right) = \pi.$$

Thus, if X_0 has distribution π, then every X_t has distribution π. Individual states still fluctuate; it is the distribution that does not change.

Definition 7.18 (Stationary distribution)**.** *Let P be a stochastic matrix on $\mathbb{X}$. A probability distribution π on $\mathbb{X}$ is a* **stationary distribution** *for P if*

$$\pi P = \pi.$$

Equivalently, for every $y \in \mathbb{X}$,

$$\pi(y) = \sum_{x \in \mathbb{X}} \pi(x) P(x, y).$$

Theorem 7.19 (Existence of a stationary distribution)**.** *Every stochastic matrix on a finite state space has at least one stationary distribution.*

Neither irreducibility nor aperiodicity is needed for existence. The next result states exactly where those assumptions enter.

Theorem 7.20 (Finite Markov chain convergence)**.** *Let P be the transition matrix of a finite irreducible Markov chain. Then P has a unique stationary distribution π, and $\pi(x) > 0$ for every $x \in \mathbb{X}$. If the chain is also aperiodic, then, for every initial distribution μ and every $y \in \mathbb{X}$,*

$$(\mu P^n)(y) \longrightarrow \pi(y) \qquad \text{as } n \to \infty.$$

Aperiodicity is needed for the convergence clause: the chain with transition matrix P_1 in the preceding section is irreducible, but its distribution alternates when it starts from one state.

7.4.1 Optional proofs*

We first prove existence. Fix any probability row vector μ. For $n \geq 1$, define the average

$$q_n := \frac{1}{n} \sum_{t=0}^{n-1} \mu P^t.$$

Each q_n is a probability vector. Hence (q_n) is a bounded sequence in $\mathbb{R}^k$ and has a convergent subsequence. Write $q_{n_j} \to \pi$. The limit π is again a probability vector. Moreover,

$$q_n P - q_n = \frac{\mu P^n - \mu}{n}.$$

Every coordinate on the right has absolute value at most $1/n$, so it converges to zero. Passing to the subsequence gives $\pi P = \pi$.

We next prove Definition 7.20 using finite sums and an elementary contraction argument.

First, let π be a stationary distribution, whose existence was proved above. Choose a state x with $\pi(x) > 0$. For any y, irreducibility gives an n such that $P^n(x, y) > 0$. Since $\pi P^n = \pi$,

$$\pi(y) \geq \pi(x) P^n(x, y) > 0.$$

Thus every coordinate of π is positive.

Suppose that ν is another stationary distribution and set

$$c := \min_{x\in\mathbb{X}} \frac{\nu(x)}{\pi(x)}, \qquad \lambda := \nu - c\pi.$$

Then $\lambda \geq 0$, $\lambda P = \lambda$, and $\lambda(x_0) = 0$ for at least one state x_0. If λ were not zero, there would be a state z with $\lambda(z) > 0$. Choose n with $P^n(z, x_0) > 0$. Then

$$0 = \lambda(x_0) = (\lambda P^n)(x_0) \geq \lambda(z)P^n(z, x_0) > 0,$$

a contradiction. Hence $\nu = c\pi$. Both vectors have coordinate sum 1, so $c = 1$ and $\nu = \pi$. This proves uniqueness without using aperiodicity.

Now assume that the chain is aperiodic. We first show that some power of P has all entries positive. For each state x, the set $\mathbb{T}(x)$ is closed under addition and has greatest common divisor 1. A finite subset $a_1, \ldots, a_\ell \in \mathbb{T}(x)$ already has greatest common divisor 1. The nonnegative integer combinations of these numbers contain every sufficiently large integer: modulo a_1, their residues form a finite additive subgroup; the greatest-common-divisor condition makes this subgroup all residue classes. Choose one nonnegative combination for each residue class. Adding a sufficiently large multiple of a_1 then gives every sufficiently large integer. It follows that every sufficiently large integer belongs to $\mathbb{T}(x)$.

For each pair x, y, choose $r(x, y) \in \mathbb{Z}_+$ with $P^{r(x,y)}(x, y) > 0$. A sufficiently long return from x followed by this path shows that $P^n(x, y) > 0$ for all sufficiently large n. The state space is finite, so there is one integer $m \geq 1$ such that

$$P^m(x, y) > 0 \qquad \text{for every } x, y \in \mathbb{X}.$$

Let $k = |\mathbb{X}|$ and

$$\varepsilon := \min_{x,y\in\mathbb{X}} P^m(x, y) > 0, \qquad \alpha := k\varepsilon \leq 1.$$

If $\alpha = 1$, every row of P^m is the uniform distribution, and the conclusion follows immediately. Suppose $\alpha < 1$ and define

$$R(x, y) := \frac{P^m(x, y) - \varepsilon}{1 - \alpha}.$$

The matrix R is stochastic. If u is the uniform probability row vector, then

$$P^m(x, \cdot) = \alpha u + (1 - \alpha)R(x, \cdot).$$

For probability row vectors μ and ν, the signed vector $\mu - \nu$ has coordinate sum zero. Therefore

$$\begin{aligned}\|\mu P^m - \nu P^m\|_1 &= (1 - \alpha)\|(\mu - \nu)R\|_1 \\ &\leq (1 - \alpha)\|\mu - \nu\|_1,\end{aligned}$$

where $\|v\|_1 := \sum_x |v(x)|$. The inequality follows from the triangle inequality and the fact that each row of R sums to 1.

Write $n = qm + r$ with $0 \leq r < m$. Since $\pi P^n = \pi$ and multiplication by a stochastic matrix does not increase the ℓ^1 norm, repeated use of the contraction gives

$$\|\mu P^n - \pi\|_1 \leq (1 - \alpha)^q\|\mu - \pi\|_1 \longrightarrow 0.$$

This proves the convergence clause of Definition 7.20.

7.5 Reversibility

Detailed balance is often an efficient way to find a stationary distribution.

Definition 7.21 (Reversible distribution)**.** *Let P be a stochastic matrix on $\mathbb{X}$. A probability distribution π on $\mathbb{X}$ is* **reversible** *for P if, for every $x, y \in \mathbb{X}$,*

$$\pi(x)P(x,y) = \pi(y)P(y,x). \tag{7.8}$$

A chain with such a distribution is called reversible with respect to π.

If $X_0 \sim \pi$, then the left-hand side of (7.8) is $\mathbb{P}(X_0 = x, X_1 = y)$. Detailed balance says that this one-step probability is unchanged when the direction is reversed.

Proposition 7.22 (Detailed balance implies stationarity)**.** *If π is reversible for a stochastic matrix P, then π is stationary for P.*

Proof. Fix $y \in \mathbb{X}$. By detailed balance and the fact that the y-th row of P sums to 1,

$$\begin{aligned}(\pi P)(y) &= \sum_{x \in \mathbb{X}} \pi(x)P(x,y) \\ &= \sum_{x \in \mathbb{X}} \pi(y)P(y,x) = \pi(y).\end{aligned}$$

Thus $\pi P = \pi$. □

7.5.1 Random walks on graphs

Definition 7.23 (Finite graphs)**.** *A finite simple undirected graph $\mathbb{G} = (\mathbb{V}, \mathbb{E})$ consists of a finite* **vertex set** *$\mathbb{V}$ and an* **edge set** *$\mathbb{E}$ whose elements are two-element subsets $\{v, w\}$ of $\mathbb{V}$. The* **neighbourhood** *and* **degree** *of v are*

$$\operatorname{nbhd}(v) := \{w \in \mathbb{V} : \{v,w\} \in \mathbb{E}\}, \qquad \deg(v) := |\operatorname{nbhd}(v)|.$$

A path is a sequence of vertices in which consecutive vertices share an edge. The graph is **connected** *if there is a path from every vertex to every other vertex.*

A finite directed graph instead has $\mathbb{E} \subseteq \mathbb{V} \times \mathbb{V}$. Its directed edge (v, w) points from v to w. The out-neighbours of v are the vertices w for which $(v, w) \in \mathbb{E}$.

Model 7.24 (Random walk on a connected undirected graph)**.** *Let $\mathbb{G} = (\mathbb{V}, \mathbb{E})$ be a finite connected undirected graph with at least two vertices. The simple random walk on $\mathbb{G}$ is the Markov chain with state space $\mathbb{V}$ and transition matrix*

$$P(v,w) = \begin{cases} \dfrac{1}{\deg(v)}, & \text{if } \{v,w\} \in \mathbb{E}, \\ 0, & \text{otherwise.} \end{cases}$$

Connectivity and the assumption $|\mathbb{V}| \geq 2$ ensure that every degree is positive. Connectivity also makes the chain irreducible.

Proposition 7.25. *Let P be the transition matrix in Definition 7.24, and set*

$$d := \sum_{v \in \mathbb{V}} \deg(v).$$

Then the distribution

$$\pi(v) := \frac{\deg(v)}{d}, \qquad v \in \mathbb{V},$$

is reversible and hence stationary. It is the unique stationary distribution of the walk.

Proof. The degree sum d is positive, and the displayed values are nonnegative and sum to 1. If $\{v, w\} \in \mathbb{E}$, then

$$\pi(v)P(v,w) = \frac{\deg(v)}{d} \frac{1}{\deg(v)} = \frac{1}{d} = \pi(w)P(w,v).$$

If $\{v, w\} \notin \mathbb{E}$, both sides are zero. Thus detailed balance holds, and Definition 7.22 gives stationarity. The chain is irreducible, so uniqueness follows from Definition 7.20. $\square$

Example 7.26 (A random-surfer model for PageRank)**.** *Let $\mathbb{V}$ be a finite set of webpages and use the directed edge (v, w) when page v links to page w. Choose a probability distribution q on $\mathbb{V}$ with $q(w) > 0$ for every page, and choose $\alpha \in (0, 1)$. Define the link-following matrix L as follows. From a page with outgoing links, choose one of them uniformly. From a page with no outgoing links, choose a page according to q. The random-surfer transition matrix is*

$$P(v,w) := (1-\alpha)L(v,w) + \alpha q(w).$$

At every step, the surfer follows the link rule with probability $1 - \alpha$ and moves according to q with probability α. Since

$$P(v,w) \geq \alpha q(w) > 0$$

for all v, w, the chain is irreducible and aperiodic. It therefore has a unique stationary distribution, and the distribution from every initial page converges to it. PageRank uses the stationary probabilities as page scores.

7.6 Exercises

Exercise 7.27. *Write the coordinate calculation that proves*

$$\mu_t(y) = \sum_{x \in \mathbb{X}} \mu_{t-1}(x) P_t(x,y).$$

Include the case $\mu_{t-1}(x) = 0$, and then derive (7.2).

Exercise 7.28. *For the weather matrix in Section 7.1, write the two interval partitions used in the proof of Definition 7.8. Explain how this is inversion sampling applied separately to each row of P.*

Exercise 7.29. *In the proof of Definition 7.9, identify the equality that can fail if U_t is allowed to depend on $U_1, \ldots, U_{t-1}$. Give a two-state example in which the one-step marginal probabilities are correct but reusing the same uniform variable gives the wrong two-step distribution.*

Exercise 7.30. *Show that a stochastic matrix can have more than one random mapping representation.*

Exercise 7.31. *Let* $\mathbb{X} = \{0, 1, 2\}$*, let* $A = \{2\}$*, and let*

$$P = \begin{pmatrix} 1/2 & 1/2 & 0 \\ 1/4 & 1/4 & 1/2 \\ 0 & 0 & 1 \end{pmatrix}.$$

Use the first-step equations to compute $h_A(0)$*,* $h_A(1)$*,* $m_A(0)$*, and* $m_A(1)$*. Then use Definition 7.16 to write* $\mathbb{P}_0(\tau_A = n)$ *in terms of the matrix* Q*.*

Exercise 7.32. *For the weather matrix, verify detailed balance directly for* $\pi = (2/3, 1/3)$ *and then check* $\pi P = \pi$*.*

Exercise 7.33. *Consider the path graph with vertices* v_1, v_2, v_3, v_4 *and edge set*

$$\{\{v_1, v_2\}, \{v_2, v_3\}, \{v_3, v_4\}\}.$$

Compute the distribution in Definition 7.25, verify detailed balance for every pair of vertices, and decide whether the chain is aperiodic.

Exercise 7.34. *In Python, use the file* `master/jp/data/pride_and_prejudice.txt`*. Add special tokens marking the beginning and end of each sentence, and treat each sentence as an observed path from one homogeneous Markov chain on the finite vocabulary. This is a modelling assumption; consecutive words in the book do not prove the Markov property or homogeneity.*

- *Estimate each nonempty row of the transition matrix by relative frequencies. Specify how you handle a word that has no observed successor.*
- *Estimate the one-step probability of moving from* `the` *to* `her`*, and compute one chosen multi-step probability* $P^n(\texttt{the}, \texttt{her})$ *by propagating a probability vector* n *times. Exploit the nonzero transitions rather than constructing a dense matrix power for the entire vocabulary.*
- *Use Definition 7.8 to simulate a sentence beginning with* `Lady`*. Draw an independent uniform random number at every step and stop at the end-of-sentence token.*
- *For this part, make the end token move to the beginning token with probability one. Simulate one long path and use Matplotlib to display the ten largest empirical state frequencies. Examine the fitted transition graph for irreducibility and aperiodicity, and explain why the plot alone does not prove convergence to a stationary distribution.*

Exercise 7.35 (A finite-capacity branching chain). *Fix $M \geq 2$. Let $Z_0 = 1$, and let $X_{n,i}$, for $n \in \mathbb{Z}_+$ and $1 \leq i \leq M$, be mutually independent random variables, independent of Z_0, with*

$$\mathbb{P}(X_{n,i} = 0) = \mathbb{P}(X_{n,i} = 1) = \mathbb{P}(X_{n,i} = 2) = \frac{1}{3}.$$

Define

$$Z_{n+1} := \min\left\{M, \sum_{i=1}^{Z_n} X_{n,i}\right\},$$

where the empty sum is zero.

- *Show that $(Z_n)_{n\in\mathbb{Z}_+}$ is a Markov chain on $\{0, 1, \ldots, M\}$, and derive its transition probabilities.*
- *Identify the absorbing state and determine which states communicate.*
- *Use Python to simulate the chain for several values of M. Estimate the expected time to reach 0, report the Monte Carlo sample size, and use Matplotlib to plot the estimated mean hitting time against M.*

Removing the capacity M gives a Galton–Watson process on a countably infinite state space, which lies outside the scope of this chapter.

Chapter 8

Pattern recognition

Course guide

Scheduled lecture(s): Lecture 11
Required for those lectures: Sections 8.1–8.3
Prerequisites: Chapter 2, Section 2.6.5; Chapter 4, Section 4.1; and basic linear algebra
Optional or advanced: Section 8.2.1 is optional. Model evaluation and performance guarantees are developed separately in Chapter 9.

Pattern recognition is the task of assigning a label to an observed feature vector. For a concrete example, let

$$\mathbb{X} = \{0, 1\}^2, \qquad \mathbb{Y} = \{-1, 1\}.$$

The coordinates of $x = (x_1, x_2) \in \mathbb{X}$ might record whether an email contains each of two words, and $y = 1$ might mean spam. Since the data space $\mathbb{X} \times \mathbb{Y}$ is finite, every subset is an event. One possible probability law is

x	y	$\mathbb{P}(\{(X, Y) = (x, y)\})$
$(0, 0)$	-1	$1/4$
$(1, 0)$	-1	$1/4$
$(0, 1)$	1	$1/4$
$(1, 1)$	1	$1/4$

The four displayed outcomes have the stated probabilities, and every other outcome in $\mathbb{X} \times \mathbb{Y}$ has probability zero. In this model the rule that predicts 1 exactly when $x_2 = 1$ makes no errors.

In general, $\mathbb{X}$ is called the **feature space** or **instance space**, and $\mathbb{Y}$ is called the **label space**. We usually take $\mathbb{X} \subseteq \mathbb{R}^d$ and a finite label space. A **classifier** is a function $g : \mathbb{X} \to \mathbb{Y}$ for which $g(X)$ is a random variable. The input X is a feature vector, Y is its label, and (X, Y) is one observation in the data space $\mathbb{X} \times \mathbb{Y}$.

Let $n \in \mathbb{N}$ with $n \geq 1$. A training sample is an ordered tuple

$$T_n = ((X_1, Y_1), \ldots, (X_n, Y_n))$$

of independent copies of (X, Y). A learning rule uses T_n to choose a classifier $\widehat{g}_n$. Its purpose is to predict the label of a fresh observation drawn from the same probability law. Performance on fresh data is called **generalization**.

Linear classifiers, the perceptron, and kernels are foundational models rather than a catalogue of current state-of-the-art systems. They make the geometry of classification and the effect of feature representations explicit. The same ideas continue to appear in modern learning methods, even when the fitted models are much larger.

8.1 Linear classifiers

We first take $\mathbb{X} = \mathbb{R}^d$ and $\mathbb{Y} = \{-1, 1\}$. For $w \in \mathbb{R}^d$ and $b \in \mathbb{R}$, define the affine classifier

$$g_{w,b}(x) = \begin{cases} 1, & w \cdot x + b > 0, \\ -1, & w \cdot x + b \le 0. \end{cases}$$

When $w \neq 0$, its decision boundary is the hyperplane

$$\{x \in \mathbb{R}^d : w \cdot x + b = 0\}.$$

The parameter w determines the direction perpendicular to the hyperplane, and b determines its position.

8.1.1 Linearly separable data

A labelled data set $(x_1, y_1), \ldots, (x_n, y_n)$ in $\mathbb{R}^d \times \{-1, 1\}$ is **linearly separable** if there are $w \in \mathbb{R}^d$ and $b \in \mathbb{R}$ such that

$$y_i(w \cdot x_i + b) > 0, \qquad i = 1, \ldots, n.$$

This single inequality says that every point lies strictly on the side of the decision boundary assigned to its label. Figure 8.1 shows one such data set.

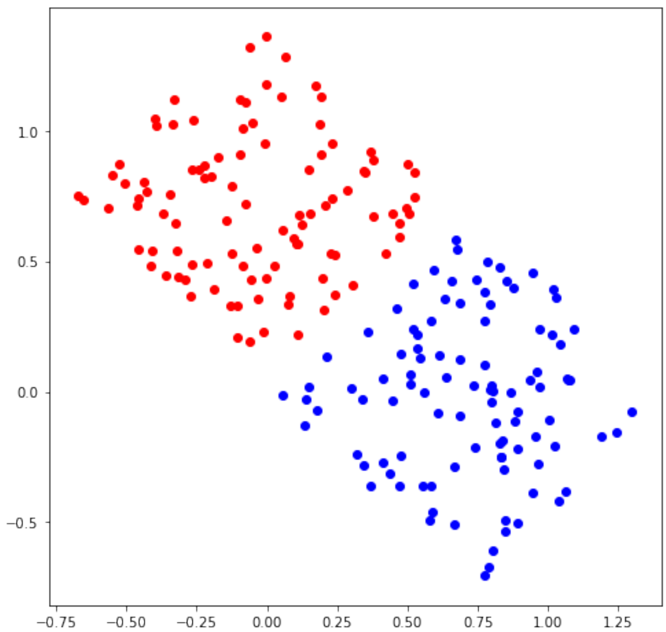


Figure 8.1: A linearly separable data set with labels 1 (red) and −1 (blue).

It is convenient to include the intercept as one extra coordinate. Set

$$\widetilde{x}_i = (x_i, 1) \in \mathbb{R}^{d+1}, \qquad \widetilde{w} = (w, b) \in \mathbb{R}^{d+1}.$$

Then $\widetilde{w} \cdot \widetilde{x}_i = w \cdot x_i + b$, so the separation condition becomes

$$y_i(\widetilde{w} \cdot \widetilde{x}_i) > 0, \qquad i = 1, \dots, n.$$

8.1.2 The perceptron algorithm

The perceptron algorithm searches for a separating vector in the augmented space. Starting from $w_0 = 0 \in \mathbb{R}^{d+1}$, repeat the following step:

$$\text{if } y_i(w_j \cdot \widetilde{x}_i) \leq 0, \qquad w_{j+1} := w_j + y_i \widetilde{x}_i.$$

Here j counts updates, and any index i with nonpositive signed margin may be chosen. The algorithm stops when every signed margin is positive.

Why the algorithm stops

The following theorem gives a finite upper bound on the number of updates. We use $|v|$ for the Euclidean norm of a vector v.

Theorem 8.1 (Perceptron convergence). *Suppose there is a vector $w^* \in \mathbb{R}^{d+1}$ such that*

$$y_i(w^* \cdot \widetilde{x}_i) \geq 1, \qquad i = 1, \dots, n.$$

Set $r := \max_{1 \leq i \leq n} |\widetilde{x}_i|$. Then the perceptron algorithm makes at most $r^2 |w^*|^2$ updates. When it stops, its vector w satisfies

$$y_i(w \cdot \widetilde{x}_i) > 0, \qquad i = 1, \dots, n.$$

Proof. Suppose update j uses the point with index $I(j)$. By the update rule,

$$w_{j+1} = w_j + y_{I(j)} \widetilde{x}_{I(j)}, \qquad y_{I(j)}(w_j \cdot \widetilde{x}_{I(j)}) \leq 0.$$

The separating assumption gives

$$\begin{aligned} w_{j+1} \cdot w^* &= w_j \cdot w^* + y_{I(j)}(\widetilde{x}_{I(j)} \cdot w^*) \\ &\geq w_j \cdot w^* + 1. \end{aligned} \tag{8.1}$$

The nonpositive signed margin at the updated point gives

$$\begin{aligned} |w_{j+1}|^2 &= |w_j|^2 + 2y_{I(j)}(w_j \cdot \widetilde{x}_{I(j)}) + |\widetilde{x}_{I(j)}|^2 \\ &\leq |w_j|^2 + r^2. \end{aligned} \tag{8.2}$$

Since $w_0 = 0$, after $M \geq 1$ updates these inequalities imply

$$w_M \cdot w^* \geq M, \qquad |w_M|^2 \leq M r^2.$$

Cauchy–Schwarz therefore gives

$$M \leq w_M \cdot w^* \leq |w_M||w^*| \leq \sqrt{M}\, r |w^*|.$$

Dividing by $\sqrt{M}$ and squaring yields $M \leq r^2 |w^*|^2$. Hence the algorithm cannot make more than this many updates. Its stopping condition gives the final strict inequalities. □

The margin level 1 in the theorem is a normalization, not an additional geometric restriction. If a finite data set has strictly positive signed margins for some vector, multiplying that vector by a sufficiently large constant makes every signed margin at least 1.

8.2 Kernelization

Linear separation may become possible after a nonlinear transformation of the features. For $r > 0$, write

$$B_r := \{x \in \mathbb{R}^2 : |x| < r\}.$$

Suppose points in B_1 have label 1, while points in $B_4 \setminus B_3$ have label -1, as illustrated in Fig. 8.2. No line can place the inner disc on one side and the entire surrounding annulus on the other.

Indeed, suppose an affine function were positive on B_1 and negative on $B_4 \setminus B_3$. It would be positive at the origin. On the other hand, for any unit vector u, it would have to be negative at both $3.5u$ and $-3.5u$. Adding these last two inequalities contradicts positivity at the origin. Reversing all signs gives the same contradiction for the opposite assignment of labels.

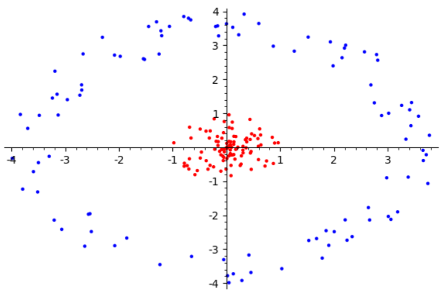


Figure 8.2: Data in two classes that cannot be separated by a line.

Define

$$\phi(x) = (x_1, x_2, x_1^2 + x_2^2) \in \mathbb{R}^3.$$

The third coordinate is less than 1 on B_1 and greater than 9 on $B_4 \setminus B_3$. Thus the plane with third coordinate equal to 5 separates the transformed classes. The transformed data are shown in Fig. 8.3.

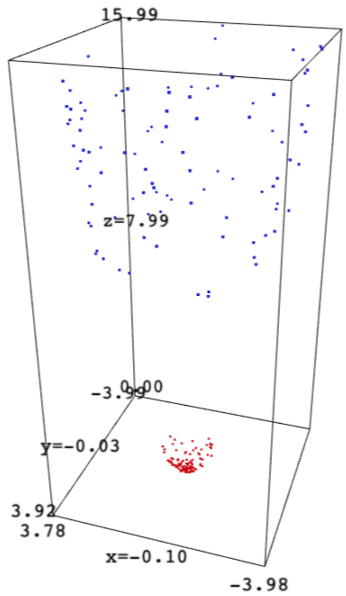


Figure 8.3: The transformed data are linearly separable in three dimensions.

An intercept can again be included by using $\widetilde{\phi}(x) = (\phi(x), 1)$. If the perceptron is run with transformed features, then after any number of updates its weight has the form

$$w = \sum_{j=1}^{n} \alpha_j y_j \widetilde{\phi}(x_j),$$

where α_j is the number of updates made using point j. Consequently,

$$w \cdot \widetilde{\phi}(x) = \sum_{j=1}^{n} \alpha_j y_j \big(\widetilde{\phi}(x_j) \cdot \widetilde{\phi}(x)\big).$$

Only inner products of transformed feature vectors are needed.

Definition 8.2 (Kernel and Gram matrix)**.** *A function $k : \mathbb{R}^d \times \mathbb{R}^d \to \mathbb{R}$ is a* **kernel** *if, for every integer $n \geq 1$ and every $x_1, \ldots, x_n \in \mathbb{R}^d$, the Gram matrix*

$$K = (k(x_i, x_j))_{i,j=1}^{n}$$

is symmetric and positive semidefinite. Thus, for all $a_1, \ldots, a_n \in \mathbb{R}$,

$$\sum_{i=1}^{n} \sum_{j=1}^{n} a_i a_j k(x_i, x_j) \geq 0.$$

This definition uses only finite matrices. It also includes kernels whose feature representations are not finite-dimensional.

Two basic examples follow directly from the definition. The constant function $k(x, y) = 1$ is a kernel because

$$\sum_{i=1}^{n} \sum_{j=1}^{n} a_i a_j = \left(\sum_{i=1}^{n} a_i \right)^2 \geq 0.$$

The linear function $k(x, y) = x \cdot y$ is a kernel because

$$\sum_{i=1}^{n}\sum_{j=1}^{n} a_i a_j (x_i \cdot x_j) = \left|\sum_{i=1}^{n} a_i x_i\right|^2 \geq 0.$$

Both Gram matrices are symmetric.

Lemma 8.3 (Finite feature representation)**.** *Let K be a symmetric positive semidefinite $n \times n$ matrix. There are vectors $v_1, \ldots, v_n \in \mathbb{R}^n$ such that*

$$K_{ij} = v_i \cdot v_j, \qquad 1 \leq i, j \leq n.$$

The vectors can be chosen in $\mathbb{R}^r$, where r is the rank of K.

Proof. By the spectral theorem,

$$K = Q\Lambda Q^T,$$

where Q is orthogonal and every diagonal entry of Λ is nonnegative. Set $B = Q\Lambda^{1/2}$. Then $K = BB^T$. Taking v_i to be row i of B gives $K_{ij} = v_i \cdot v_j$. Coordinates corresponding to zero eigenvalues may be removed, leaving r coordinates. □

If $k(x, y) = \phi(x) \cdot \phi(y)$ for an explicit finite-dimensional map, then k is a kernel by the same calculation. Conversely, the lemma gives a finite feature representation on every fixed finite collection of points. The computation of the perceptron may therefore use the values $k(x_i, x_j)$ without explicitly constructing transformed features. This is the **kernel trick**.

Theorem 8.4 (Rules for constructing kernels)**.** *Suppose $k_1, k_2 : \mathbb{R}^d \times \mathbb{R}^d \to \mathbb{R}$ are kernels. Then:*

1. *ck_1 is a kernel for every constant $c \geq 0$;*
2. *$(x, y) \mapsto f(x)f(y)k_1(x, y)$ is a kernel for every function $f : \mathbb{R}^d \to \mathbb{R}$;*
3. *$k_1 + k_2$ is a kernel;*
4. *the pointwise product k_1k_2 is a kernel.*

Proof. Fix $x_1, \ldots, x_n$ and $a = (a_1, \ldots, a_n)^T$. Let K_1 and K_2 be the two Gram matrices. The first and third claims follow from

$$a^T(cK_1)a = c(a^TK_1a), \qquad a^T(K_1 + K_2)a = a^TK_1a + a^TK_2a.$$

For the second claim, set $D = \mathrm{diag}(f(x_1), \ldots, f(x_n))$. Its Gram matrix is DK_1D, and

$$a^TDK_1Da = (Da)^TK_1(Da) \geq 0.$$

All three matrices are symmetric.

For the product, use the preceding lemma to write $(K_1)_{ij} = u_i \cdot u_j$ and $(K_2)_{ij} = v_i \cdot v_j$. Form a vector z_i whose coordinates are all products $(u_i)_p(v_i)_q$. Then

$$z_i \cdot z_j = (u_i \cdot u_j)(v_i \cdot v_j) = (K_1)_{ij}(K_2)_{ij}.$$

Hence the product Gram matrix is positive semidefinite. □

Corollary 8.5 (Polynomial kernels)**.** *Let $\gamma, r \geq 0$ and let $q \in \mathbb{N}$ with $q \geq 1$. Then*

$$k(x, y) = (\gamma x \cdot y + r)^q$$

is a kernel.

Proof. The linear kernel, its nonnegative multiple by γ, and the constant kernel r are kernels. Apply the sum rule and then apply the product rule $q - 1$ times. □

8.2.1 Optional: an infinite-dimensional kernel

For $\gamma > 0$, the Gaussian radial basis function kernel is

$$k(x, y) = \exp(-\gamma|x - y|^2).$$

To see why it is a kernel, write

$$k(x, y) = e^{-\gamma|x|^2} e^{-\gamma|y|^2} \sum_{q=0}^{\infty} \frac{(2\gamma)^q}{q!} (x \cdot y)^q.$$

Every finite partial sum is a kernel by Definition 8.4. For a fixed finite collection of points, its Gram matrices converge entry by entry to the Gaussian Gram matrix. Taking the limit in $a^T K a \geq 0$ shows that the limiting matrix is positive semidefinite. The infinite series corresponds to an infinite collection of features, but kernel computations require only the displayed closed formula.

8.3 Exercises

Exercise 8.6. *Suppose $(x_1, y_1), \ldots, (x_n, y_n)$ is linearly separable and $\widetilde{w}$ has strictly positive signed margins $y_i(\widetilde{w} \cdot \widetilde{x}_i)$. Show that*

$$c := \frac{1}{\min_{1 \leq i \leq n} y_i(\widetilde{w} \cdot \widetilde{x}_i)}$$

is well-defined and that $w^* = c\widetilde{w}$ satisfies the normalized margin assumption in the perceptron convergence theorem. Explain why finiteness of the data set is used.

Exercise 8.7. *For $\phi(x) = (x_1, x_2, x_1^2 + x_2^2)$, compute $k(x, y) = \phi(x) \cdot \phi(y)$. For arbitrary $a_1, \ldots, a_n \in \mathbb{R}$, verify directly that*

$$\sum_{i=1}^{n} \sum_{j=1}^{n} a_i a_j k(x_i, x_j) = \left| \sum_{i=1}^{n} a_i \phi(x_i) \right|^2.$$

Conclude that k is a kernel.

Exercise 8.8. *Reproduce directly from the definition the proof that the constant function $k(x, y) = 1$ and the linear function $k(x, y) = x \cdot y$ are kernels. Show that a negative constant function is not a kernel.*

Exercise 8.9 (Perceptron in Python)**.** *Choose an affine separator, generate finitely many points in $\mathbb{R}^2$ that do not lie on its boundary, and assign their labels in $\{-1, 1\}$ using that separator. Retain its augmented vector for comparison. Then complete the following tasks.*

- *Implement the augmented perceptron update using NumPy. Cycle through the observations in a fixed order until an entire pass makes no update, and record the total number of updates.*
- *Use Matplotlib to plot the labelled observations and the resulting decision boundary. Verify every signed margin numerically.*
- *Find a vector satisfying the normalized margin assumption, compute $r^2|w^*|^2$, and compare the theorem's upper bound with the observed number of updates.*
- *Repeat the experiment on the four corners of a square with the XOR labelling, which is not linearly separable. Impose a fixed update limit and explain why the convergence theorem no longer applies.*

Exercise 8.10 (A nonlinear decision boundary)**.** *Use NumPy to generate labelled points from the disc-and-annulus example in this chapter, and use Matplotlib to display them. Transform the points by*

$$\phi(x) = (x_1, x_2, x_1^2 + x_2^2)$$

and fit the augmented perceptron from the preceding exercise to the transformed data. Then fit `sklearn.svm.SVC(kernel="poly", degree=2)` *to the original features. Plot the two predicted decision regions on a grid. This exercise compares feature representations; do not interpret performance on the training points as a generalization guarantee.*

Chapter 9

Model evaluation

Course guide

Scheduled lecture(s): Lecture 12 (revisited in Lecture 15)
Required for those lectures: Lecture 12: Sections 9.1–9.3 and corresponding exercises in Section 9.6; Lecture 15: Sections 9.4–9.5 and corresponding exercises in Section 9.6
Prerequisites: Chapter 2, Section 2.6.5; Chapter 3, Section 3.1; and Chapter 4, Section 4.1
Optional or advanced: The unbounded-loss discussion in Section 9.4.2 is optional. Formal concentration theory for cross-validation is outside the scope of the course.

Model evaluation compares predictions with observations that were not used to fit the predictor. To begin with a finite example, let $X, Y \in \{0, 1\}$ and suppose

$$\mathbb{P}(X = 0) = \mathbb{P}(X = 1) = \frac{1}{2},$$
$$\mathbb{P}(Y = 0 \mid X = 0) = \mathbb{P}(Y = 1 \mid X = 1) = \frac{3}{4}.$$

The classifier $g(x) = x$ therefore has error probability $1/4$. If five independent test observations give losses $0, 1, 0, 0, 1$, their average is $2/5$. This is an observed test error, not the population error $1/4$. Concentration inequalities quantify the difference between these two quantities.

The same distinction applies to classification, regression, and other prediction problems. The central requirement is that the observations used for the final evaluation have not influenced the fitted predictor or the evaluation procedure.

9.1 Evaluation on an independent test set

Let $Z = (X, Y)$ be one observation, and let $\ell(y, u)$ be a nonnegative loss. For a fixed prediction function g, define its **risk** by

$$R(g) := \mathbb{E}\left[\ell(Y, g(X))\right].$$

When a fitted function $\widehat{g}$ is inserted into this deterministic risk functional, $R(\widehat{g})$ is random through the training data. The expectation defining $R(g)$ is over a fresh observation with g held fixed; it is not a training-sample average.

Let $n \in \mathbb{N}$ with $n \geq 1$, and let

$$T_n = (Z_1, \ldots, Z_n)$$

be a training sample of independent copies of Z. A learning rule uses only T_n to produce a fitted prediction function $\widehat{g}$. Its training risk is

$$\widehat{R}_n^{\text{train}}(\widehat{g}) := \frac{1}{n}\sum_{i=1}^{n} \ell(Y_i, \widehat{g}(X_i)).$$

Because the same observations were used to choose $\widehat{g}$, this average need not be close to $R(\widehat{g})$.

Let $m \in \mathbb{N}$ with $m \geq 1$, and let

$$V_m = (Z_{n+1}, \dots, Z_{n+m}),$$

be a test sample of independent copies of Z, independent of T_n. Define the test risk by

$$\widehat{R}_m^{\text{test}}(\widehat{g}) := \frac{1}{m}\sum_{j=1}^{m} \ell(Y_{n+j}, \widehat{g}(X_{n+j})).$$

Theorem 9.1 (Held-out guarantee for bounded loss)**.** *Let T_n and V_m be the training and test samples above, with $m \geq 1$, and suppose V_m is independent of T_n. Let the learning rule use only T_n to produce $\widehat{g}$. Assume that the model, learning rule, and loss are such that the risks and conditional probabilities below are defined in the discrete or joint-density framework of Section 2.6.5. Suppose there is a constant $B > 0$ such that*

$$0 \leq \ell(Y, g(X)) \leq B$$

almost surely for every prediction function g that the learning rule can produce. Then, for every $\epsilon > 0$,

$$\mathbb{P}\left(\left|\widehat{R}_m^{\text{test}}(\widehat{g}) - R(\widehat{g})\right| > \epsilon \;\middle|\; T_n\right) \leq 2\exp\left(-\frac{2m\epsilon^2}{B^2}\right), \tag{9.1}$$

$$\mathbb{P}\left(\left|\widehat{R}_m^{\text{test}}(\widehat{g}) - R(\widehat{g})\right| > \epsilon\right) \leq 2\exp\left(-\frac{2m\epsilon^2}{B^2}\right). \tag{9.2}$$

Proof. After the training data are fixed, $\widehat{g}$ is fixed. The m test losses are then independent, belong to $[0, B]$, and have common mean $R(\widehat{g})$. The conditional inequality follows from Definition 3.5. Taking the expectation of the conditional probability gives the unconditional inequality. □

Here conditioning on T_n means that the observed training sample is held fixed. In the discrete and joint-density models developed in Section 2.6.5, the tower property justifies the passage from the conditional statement to the unconditional one. This notation does not mean conditioning on an event of probability zero by Definition 1.12.

For a confidence level $1 - \alpha$, where $\alpha \in (0, 1)$, set

$$\delta_{m,B}(\alpha) := B\sqrt{\frac{\log(2/\alpha)}{2m}}.$$

The theorem gives the confidence interval

$$\left[\max\left\{0, \widehat{R}_m^{\text{test}}(\widehat{g}) - \delta_{m,B}(\alpha)\right\}, \min\left\{B, \widehat{R}_m^{\text{test}}(\widehat{g}) + \delta_{m,B}(\alpha)\right\}\right]$$

for $R(\widehat{g})$ with probability at least $1-\alpha$.

The test size should be chosen from the desired confidence and precision. A fixed percentage such as 30% does not by itself guarantee a useful interval. The guarantee also requires the fitted predictor and evaluation procedure to be fixed without using the test results. Repeatedly inspecting the test results while changing the model turns the test set into part of the fitting procedure.

A standard workflow uses training data to fit candidate models, validation data to choose among them, and an independent test set once for the final evaluation. Every data-dependent step, including feature selection, normalization, and tuning, belongs to the training and validation stages.

9.2 Classification metrics

For labels in $\{0,1\}$, the 0–1 loss is

$$\ell(y,u) := \mathbb{1}_{\{y\neq u\}}.$$

It belongs to $[0,1]$, so Definition 9.1 applies with $B=1$. In this case $R(g)=\mathbb{P}(Y\neq g(X))$, and the test risk is the fraction of incorrectly classified test observations.

Other classification metrics are conditional probabilities. Let $g:\mathbb{X}\to\{0,1\}$ be fixed and assume that $g(X)$ is a random variable. If

$$\mathbb{P}(g(X)=1)>0, \qquad \mathbb{P}(Y=1)>0,$$

the **precision** and **recall** for class 1 are

$$\pi := \mathbb{P}(Y=1 \mid g(X)=1), \qquad \rho := \mathbb{P}(g(X)=1 \mid Y=1),$$

respectively. Recall is also called sensitivity. The positivity assumptions are exactly what makes the conditional probabilities defined.

Let $m\in\mathbb{N}$ with $m\geq 1$, and let $(X_1,Y_1),\ldots,(X_m,Y_m)$ be an independent test sample. Define

$$N_{\mathrm{pred}} := \sum_{i=1}^{m}\mathbb{1}_{\{g(X_i)=1\}}, \qquad N_{\mathrm{pos}} := \sum_{i=1}^{m}\mathbb{1}_{\{Y_i=1\}},$$
$$N_{\mathrm{tp}} := \sum_{i=1}^{m}\mathbb{1}_{\{g(X_i)=1,\ Y_i=1\}}.$$

When the denominators are positive, the empirical metrics are

$$\widehat{\pi} := \frac{N_{\mathrm{tp}}}{N_{\mathrm{pred}}}, \qquad \widehat{\rho} := \frac{N_{\mathrm{tp}}}{N_{\mathrm{pos}}}.$$

The effective test size is N_{pred} for precision and N_{pos} for recall. Both are random.

Proposition 9.2 (Confidence intervals for precision and recall)**.** *Under the assumptions above, N_{pred} has the binomial distribution with parameters m and $\mathbb{P}(g(X)=1)$, and N_{pos} has the binomial distribution with parameters m and $\mathbb{P}(Y=1)$.*

Fix $\alpha\in(0,1)$. If $N_{\mathrm{pred}}\geq 1$, define

$$I_\pi := \left[\max\left\{0,\widehat{\pi}-\sqrt{\frac{\log(4/\alpha)}{2N_{\mathrm{pred}}}}\right\},\min\left\{1,\widehat{\pi}+\sqrt{\frac{\log(4/\alpha)}{2N_{\mathrm{pred}}}}\right\}\right],$$

and set $I_\pi = [0,1]$ *when* $N_{\text{pred}} = 0$. *Similarly, if* $N_{\text{pos}} \geq 1$, *define*

$$I_\rho := \left[\max\left\{0, \widehat{\rho} - \sqrt{\frac{\log(4/\alpha)}{2N_{\text{pos}}}}\right\}, \min\left\{1, \widehat{\rho} + \sqrt{\frac{\log(4/\alpha)}{2N_{\text{pos}}}}\right\}\right],$$

and set $I_\rho = [0,1]$ *when* $N_{\text{pos}} = 0$. *Then*

$$\mathbb{P}(\pi \in I_\pi,\ \rho \in I_\rho) \geq 1 - \alpha.$$

Proof. For precision, retain the test pairs for which $g(X_i) = 1$. By Definition 2.56, conditional on $N_{\text{pred}} = k$, the retained labels are i.i.d. Bernoulli(π) random variables whenever $k \geq 1$ and $\mathbb{P}(N_{\text{pred}} = k) > 0$. For recall, retain the pairs for which $Y_i = 1$. Conditional on $N_{\text{pos}} = k$, the retained predictions are i.i.d. Bernoulli(ρ) random variables whenever $k \geq 1$ and $\mathbb{P}(N_{\text{pos}} = k) > 0$. The binomial count distributions follow from the same theorem.

Apply Definition 3.8 to each metric with error probability $\alpha/2$. The union bound gives simultaneous coverage at least $1 - \alpha$; the two intervals need not be independent. □

If g was fitted using independent training data, apply the proposition after fixing those data. If a fitted classifier never predicts class 1, its population precision is undefined. If a retained test count is zero while the corresponding population metric is defined, the interval $[0,1]$ records that the test sample supplies no information about that metric.

9.3 Cross-validation: data reuse and stability

An independent test set gives a clean guarantee, but reserving many observations for testing leaves fewer observations for fitting and model selection. Cross-validation reuses a development data set while ensuring that each recorded loss is computed using a predictor that was not fitted on that particular observation.

For a concrete example, suppose $n = 6$ and split the indices into three folds,

$$F_1 = \{1,2\}, \qquad F_2 = \{3,4\}, \qquad F_3 = \{5,6\}.$$

Fit three predictors. The first is fitted using observations $3,4,5,6$ and evaluated on observations $1,2$; the other two are obtained by rotating the held-out fold. All six observations contribute one validation loss, while each fit uses four observations.

For the basic construction, let $K \in \mathbb{N}$ satisfy $2 \leq K \leq n$ and choose a fixed partition $F_1, \ldots, F_K$ before fitting any predictor. Suppose that its members are nonempty, disjoint subsets whose union is $\{1, \ldots, n\}$. Let $\widehat{g}^{(-r)}$ be the predictor fitted without the observations indexed by F_r. The K-**fold cross-validation score** is

$$\widehat{R}_{\text{CV}} := \frac{1}{n} \sum_{r=1}^{K} \sum_{i \in F_r} \ell(Y_i, \widehat{g}^{(-r)}(X_i)).$$

When $K = n$ and $F_i = \{i\}$, this is **leave-one-out cross-validation**. Stratified folds may use the labels to balance class counts. In that case the partition is data-dependent and the stratification rule is part of the evaluation procedure. Stratification does not make the validation losses independent.

The terms in $\widehat{R}_{\text{CV}}$ are not independent test losses. The fitted predictors use overlapping training observations, so changing one observation may alter several terms in the average. Consequently, the proof of Definition 9.1 does not apply to the cross-validation score.

Cross-validation is most reliable when the learning procedure is stable. In this context, stability means that removing or replacing a small part of the training data changes the loss on a new observation only slightly. If a single observation can substantially change the fitted predictor, the cross-validation score can also change substantially. Averaging the folds does not by itself remove this instability.

There is also a difference between the predictors used in cross-validation and the final predictor. A K-fold score evaluates predictors fitted on roughly $(K-1)n/K$ observations. After model selection, one commonly refits the chosen procedure on all n development observations. Stability is the reason one may expect the behavior of these fits to be similar; equality is not automatic.

Cross-validation is useful for comparing procedures, choosing tuning parameters, and diagnosing sensitivity to the data split. The following workflow keeps model selection separate from final assessment:

1. Set aside an independent final test sample.
2. On the remaining development data, specify the candidate procedures, folds, and comparison metric.
3. Repeat every fitted operation inside each training fold. This includes normalization, feature selection, and tuning.
4. Select a procedure from its cross-validation results and refit it on all development data.
5. Use the final test sample once to assess the selected predictor.

Repeated K-fold cross-validation can reveal sensitivity to the random fold partition, but it does not turn the dependent validation losses into an independent test sample. A theoretical treatment of cross-validation requires additional assumptions on stability and tails; see [AV] for further reading.

9.4 Regression metrics

For a real-valued response, the squared-error loss is

$$\ell(y, u) := (y - u)^2.$$

For a fixed prediction function g, its mean squared prediction error is

$$R(g) = \mathbb{E}\,[(Y - g(X))^2].$$

For $m \in \mathbb{N}$ with $m \geq 1$, on an independent test sample $(X'_1, Y'_1), \dots, (X'_m, Y'_m)$, define

$$\widehat{\mathrm{MSE}}_m(g) := \frac{1}{m}\sum_{j=1}^{m}(Y'_j - g(X'_j))^2.$$

9.4.1 Bounded responses and predictions

Suppose $M > 0$ and, after the training data are fixed,

$$|Y| \leq M, \qquad |\widehat{g}(X)| \leq M$$

almost surely. The squared test losses then belong to $[0, 4M^2]$. Definition 9.1 gives

$$\mathbb{P}\left(\left|\widehat{\mathrm{MSE}}_m(\widehat{g}) - R(\widehat{g})\right| > \epsilon \;\middle|\; T_n\right) \le 2\exp\left(-\frac{m\epsilon^2}{8M^4}\right). \tag{9.3}$$

If $Y \in [-M, M]$, clipping predictions to this interval cannot increase their squared error. Thus the prediction bound can be enforced without worsening the loss.

Let

$$\overline{Y}'_m := \frac{1}{m}\sum_{j=1}^m Y'_j,$$

and define

$$\mathrm{SSE}_m(g) := \sum_{j=1}^m (Y'_j - g(X'_j))^2,$$
$$\mathrm{SST}_m := \sum_{j=1}^m (Y'_j - \overline{Y}'_m)^2.$$

If $\mathrm{SST}_m > 0$, the test-set fraction of variance unexplained and coefficient of determination are

$$\widehat{\mathrm{FVU}}_m(g) := \frac{\mathrm{SSE}_m(g)}{\mathrm{SST}_m}, \qquad \widehat{R}^2_m(g) := 1 - \widehat{\mathrm{FVU}}_m(g).$$

If all test responses are equal, then $\mathrm{SST}_m = 0$ and these ratios are undefined. The FVU can exceed 1, so $\widehat{R}^2_m$ can be negative. The notation R^2 does not mean that this evaluation metric must be the square of another quantity.

If $0 < \mathbb{V}[Y] < \infty$ and $R(g) < \infty$, the population analogue is

$$R^2_{\mathrm{pop}}(g) := 1 - \frac{R(g)}{\mathbb{V}[Y]}.$$

The test statistic $\widehat{R}^2_m(g)$ is a ratio with a random denominator. The concentration bound for test MSE does not by itself give a confidence interval for this ratio, especially when $\mathbb{V}[Y]$ is small.

9.4.2 Optional: unbounded loss

Squared loss need not be bounded. One possible replacement is a tail assumption stated directly for the loss. Suppose that, after fixing the training data, the random variable

$$(Y - \widehat{g}(X))^2$$

is sub-exponential with parameter $\lambda(\widehat{g}) > 0$. Then Definition 3.13 gives

$$\mathbb{P}\left(\left|\widehat{\mathrm{MSE}}_m(\widehat{g}) - R(\widehat{g})\right| > \epsilon \;\middle|\; T_n\right)$$
$$\le 2\exp\left(-\frac{m}{2}\min\left\{\frac{\epsilon^2}{\lambda(\widehat{g})^2}, \frac{\epsilon}{\lambda(\widehat{g})}\right\}\right).$$

To obtain a bound with a deterministic right-hand side before training, one needs a deterministic upper bound for $\lambda(\widehat{g})$. Squared Gaussian-type quantities are typically sub-exponential rather than sub-Gaussian.

9.5 A variance-sensitive bound

Hoeffding's inequality uses only the range of the loss. Bennett's inequality also uses its variance and can be sharper when that variance is small.

Theorem 9.3 (Bennett's inequality)**.** *Let $m \in \mathbb{N}$ with $m \geq 1$, and let $W_1, \ldots, W_m$ be i.i.d. real-valued random variables with mean μ and variance $v > 0$. Suppose that $W_i - \mu \leq b$ almost surely for every $i = 1, \ldots, m$, where $b > 0$. Then, for every $\epsilon > 0$,*

$$\mathbb{P}(\overline{W}_m - \mu \geq \epsilon) \leq \exp\left(-\frac{mv}{b^2} h\left(\frac{b\epsilon}{v}\right)\right),$$

where

$$\overline{W}_m := \frac{1}{m}\sum_{i=1}^{m} W_i, \qquad h(u) := (1+u)\log(1+u) - u, \quad u \geq 0.$$

Proof. Set $X_i = W_i - \mu$. For $s \geq 0$ and $x \leq b$, elementary calculus gives

$$e^{sx} \leq 1 + sx + \frac{e^{sb} - 1 - sb}{b^2} x^2.$$

Since $\mathbb{E}[X_i] = 0$ and $\mathbb{E}[X_i^2] = v$,

$$\mathbb{E}[e^{sX_i}] \leq \exp\left(\frac{v}{b^2}(e^{sb} - 1 - sb)\right).$$

Markov's inequality and independence therefore give

$$\mathbb{P}\left(\sum_{i=1}^{m} X_i \geq m\epsilon\right) \leq \exp\left(-sm\epsilon + \frac{mv}{b^2}(e^{sb} - 1 - sb)\right).$$

Choose

$$s = \frac{1}{b}\log\left(1 + \frac{b\epsilon}{v}\right).$$

Substitution gives the stated exponent. □

Corollary 9.4 (Bennett bound for test risk)**.** *Suppose the hypotheses of Definition 9.1 hold. After fixing the training data, set*

$$v(\widehat{g}) := \mathbb{V}[\ell(Y, \widehat{g}(X))].$$

If $v(\widehat{g}) > 0$, then

$$\mathbb{P}\left(\left|\widehat{R}_m^{\text{test}}(\widehat{g}) - R(\widehat{g})\right| \geq \epsilon \;\middle|\; T_n\right) \leq 2\exp\left(-\frac{mv(\widehat{g})}{B^2} h\left(\frac{B\epsilon}{v(\widehat{g})}\right)\right).$$

If $v(\widehat{g}) = 0$, the test risk equals $R(\widehat{g})$ almost surely.

Proof. Conditional on the training data, the test losses are i.i.d. and belong to $[0, B]$. Their centered versions and their negatives are bounded above by B. Apply Definition 9.3 to both tails and use the union bound. □

The variance $v(\widehat{g})$ is a population quantity. Replacing it by the sample variance computed on the same test set requires an empirical Bennett or empirical Bernstein inequality; it is not justified by the theorem above.

9.6 Exercises

Exercise 9.5. *Derive* (9.2) *from* (9.1) *using the tower property. Identify where the proof uses independence of the test sample, the common data-generating law, and boundedness of the loss.*

Exercise 9.6. *In a data-science competition, teams may submit repeatedly and are ranked using the same hidden data. Explain why the best displayed score need not identify the predictor with the smallest population risk. Would keeping a second hidden set for one final evaluation address the same problem?*

Exercise 9.7. *Let* $B > 0$*,* $\alpha \in (0, 1)$*, and* $\epsilon > 0$*. Use Definition* 9.1 *to find an integer test size* m *that guarantees a confidence-interval half-width at most* ϵ *with confidence at least* $1 - \alpha$*. Compute the required size when* $B = 1$*,* $\alpha = 0.05$*, and* $\epsilon = 0.02$*.*

Exercise 9.8. *Analyze precision and recall when* g *always returns* 1*, when* g *always returns* 0*, and when* $\mathbb{P}(Y = 1)$ *is close to zero. Identify undefined population metrics and test counts that are likely to be small. Repeat the analysis for class* 0*.*

Exercise 9.9 (Held-out classification in scikit-learn)**.** *Use* `sklearn.datasets.load_digits` *and turn the labels into the binary target* $Y = \mathbb{1}_{\{digit \geq 5\}}$*. Split the observations once with* `train_test_split(...,random_state=0)` *and fit* `sklearn.svm.SVC(kernel="linear")` *using only the training part.*

- *From the test predictions, compute the error rate, precision, and recall directly from indicator sums. Check the values with* `sklearn.metrics.classification_report`*.*
- *Construct the simultaneous confidence intervals in Definition* 9.2*. Report both random effective sample sizes.*
- *Use Matplotlib to display the four confusion-matrix counts. Explain why the test observations may be treated as independent after the fitted classifier is held fixed.*

Exercise 9.10. *Suppose a preprocessing rule standardizes each feature using its sample mean and standard deviation. Explain why computing these quantities once from all development data leaks validation information. Describe the correct computation within one fold.*

Exercise 9.11 (Held-out regression in scikit-learn)**.** *Load the diabetes data with* `sklearn.datasets.load_diabetes`*. Use* `train_test_split` *to create one training set and one test set, fit* `sklearn.linear_model.LinearRegression` *on the training set, and keep the test set untouched until fitting is complete.*

- *Compute the test MSE and* R^2 *directly from their definitions. Check the latter against* `LinearRegression.score`*.*
- *Use Matplotlib to plot predicted against observed test responses and to plot residuals against predicted responses. State what each plot can reveal and what it cannot prove.*

- *Explain why observing bounded responses and predictions in this one data set does not establish the almost-sure bound required for* (9.3).

Exercise 9.12. *Let W take values in $[0, B]$ and suppose $\mathbb{V}[W] = 0$. Prove directly that $W = \mathbb{E}[W]$ almost surely. Deduce the zero-variance clause in the Bennett bound for test risk without applying Definition 9.3 with $v = 0$.*

Chapter 10

High dimension

Course guide

Scheduled lecture(s): Lecture 13
Required for those lectures: Sections 10.1–10.4 and the exercises in Section 10.6
Prerequisites: Chapter 2, Section 2.6, Chapter 3, Section 3.1, and Chapter 6, Section 6.2
Optional or advanced: Section 10.5 is optional; it also uses the Gamma function and a higher-dimensional polar-coordinate calculation.

10.1 Balls, spheres, and volume

Geometry in many dimensions is difficult to infer from two- and three-dimensional pictures. We begin with volume because its scaling already shows one important effect: a fixed-width interior occupies a rapidly decreasing fraction of a high-dimensional body.

Throughout this chapter, d is a positive integer and $|x|$ denotes the Euclidean norm of $x \in \mathbb{R}^d$.

Definition 10.1 (Balls and spheres)**.** *For $x \in \mathbb{R}^d$ and $r > 0$, the open ball and sphere with centre x and radius r are*

$$B_r(x) := \{y \in \mathbb{R}^d : |x - y| < r\},$$
$$S_r(x) := \{y \in \mathbb{R}^d : |x - y| = r\}.$$

We write $B_r := B_r(0)$ and $S_r := S_r(0)$. The sets $B_1(x)$ and $S_1(x)$ are called a unit ball and a unit sphere, respectively.

On $\mathbb{R}^d$, we use the standard event collection $\mathcal{B}(\mathbb{R}^d)$ from Section 1.4. It contains the open and closed sets, and hence all balls, spheres, shells, and cubes used below. We treat this event collection as part of the model; students are not expected to construct it. For a set $E \in \mathcal{B}(\mathbb{R}^d)$ whose usual multivariable integral is finite, its volume is

$$|E| := \int_{\mathbb{R}^d} \mathbb{1}_E(x)\,dx.$$

Here $\mathbb{1}_E(x)$ equals 1 when $x \in E$ and 0 otherwise.

Lemma 10.2 (Scaling of volume)**.** *Let $E \in \mathcal{B}(\mathbb{R}^d)$ have finite volume and let $t > 0$. Then*

$$|tE| = t^d|E|, \qquad tE := \{tx : x \in E\}.$$

Proof. Use the change of variables $y = tx$. Its Jacobian determinant is t^d, so

$$|tE| = \int_{\mathbb{R}^d} \mathbb{1}_{tE}(y)\,dy = t^d \int_{\mathbb{R}^d} \mathbb{1}_E(x)\,dx = t^d|E|.$$

□

For example, if $Q = [-1,1]^d$ and $0 < t < 1$, then

$$\frac{|tQ|}{|Q|} = t^d.$$

The same ratio holds for B_1 because $tB_1 = B_t$.

Example 10.3 (Volume near the boundary)**.** *The fraction of the unit ball lying in the outer shell $B_1 \setminus B_{1-\delta}$ is*

$$\frac{|B_1 \setminus B_{1-\delta}|}{|B_1|} = 1 - (1-\delta)^d, \qquad 0 < \delta < 1.$$

In dimension $d = 100$, the outer shell of width 0.03 contains the fraction

$$1 - 0.97^{100} \approx 0.952.$$

Thus more than 95% of the volume lies within distance 0.03 of the sphere S_1.

We next show, without using the exact volume formula, that the volume of the unit ball tends to zero as d increases.

Model 10.4 (Spherical Gaussian)**.** *A random vector $Z = (Z_1, \dots, Z_d)$ is a* **spherical Gaussian** *if it has density*

$$f_Z(z) = \frac{1}{(2\pi)^{d/2}} e^{-|z|^2/2}, \qquad z \in \mathbb{R}^d.$$

Equivalently, its coordinates are independent standard Gaussian random variables.

Model 10.5 (Normalized Gaussian)**.** *Let Z be a spherical Gaussian and set*

$$Y := \frac{Z}{\sqrt{2\pi}}.$$

Then Y has density

$$f_Y(y) = e^{-\pi|y|^2}, \qquad y \in \mathbb{R}^d.$$

Its coordinates are independent Gaussian random variables with mean 0 and variance $1/(2\pi)$.

Lemma 10.6. *If $d > 4\pi$, then*

$$|B_1| \le \frac{8e^{\pi}}{d}.$$

In particular, $|B_1| \to 0$ as $d \to \infty$.

Proof. Let Y be the normalized Gaussian in Definition 10.5. Since $\mathbb{E}[Y_i^2] = 1/(2\pi)$ and $\mathbb{V}(Y_i^2) = 1/(2\pi^2)$, independence gives

$$\mathbb{E}[|Y|^2] = \frac{d}{2\pi}, \qquad \mathbb{V}(|Y|^2) = \frac{d}{2\pi^2}.$$

Set $u = d/(2\pi) - 1$, which is positive because $d > 4\pi$. Chebyshev's inequality gives

$$\mathbb{P}\left(\left||Y|^2 - \frac{d}{2\pi}\right| \geq u\right) \leq \frac{d/(2\pi^2)}{(d/(2\pi) - 1)^2} = \frac{2d}{(d - 2\pi)^2} \leq \frac{8}{d}. \tag{10.1}$$

If $Y \in B_1$, then $d/(2\pi) - |Y|^2 > u$. Therefore (10.1) implies

$$\mathbb{P}(Y \in B_1) \leq \frac{8}{d}.$$

On the other hand, $f_Y(y) \geq e^{-\pi}$ for $y \in B_1$, so

$$e^{-\pi}|B_1| \leq \int_{B_1} f_Y(y)\, dy = \mathbb{P}(Y \in B_1).$$

Combining the last two estimates proves the result. □

The bound is intentionally elementary and is not sharp. The exact formula in optional Definition 10.14 shows much faster decay.

10.2 Geometry in high dimension

The calculation above gives two distinct conclusions. First, $|B_1|$ tends to zero even though the radius remains 1. Second, within the ball itself, most volume lies near its boundary. Neither statement says that every point is near the boundary; the statement concerns the proportion of volume.

The same scaling explains why rejection sampling from a cube becomes inefficient. Since $B_1 \subseteq [-1, 1]^d$, a point drawn uniformly from the cube lands in B_1 with probability

$$\frac{|B_1|}{2^d}.$$

By Definition 10.6, this probability is at most $8e^{\pi}/(d2^d)$ when $d > 4\pi$.

10.3 Sampling uniformly from balls and spheres

For a sphere, "uniform" refers to surface area. For a ball, it refers to ordinary d-dimensional volume. These are different geometric objects and therefore require separate definitions.

Model 10.7 (Uniform distribution on the unit sphere)**.** *Let $d \geq 2$. A random vector Θ is* **uniform on the unit sphere** *if $\Theta \in S_1$ with probability 1 and*

$$\mathbb{P}(\Theta \in A) = \frac{1}{|S_1|}\int_{S_1} \mathbb{1}_A(\theta)\, d\Omega(\theta)$$

for every $A \in \mathcal{B}(\mathbb{R}^d)$. Here $d\Omega$ is the usual surface-area element and $|S_1| := \int_{S_1} d\Omega$. We write $\Theta \sim \mathrm{Uniform}(S_1)$.

Model 10.8 (Uniform distribution on the unit ball)**.** *A random vector X is* **uniform on the unit ball** *if*

$$\mathbb{P}(X \in A) = \frac{|A \cap B_1|}{|B_1|}$$

for every $A \in \mathcal{B}(\mathbb{R}^d)$. We write $X \sim \mathrm{Uniform}(B_1)$.

10.3.1 Why projecting a square is not uniform

Consider the radial projection

$$\pi(z) := \frac{z}{|z|}, \qquad z \in \mathbb{R}^2 \setminus \{0\}.$$

Let (X, Y) have joint density p. The polar-coordinate formula shows that the angle Φ of $\pi(X, Y)$ has density

$$f_\Phi(\phi) = \int_0^\infty p(r\cos\phi, r\sin\phi) r\, dr, \qquad 0 \le \phi < 2\pi. \tag{10.2}$$

The factor r is the polar Jacobian.

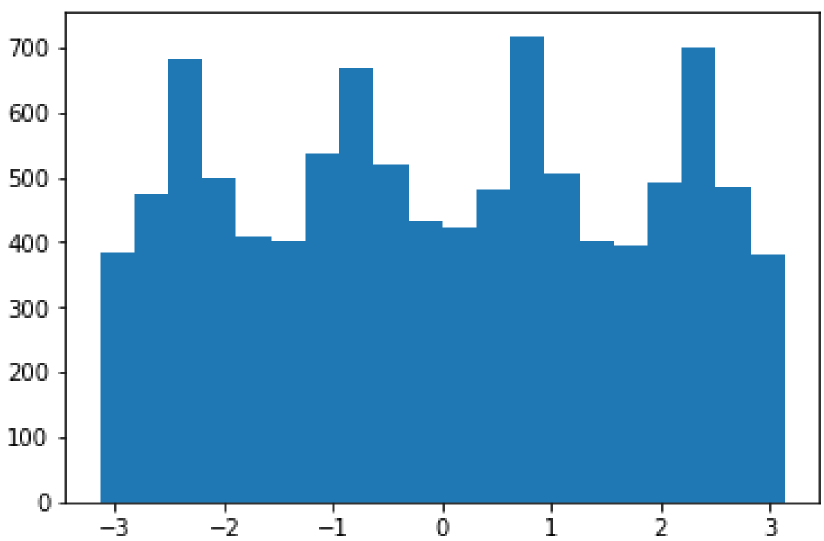
Suppose now that (X, Y) is uniform on $[-1, 1]^2$. Along the direction ϕ, the square ends at radius

$$\rho(\phi) = \frac{1}{\max\{|\cos\phi|, |\sin\phi|\}}.$$

Since $p = 1/4$ inside the square, (10.2) gives

$$f_\Phi(\phi) = \frac{1}{8\max\{\cos^2\phi, \sin^2\phi\}}.$$

This density is largest in the directions of the four corners, as the simulation below illustrates.



Definition 10.9 (Rotational symmetry). *A function $f : \mathbb{R}^d \to \mathbb{R}$ is* **rotationally symmetric** *if there is a function $g : [0, \infty) \to \mathbb{R}$ such that*

$$f(x) = g(|x|), \qquad x \in \mathbb{R}^d.$$

Remark 10.10. *If a density p on $\mathbb{R}^2$ is rotationally symmetric, then (10.2) does not depend on ϕ. Hence the projected angle is uniform. The same conclusion holds in $\mathbb{R}^d$: a rotationally symmetric density has a uniform direction on S_1.*

10.3.2 Sampling from the sphere

The spherical Gaussian provides a direct sampling method.

Lemma 10.11 (Gaussian sphere sampler). *Let $d \geq 2$, let Z be a spherical Gaussian, and define*

$$\Theta := \frac{Z}{|Z|}.$$

Then $\Theta \sim \mathrm{Uniform}(S_1)$. Moreover, $|Z|$ and Θ are independent.

Proof. A continuous Gaussian vector equals 0 with probability 0, so Θ is defined with probability 1. In polar coordinates $z = r\theta$, its probability element is proportional to

$$e^{-r^2/2} r^{d-1}\, dr\, d\Omega(\theta).$$

The first factor depends only on r, and the second is surface area on S_1. Thus the direction is uniform and the radius and direction are independent. □

One may instead draw uniformly from $[-1, 1]^d$, accept only points in B_1, and project the accepted points onto S_1. Conditioning on acceptance gives the uniform ball model, and Definition 10.10 then gives a uniform direction. This is an instance of accept–reject sampling from Algorithm 1.

10.3.3 Sampling from the ball

A uniform radius does not give a uniform point in the ball. Indeed, if $U \sim \mathrm{Uniform}([0, 1])$ and $\Theta \sim \mathrm{Uniform}(S_1)$, then

$$\mathbb{P}(|U\Theta| \leq r) = r, \qquad 0 \leq r \leq 1,$$

whereas volume scaling requires the probability r^d.

Theorem 10.12 (Uniform ball sampler). *Fix $d \geq 2$. Let U be uniform on $[0, 1]$, and let Θ be uniform on S_1 and independent of U. Then*

$$U^{1/d}\Theta \sim \mathrm{Uniform}(B_1).$$

Proof. Set $R = U^{1/d}$. For $0 \leq r \leq 1$,

$$\mathbb{P}(R \leq r) = \mathbb{P}(U \leq r^d) = r^d = \frac{|B_r|}{|B_1|},$$

where the last equality follows from Definition 10.2. Independence gives a uniform direction at every radius. Consequently, for $0 \le a < b \le 1$ and a standard surface patch $C \subseteq S_1$,

$$\mathbb{P}(a < R \le b,\ \Theta \in C) = (b^d - a^d)\frac{\int_C d\Omega}{|S_1|}.$$

By polar integration, the expression on the right is exactly the fraction of the volume of B_1 contained in the corresponding polar sector. This is the uniform model in Definition 10.8. $\square$

The transformation $U^{1/d}$ is also obtained by the inversion method in Definition 6.12. The independence assumption in Definition 10.12 is essential: specifying the two marginal distributions alone does not determine their joint distribution.

10.4 Concentration in a high-dimensional annulus

For a spherical Gaussian $X = (X_1, \dots, X_d)$,

$$\mathbb{E}\left[|X|^2\right] = \sum_{i=1}^{d} \mathbb{E}\left[X_i^2\right] = d.$$

The following theorem makes precise the statement that $|X|$ is usually close to $\sqrt{d}$. Its more general form is needed for random projection in the next chapter.

Theorem 10.13 (High-dimensional annulus). *Let $X = (X_1, \dots, X_d)$ have independent, mean-zero components. Suppose that every X_i is sub-Gaussian with parameter 1 and has the same variance $a^2 > 0$. Then, for every $\epsilon > 0$,*

$$\mathbb{P}\left(\left|\frac{|X|^2}{d} - a^2\right| \ge \epsilon\right) \le 2\exp\left(-\frac{d}{2}\min\left\{\frac{\epsilon^2}{128}, \frac{\epsilon}{8\sqrt{2}}\right\}\right).$$

Consequently, for every $\beta > 0$,

$$\mathbb{P}\left(\left||X| - a\sqrt{d}\right| \ge \beta\right) \le 2\exp\left(-\frac{d}{2}\min\left\{\frac{a^2\beta^2}{128d}, \frac{a\beta}{8\sqrt{2d}}\right\}\right).$$

Proof. Since X_i is centred and sub-Gaussian with parameter 1, Definition 3.14 shows that X_i^2 is sub-exponential with parameter $8\sqrt{2}$. The variables X_i^2 are independent and $\mathbb{E}\left[X_i^2\right] = a^2$. The first estimate is therefore Definition 3.13 applied to $d^{-1}\sum_{i=1}^d X_i^2$.

Set $r_0 = a\sqrt{d}$. If $\left||X| - r_0\right| \ge \beta$, then

$$\left||X|^2 - r_0^2\right| = \left||X| - r_0\right|(|X| + r_0) \ge \beta r_0.$$

Hence this event is contained in

$$\left\{\left|\frac{|X|^2}{d} - a^2\right| \ge \frac{a\beta}{\sqrt{d}}\right\}.$$

Substitute $\epsilon = a\beta/\sqrt{d}$ into the first estimate. $\square$

For a spherical Gaussian, $a = 1$. Taking $\beta = \delta\sqrt{d}$ for a fixed $\delta > 0$ shows that

$$\mathbb{P}\left(1 - \delta \le \frac{|X|}{\sqrt{d}} \le 1 + \delta\right) \longrightarrow 1$$

as $d \to \infty$. This is concentration of the relative radius; it does not say that all Gaussian samples have exactly the same length.

10.5 Optional: Exact volume of the unit ball

This section assumes the polar-coordinate formula in $\mathbb{R}^d$. It is not needed for the annulus theorem or for the random-projection argument in the next chapter.

For $s > 0$, the Gamma function is

$$\Gamma(s) := \int_0^\infty e^{-t} t^{s-1}\, dt.$$

For a positive integer k, integration by parts gives $\Gamma(k) = (k-1)!$.

Theorem 10.14 (Volume of the unit ball)**.** *For every integer $d \geq 1$,*

$$|B_1| = \frac{2\pi^{d/2}}{d\,\Gamma(d/2)} = \frac{\pi^{d/2}}{\Gamma(d/2+1)}.$$

Proof. Let

$$I_d := \int_{\mathbb{R}^d} e^{-|x|^2}\, dx.$$

Repeated one-dimensional integration and the Gaussian integral give

$$I_d = \prod_{i=1}^d \int_{-\infty}^\infty e^{-x_i^2}\, dx_i = \pi^{d/2}.$$

Polar integration gives a second expression:

$$\begin{aligned} I_d &= |S_1| \int_0^\infty e^{-r^2} r^{d-1}\, dr \\ &= \frac{|S_1|}{2} \int_0^\infty e^{-t} t^{d/2-1}\, dt = \frac{|S_1|}{2} \Gamma(d/2), \end{aligned}$$

where $t = r^2$ in the second line. The same polar formula gives

$$|B_1| = |S_1| \int_0^1 r^{d-1}\, dr = \frac{|S_1|}{d}.$$

Eliminating $|S_1|$ proves the first formula. The identity $\Gamma(s+1) = s\Gamma(s)$ gives the second. □

For even d, this becomes

$$|B_1| = \frac{\pi^{d/2}}{(d/2)!}.$$

The exact formula confirms that, as $d \to \infty$, $|B_1|$ decreases much faster than the $1/d$ bound obtained from Chebyshev's inequality.

Further reading

The geometric viewpoint and sampling arguments in this chapter follow the treatment in [BlHo].

10.6 Exercises

Exercise 10.15. *Compute* $1-(1-\delta)^d$ *for* $(d,\delta) = (10, 0.1)$, $(100, 0.1)$, *and* $(100, 0.03)$. *Explain the difference between keeping the shell width fixed and choosing a width of the form* c/d, *where* $c > 0$ *is fixed. Then use NumPy and Matplotlib to plot the outer-shell fraction for* $4 \le d \le 200$ *with* $\delta = 0.03$ *and with* $\delta = 3/d$.

Exercise 10.16. *Write a Python function that uses NumPy to generate* N *independent points from the Gaussian sphere sampler in Definition 10.11. Verify numerically that every returned vector has norm* 1. *For* $d = 2$, *use* `numpy.arctan2` *and Matplotlib to plot a histogram of the sampled angles. Explain why the method requires no rejection step.*

Exercise 10.17. *Prove that a point drawn uniformly from* $[-1,1]^d$, *conditional on belonging to* B_1, *is uniform on* B_1. *What is the acceptance probability? Use Definition 10.6 to explain why the Gaussian sphere sampler is preferable in high dimension.*

Exercise 10.18 (Sampling the radius)**.** *Let* $X \sim \mathrm{Uniform}(B_1)$ *and put* $R = |X|$.

1. *Use Definition 10.2 to find the distribution function and density of* R *on* $[0,1]$.
2. *Generate* 20,000 *samples using Definition 10.12 for* $d = 2, 10, 100$. *Use Matplotlib to compare histograms of the sampled radii with the corresponding densities.*
3. *Explain why drawing the radius uniformly on* $[0,1]$ *produces too many points near the centre when* $d > 1$.

Exercise 10.19. *Apply Definition 10.13 to the normalized Gaussian from Definition 10.5. Show that it gives an exponentially decreasing upper bound for* $|B_1|$ *when* d *is sufficiently large, thereby improving the Chebyshev bound in Definition 10.6. State the value of* a *and a valid choice of* β *before applying the theorem.*

Exercise 10.20 (Computational radius check)**.** *For* $d = 2, 10, 100$, *use NumPy to generate* 10,000 *spherical Gaussian vectors. For each* d, *report the empirical mean and the* 5% *and* 95% *empirical quantiles of* $|X|/\sqrt{d}$. *Plot the three empirical distributions with Matplotlib and explain how their spread changes with dimension. Use a fixed random-number generator seed so that the experiment is reproducible.*

Chapter 11

Dimension reduction and clustering

Course guide

Scheduled lecture(s): Lecture 14
Required for those lectures: Sections 11.1–11.5 and the core exercises in Section 11.9
Prerequisites: Chapter 10, especially Section 10.4; Chapter 9, Section 9.1; Chapter 3, Section 3.1; and basic matrix algebra
Optional or advanced: Sections 11.6–11.7 are advanced and are not required for Lecture 14. Section 11.8 is bibliographic only; the last exercise in Section 11.9 is advanced.

The previous chapter studied probability in high dimension. This chapter uses its concentration result to construct a random projection. It then turns to a different question: how can a data matrix be represented by a low-dimensional subspace? Singular value decomposition (SVD) and principal component analysis (PCA) answer this question using linear algebra. The two chapters are therefore consecutive but separate: concentration supplies one randomized method, while SVD and PCA are deterministic matrix methods. The final required section introduces K-means as a simple method for grouping unlabelled observations.

11.1 Random projection

Let $U_1, \dots, U_k$ be random vectors in $\mathbb{R}^d$. Their dot products with a fixed vector v define a linear map from $\mathbb{R}^d$ to $\mathbb{R}^k$. When the components of the U_i are independent and sub-Gaussian, the high-dimensional annulus theorem controls the length of the image.

Here is a finite example of such a map. Take $\Omega = \{-1, 1\}^{k\times d}$ and $\mathcal{F} = 2^{\Omega}$, and give each of the 2^{kd} sign arrays probability 2^{-kd}. Thus all entries are independent and uniform on $\{-1, 1\}$. If U_i is row i, define

$$F(x) := \frac{1}{\sqrt{k}}(U_1 \cdot x, \dots, U_k \cdot x).$$

Once the signs have been chosen, F is an ordinary linear map. For every fixed $x \in \mathbb{R}^d$, independence and centring give

$$\mathbb{E}\left[|F(x)|^2\right] = \frac{1}{k}\sum_{i=1}^{k}\mathbb{E}\left[(U_i \cdot x)^2\right] = \frac{1}{k}\sum_{i=1}^{k}\sum_{j=1}^{d} x_j^2 = |x|^2.$$

The next theorem controls the probability that $|F(x)|$ differs substantially from $|x|$. The random-sign construction is the case $a = 1$.

Theorem 11.1 (Random projection). *Let $d, k \geq 1$ be integers, let $v \in \mathbb{R}^d$ be fixed, and let $\epsilon \in (0, 1)$. Let $U_i = (U_{i1}, \ldots, U_{id}) \in \mathbb{R}^d$ for $1 \leq i \leq k$. Suppose that the kd random variables U_{ij} are mutually independent, centred, sub-Gaussian with parameter 1, and have variance $a^2 > 0$. Define the random linear map $F : \mathbb{R}^d \to \mathbb{R}^k$ by, for $x \in \mathbb{R}^d$,*

$$F(x) := \frac{1}{a\sqrt{k}}(U_1 \cdot x, \ldots, U_k \cdot x).$$

Then

$$\mathbb{P}\left(\big||F(v)| - |v|\big| \geq \epsilon|v|\right) \leq 2\exp\left(-\frac{k}{2}\min\left\{\frac{a^4\epsilon^2}{128}, \frac{a^2\epsilon}{8\sqrt{2}}\right\}\right).$$

Proof. The result is immediate when $v = 0$, so suppose that $v \neq 0$. For each i, independence of the components gives

$$\mathbb{E}\left[\exp\left(s\frac{U_i \cdot v}{|v|}\right)\right] \leq \prod_{j=1}^{d}\exp\left(\frac{s^2 v_j^2}{2|v|^2}\right) = \exp(s^2/2).$$

Thus the variables $U_i \cdot v/|v|$ are independent, centred, sub-Gaussian with parameter 1, and have variance a^2. Apply Definition 10.13 in dimension k with $\beta = \epsilon a\sqrt{k}$. Finally, use

$$|F(v)| = \frac{|v|}{a\sqrt{k}}\left|\left(\frac{U_1 \cdot v}{|v|}, \ldots, \frac{U_k \cdot v}{|v|}\right)\right|.$$

□

The theorem controls one fixed vector. Applying it to every difference in a finite data set gives simultaneous control of all pairwise distances.

Theorem 11.2 (Johnson–Lindenstrauss). *Let $d \geq 1$ and $n \geq 2$ be integers, let $0 < \epsilon < 1$, and let $k \geq 1$ be an integer satisfying*

$$k > \frac{768\log n}{\epsilon^2}.$$

Let $v_1, \ldots, v_n \in \mathbb{R}^d$, and use the random map F from Definition 11.1 with $a = 1$. Then, with probability at least $1 - 1/n$,

$$(1-\epsilon)|v_i - v_j| \leq |F(v_i) - F(v_j)| \leq (1+\epsilon)|v_i - v_j|$$

for all $1 \leq i, j \leq n$.

Proof. If $v_i = v_j$, the bounds hold deterministically. For every pair $i < j$ with $v_i \neq v_j$, linearity of F and Definition 11.1 give

$$\mathbb{P}\left(\big||F(v_i - v_j)| - |v_i - v_j|\big| \geq \epsilon |v_i - v_j|\right) \leq 2e^{-k\epsilon^2/256}.$$

There are fewer than $n^2/2$ pairs. The union bound therefore gives

$$\mathbb{P}(\text{at least one pair violates the bounds}) \leq n^2 e^{-k\epsilon^2/256} < \frac{1}{n}.$$

$\square$

Example 11.3 (A random-sign projection)**.** *Consider the three points*

$$v_1 = (0,0,0)^T, \qquad v_2 = (1,0,0)^T, \qquad v_3 = (0,1,0)^T.$$

One possible outcome of the random-sign construction with $d = 3$ and $k = 2$ is

$$F(x) = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 & -1 \\ 1 & -1 & 1 \end{bmatrix} x.$$

The images are

$$F(v_1) = (0,0)^T, \quad F(v_2) = \frac{1}{\sqrt{2}}(1,1)^T, \quad F(v_3) = \frac{1}{\sqrt{2}}(1,-1)^T.$$

The three original pairwise distances are 1, 1, and $\sqrt{2}$. The three projected distances have the same values. This particular map preserves these distances exactly, even though the theorem does not guarantee exact preservation and its stated dimension bound is not met in this small example.

The lower bound on k guarantees simultaneous preservation of every pairwise distance. It is deliberately conservative for a finite example: some individual distances may be preserved well in much smaller dimensions, as illustrated in Fig. 11.1.

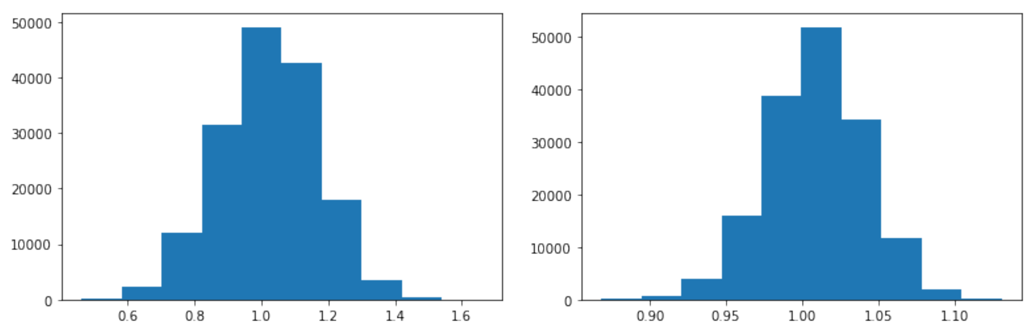


Figure 11.1: Relative errors for random projections of the Olivetti faces data, using target dimensions $k = 20$ and $k = 400$. The theorem controls the worst pair; the plots show the distribution across pairs.

11.1.1 Applications of random projection

Random projection is useful when an analysis depends mainly on distances but the original dimension d makes storage or computation expensive. For example, one may first project document vectors or image features and then apply a nearest-neighbour search or K-means. Computing a distance in $\mathbb{R}^k$ costs less than computing one in $\mathbb{R}^d$ when $k \ll d$.

The guarantee applies to a fixed finite data set and one common random map. The data points must be fixed independently of the random signs used to draw F, and the same realization of F must be applied to every point. The theorem controls distances, not the meaning of the new coordinates. A two-dimensional random projection can therefore be useful for exploration, but the stated Johnson–Lindenstrauss guarantee applies only when its lower bound on k is satisfied.

11.2 Singular value decomposition

Consider first the matrix

$$A = \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix}.$$

For a unit vector $v = (v_1, v_2)^T$,

$$|Av|^2 = 4v_1^2 + v_2^2 = 1 + 3v_1^2.$$

The maximum squared length is 4, so the maximum length is 2; both are attained in the direction $(1, 0)^T$. On the orthogonal direction $(0, 1)^T$, the squared length is 1. The construction below extends this calculation to every real matrix.

Let $n, m \geq 1$ and let A be an $n \times m$ real matrix. In a data application, its rows $x_1^T, \dots, x_n^T$ are observations in $\mathbb{R}^m$. For a unit vector $v \in \mathbb{R}^m$, the vector Av contains the coordinates $x_1 \cdot v, \dots, x_n \cdot v$. Hence

$$|Av|^2 = \sum_{i=1}^{n} (x_i \cdot v)^2.$$

The direction that maximizes this quantity is the direction in which the rows have the largest total squared coordinate.

The matrix $A^T A$ is symmetric and positive semidefinite. We use the following standard form of the spectral theorem: a real symmetric matrix has an orthonormal basis of eigenvectors, and all its eigenvalues are real. No proof of this linear-algebra result is required here. Thus $A^T A$ has an orthonormal basis of eigenvectors $v_1, \dots, v_m$ with eigenvalues

$$\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_m \geq 0.$$

The vectors v_j are called **right singular vectors**, and

$$\sigma_j := \sqrt{\lambda_j}$$

are the **singular values** of A. Let r be the rank of A. Then $\sigma_j > 0$ exactly for $1 \leq j \leq r$.

For the diagonal matrix above, the right singular vectors are $(1, 0)^T$ and $(0, 1)^T$, and the singular values are 2 and 1.

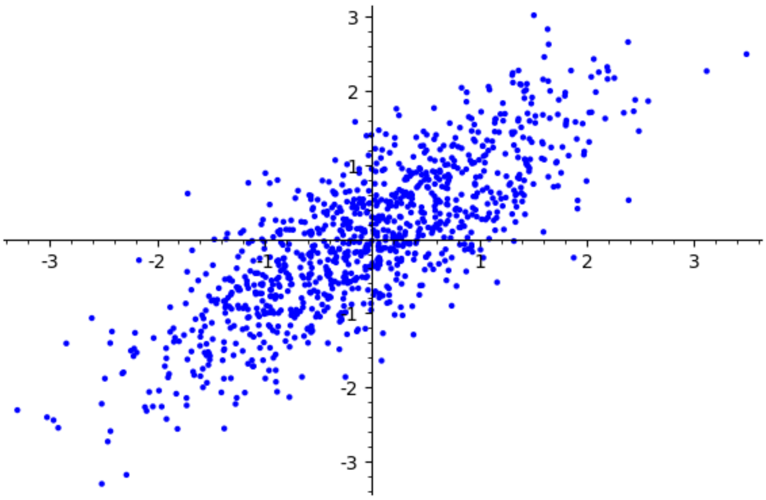


Figure 11.2: A two-dimensional data set whose dominant direction is close to the line $y = x$.

Lemma 11.4. *With the notation above,*

$$\max_{|v|=1} |Av| = \sigma_1.$$

The maximizing unit vectors are exactly the unit vectors in the eigenspace of $A^T A$ corresponding to λ_1.

Proof. Write a unit vector as $v = \sum_{j=1}^m c_j v_j$, where $\sum_{j=1}^m c_j^2 = 1$. Then

$$|Av|^2 = v^T A^T A v = \sum_{j=1}^m \lambda_j c_j^2 \le \lambda_1.$$

Equality holds precisely when $c_j = 0$ for every j with $\lambda_j < \lambda_1$. Taking square roots proves the result. □

If the largest eigenvalue is repeated, the leading eigenspace is unique but a first singular vector within it is not. Different choices can give different one-dimensional subspaces, all with the same maximizing value. When the largest eigenvalue is simple, the first singular vector is unique up to sign; that sign ambiguity does not change its span or the orthogonal projection onto it.

For $1 \le j \le r$, define

$$u_j := \frac{Av_j}{\sigma_j} \in \mathbb{R}^n.$$

These vectors are orthonormal because

$$u_i^T u_j = \frac{v_i^T A^T A v_j}{\sigma_i \sigma_j} = \frac{\lambda_j v_i^T v_j}{\sigma_i \sigma_j} = \begin{cases} 1, & i = j, \\ 0, & i \ne j. \end{cases}$$

For any $x \in \mathbb{R}^m$, expand x in the basis $v_1, \ldots, v_m$. The vectors v_j with $j > r$ lie in the null space of A, so

$$Ax = \sum_{j=1}^r \sigma_j u_j (v_j^T x).$$

Consequently,

$$A = \sum_{j=1}^{r} \sigma_j u_j v_j^T. \tag{11.1}$$

This is the **singular value decomposition**. If U_r and V_r have columns $u_1, \dots, u_r$ and $v_1, \dots, v_r$, and $D_r = \operatorname{diag}(\sigma_1, \dots, \sigma_r)$, then the same identity is

$$A = U_r D_r V_r^T.$$

This reduced form avoids defining Av_j/σ_j when $\sigma_j = 0$.

11.2.1 Best-fitting subspaces

For a subspace $W \subseteq \mathbb{R}^m$, let $P_W x$ denote the orthogonal projection of x onto W. If $w_1, \dots, w_k$ is an orthonormal basis of W, then

$$P_W x = \sum_{\ell=1}^{k} (x \cdot w_\ell) w_\ell.$$

Define the squared reconstruction error of W on the rows of A by

$$E_A(W) := \sum_{i=1}^{n} |x_i - P_W x_i|^2.$$

Theorem 11.5 (Best-fitting subspace). *Let k be an integer with $1 \le k \le m$. Define $V_k = \operatorname{span}\{v_1, \dots, v_k\}$. Then*

$$E_A(V_k) \le E_A(W)$$

for every k-dimensional subspace $W \subseteq \mathbb{R}^m$. Moreover,

$$E_A(V_k) = \sum_{j=k+1}^{r} \sigma_j^2,$$

where the sum is zero when $k \ge r$.

Proof. Let $w_1, \dots, w_k$ be an orthonormal basis of W. The Pythagorean theorem gives

$$\begin{aligned} E_A(W) &= \sum_{i=1}^{n} |x_i|^2 - \sum_{\ell=1}^{k} \sum_{i=1}^{n} (x_i \cdot w_\ell)^2 \\ &= \sum_{i=1}^{n} |x_i|^2 - \sum_{\ell=1}^{k} |Aw_\ell|^2. \end{aligned}$$

Write

$$c_j := \sum_{\ell=1}^{k} (v_j \cdot w_\ell)^2.$$

Orthogonal projection gives $0 \le c_j \le 1$, and Parseval's identity gives $\sum_{j=1}^m c_j = k$. Since the eigenvalues are decreasing,

$$\sum_{j=k+1}^{m} \lambda_j c_j \le \lambda_k \sum_{j=k+1}^{m} c_j = \lambda_k \sum_{j=1}^{k} (1 - c_j)$$
$$\le \sum_{j=1}^{k} \lambda_j (1 - c_j).$$

Consequently,

$$\sum_{\ell=1}^{k} |Aw_\ell|^2 = \sum_{j=1}^{m} \lambda_j c_j \le \sum_{j=1}^{k} \lambda_j.$$

The last upper bound is attained by $W = V_k$. Thus V_k minimizes the reconstruction error. For this choice, (11.1) and orthonormality give the stated tail sum. □

For the data in Fig. 11.2, a numerical calculation gives a first right singular vector approximately

$$v_1 = (-0.712, -0.702)^T.$$

Its span is close to the line $y = x$. Projecting onto the orthogonal complement leaves the residual data shown in Fig. 11.3.

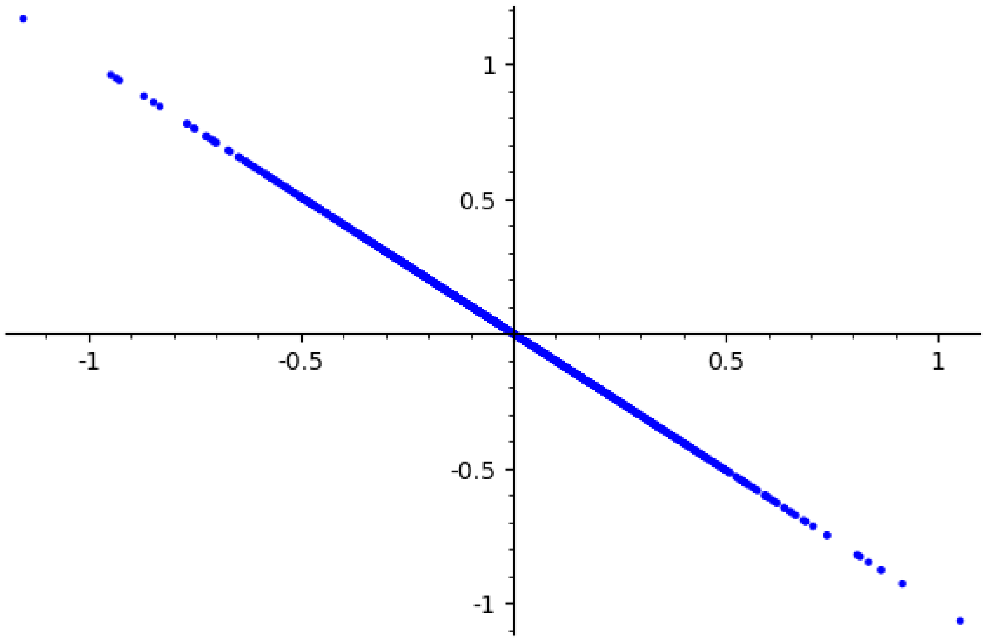


Figure 11.3: The residuals after projecting the data in Fig. 11.2 onto the orthogonal complement of the first right singular vector.

11.2.2 The power method

The power method approximates a leading eigenvector without computing the full decomposition. Suppose that $m \geq 2$. Let $B = A^T A$ and suppose $\lambda_1 > \lambda_2$. Choose a unit vector $q_0 = \sum_{j=1}^{m} c_j v_j$ with $c_1 \neq 0$, and repeatedly set

$$q_t := \frac{Bq_{t-1}}{|Bq_{t-1}|}.$$

Then $\lambda_1 > 0$, and

$$B^t q_0 = \lambda_1^t \left(c_1 v_1 + \sum_{j=2}^{m} c_j \left(\frac{\lambda_j}{\lambda_1} \right)^t v_j \right).$$

After normalization, the terms with $j \geq 2$ converge to zero. Hence q_t converges to v_1 when $c_1 > 0$ and to $-v_1$ when $c_1 < 0$. The assumptions $\lambda_1 > \lambda_2$ and $c_1 \neq 0$ guarantee this conclusion. Without the eigenvalue gap, convergence to a chosen leading eigenvector is not guaranteed. If $c_1 = 0$, every iterate remains orthogonal to v_1.

11.3 Principal component analysis

PCA applies SVD to centred data. Let $n, m \geq 1$ be integers. Given observations $x_1, \ldots, x_n \in \mathbb{R}^m$, define

$$\bar{x} := \frac{1}{n} \sum_{i=1}^{n} x_i,$$

and let A have rows $(x_i - \bar{x})^T$. Then every column of A has sample mean zero. The empirical covariance matrix, using the convention $1/n$, is

$$\widehat{\Sigma} := \frac{1}{n} A^T A.$$

Thus the PCA directions are the right singular vectors v_j of A. The variance of the data coordinate in direction v_j is

$$\frac{1}{n} \sum_{i=1}^{n} \big((x_i - \bar{x}) \cdot v_j \big)^2 = \frac{\sigma_j^2}{n}.$$

In particular, $\sigma_j/\sqrt{n}$, rather than σ_j itself, is the empirical standard deviation under this convention.

For an integer k with $1 \leq k \leq r$, let V_k be the matrix with columns $v_1, \ldots, v_k$. The matrix of the first k **principal component scores** is

$$Z_k := AV_k.$$

Row i of Z_k gives the k coordinates of the centred observation $x_i - \bar{x}$ in the principal-component basis. The corresponding reconstruction in the original coordinates is

$$A_k := Z_k V_k^T = AV_k V_k^T = \sum_{j=1}^{k} \sigma_j u_j v_j^T.$$

The original data scale is recovered by adding $\bar{x}$ to each reconstructed row. Without centring, the leading direction may describe the location of the data rather than its variation.

11.4 SVD in action

11.4.1 PCA and factor models

PCA can reveal a small number of directions along which many observed variables move. This makes it useful for exploratory summaries and for constructing lower-dimensional features. It should not be identified with factor analysis. Factor analysis specifies a probabilistic latent-variable model with an explicit noise term; PCA is the deterministic matrix decomposition described above.

11.4.2 Example: compressing image data

The data used in Fig. 11.4 contain 1797 images of handwritten digits. Each image has 8×8 pixels, so each observation is a vector in $\mathbb{R}^{64}$. This is the scikit-learn digits data set, not the MNIST data set, whose images have a different resolution. After centring the images, assemble them as the rows of a 1797×64 matrix A.

For these data, $r \geq 10$. The rank-10 reconstruction is

$$A_{10} = \sum_{j=1}^{10} \sigma_j u_j v_j^T \,.$$

The original sample images appear in Fig. 11.4; the corresponding reconstructions appear in Fig. 11.5. Each reconstruction uses only ten principal-component scores instead of 64 pixel values, together with the shared basis and mean image.

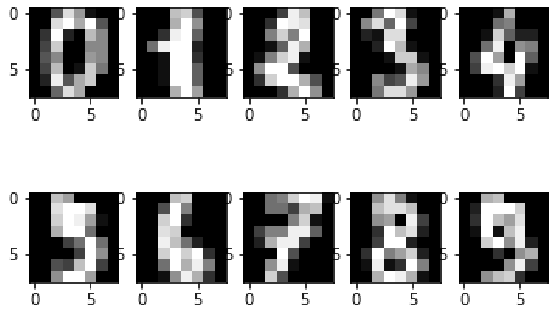


Figure 11.4: Ten sample images from the handwritten digits data.

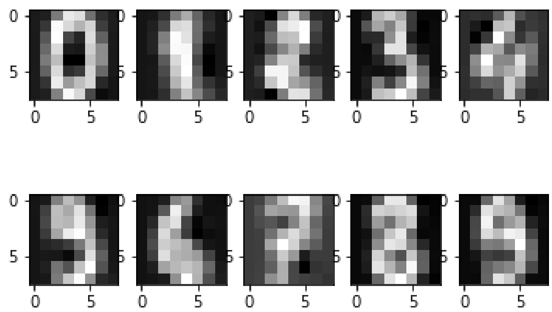


Figure 11.5: The images in Fig. 11.4 reconstructed using the first ten principal components.

Reconstruction error

For an $n \times m$ matrix M, its **Frobenius norm** is

$$\|M\|_{\mathrm{F}} := \left(\sum_{i=1}^{n} \sum_{j=1}^{m} M_{ij}^2 \right)^{1/2}.$$

By Definition 11.5, the reconstruction error satisfies

$$\|A - A_k\|_{\mathrm{F}}^2 = \sum_{j=k+1}^{r} \sigma_j^2.$$

The squared singular values therefore show exactly how the reconstruction error decreases as components are added.

Explained variance

For centred data with at least one positive singular value, the fraction of empirical variance explained by the first k principal components, where $1 \leq k \leq r$, is

$$\frac{\sum_{j=1}^{k} \sigma_j^2}{\sum_{j=1}^{r} \sigma_j^2}.$$

This ratio is undefined when every observation equals the sample mean, because then both the total variance and every singular value are zero.

11.4.3 Anomaly scores from reconstruction error

Suppose that the training observations have mean $\bar{x}$ and fitted PCA subspace V_k. For a new observation x, the residual

$$|(x - \bar{x}) - P_{V_k}(x - \bar{x})|$$

can be used as an anomaly score when the training data are representative of non-anomalous observations. This is a heuristic, not a guarantee that every large residual is anomalous. The dimension k and the decision threshold should be chosen using training or validation data. An independent test sample should be reserved for final evaluation, following the principles in Section 9.1.

11.5 Clustering with K-means

In a clustering problem, the observations have no given response labels. The aim is to divide them into groups using their features. The word "cluster" does not by itself specify what a good group is. The K-means method makes one particular choice: observations in the same group should have a small total squared distance from their group centre.

Example 11.6. *Consider the six observations*

$$\begin{aligned} x_1 &= (-2, 0), & x_2 &= (-1, 0), & x_3 &= (-1, 1), \\ x_4 &= (3, 0), & x_5 &= (4, 0), & x_6 &= (4, 1) \end{aligned}$$

in $\mathbb{R}^2$*, and choose* $K = 2$ *groups. Start with centres* $\mu_1 = (-2, 0)$ *and* $\mu_2 = (3, 0)$*. Nearest-centre assignment puts* x_1, x_2, x_3 *in the first group and the other observations in the second. The group means are*

$$\bar{x}_1 = \left(-\frac{4}{3}, \frac{1}{3}\right) \quad \text{and} \quad \bar{x}_2 = \left(\frac{11}{3}, \frac{1}{3}\right).$$

Replacing the centres by these means reduces the total squared distance from 6 *to* 8/3*. The assignment does not change, so the procedure stops.*

We now state the criterion used in the example. Let $n, d \geq 1$ be integers, let $x_1, \ldots, x_n \in \mathbb{R}^d$ be fixed observations, and let K be an integer with $1 \leq K \leq n$. An assignment is a vector $c = (c_1, \ldots, c_n) \in \{1, \ldots, K\}^n$, where $c_i = j$ means that x_i is assigned to group j.

Definition 11.7 (K**-means objective**)**.** *For an assignment* c *and centres* $\mu_1, \ldots, \mu_K \in \mathbb{R}^d$*, define the within-cluster sum of squares by*

$$W(c, \mu_1, \ldots, \mu_K) := \sum_{i=1}^{n} |x_i - \mu_{c_i}|^2.$$

A K*-means solution is an assignment and a tuple of centres that jointly minimize this objective. For fixed centres, minimizing over the assignments gives*

$$\Phi(\mu_1, \ldots, \mu_K) := \sum_{i=1}^{n} \min_{1 \leq j \leq K} |x_i - \mu_j|^2.$$

Thus joint minimization of W *is equivalent to minimizing* Φ *over the centres.*

The next identity explains the update used in Definition 11.6.

Lemma 11.8 (Centroid identity)**.** *Let* I *be a nonempty subset of* $\{1, \ldots, n\}$*, and define*

$$\bar{x}_I := \frac{1}{|I|} \sum_{i \in I} x_i.$$

Then, for every $u \in \mathbb{R}^d$*,*

$$\sum_{i \in I} |x_i - u|^2 = \sum_{i \in I} |x_i - \bar{x}_I|^2 + |I| \, |u - \bar{x}_I|^2.$$

Consequently, $\bar{x}_I$ *is the unique centre that minimizes the sum on the left.*

Proof. Write $x_i - u = (x_i - \bar{x}_I) + (\bar{x}_I - u)$ and expand the squared norm. The cross terms sum to zero because

$$\sum_{i \in I} (x_i - \bar{x}_I) = 0.$$

This proves the identity. Its final term is nonnegative and is zero exactly when $u = \bar{x}_I$. $\square$

Finding a global minimizer of the K-means objective can be difficult. **Lloyd's algorithm** instead alternates two elementary improvements:

1. Choose initial centres $\mu_1, \ldots, \mu_K$.

2. Assign each x_i to a nearest centre. If several centres are tied, choose the one with the smallest index.

3. Replace the centre of each nonempty group by its mean. If a group is empty, retain its previous centre.

4. Repeat steps 2 and 3 until the assignment no longer changes.

The tie convention and the empty-group convention make every step defined. Other implementations may use different conventions.

Proposition 11.9 (Descent of Lloyd's algorithm)**.** *Every assignment step and every centre-update step of Lloyd's algorithm leaves the K-means objective unchanged or decreases it. Suppose, in addition, that no group becomes empty and every nearest centre is unique at every assignment step. Then the algorithm reaches a fixed assignment after finitely many iterations.*

Proof. With the centres fixed, assigning x_i to a nearest centre minimizes its summand in the objective. Hence an assignment step cannot increase the objective. With the assignment fixed, Definition 11.8 shows that replacing each nonempty group's centre by its mean cannot increase the objective. Retaining the centre of an empty group does not change any summand.

Under the additional hypotheses, let $c^{(t)}$ be the assignment and let $\mu^{(t)} = (\mu_1^{(t)}, \ldots, \mu_K^{(t)})$ be the centres after update t. Write $W(c, \mu)$ for $W(c, \mu_1, \ldots, \mu_K)$. If $c^{(t+1)} \neq c^{(t)}$, uniqueness in the assignment step gives

$$W(c^{(t+1)}, \mu^{(t)}) < W(c^{(t)}, \mu^{(t)}).$$

The subsequent centre update cannot increase the left-hand side. Since no group is empty, the updated centres are determined by $c^{(t)}$. Thus an assignment cannot recur after a strict decrease. There are only K^n assignments, so the algorithm reaches a fixed assignment after finitely many iterations. □

The proposition gives a local guarantee, not a global one. Different initial centres can lead to different fixed assignments, so it is common to run the algorithm from several initial choices and retain the result with the smallest objective. Cluster numbers have no intrinsic meaning: permuting the numbers gives the same grouping.

Example 11.10 (Colour compression with K-means)**.** *Represent each pixel of a colour image by its red, green, and blue values $x_i \in \{0, 1, \ldots, 255\}^3$. Choose, for example, $K = 16$. Lloyd's algorithm assigns every pixel to a nearest centre in $\mathbb{R}^3$ and replaces each nonempty group's centre by its mean colour. After the algorithm stops, replace every pixel by its assigned centre.*

The result uses a palette of at most 16 colours. Apart from the palette, each pixel needs only its cluster number. The K-means objective is the total squared change in the RGB values, so the method gives a precise criterion for this compression. It does not account for the positions of neighbouring pixels, and Lloyd's algorithm need not find the globally best palette.

Other applications use the assignments themselves rather than the centres as a compressed representation. For example, observations may be standardized daily electricity-use profiles, with one coordinate for each hour. The cluster means then describe typical profiles, while the assignments group days that are close under squared Euclidean distance. Such groups are exploratory summaries; K-means alone does not show that the data came from K distinct populations.

The squared Euclidean distance also determines what K-means can find. Features measured on larger numerical scales can dominate the result, so any centering or scaling must be chosen as part of the data-analysis procedure. Outliers can strongly affect the means. The method is most natural for compact, well-separated groups and can poorly represent elongated, nested, or unequal-density groups.

PCA may be applied before K-means when the original feature dimension is large. This can reduce computation and noise, but PCA preserves directions of large variation rather than directions that necessarily separate clusters. When the clustering is part of a procedure whose performance will be assessed on a final test sample, both the PCA transformation and the clustering must be fitted without using that sample.

Finally, the training objective alone cannot select K. For $K = 1$, the centre is the overall sample mean, while $K = n$ permits objective value zero by placing one centre at each observation. Domain constraints, stability across development samples, or performance in a specified downstream task can guide the choice. Any final performance claim should follow the held-out evaluation principles in Section 9.1.

11.6 Advanced: population principal components

Let $d \geq 1$ be an integer, and let X be a random vector in $\mathbb{R}^d$ with finite second moments. Its second-moment matrix is

$$\Sigma := \mathbb{E}\,[XX^T].$$

If X is centred, then Σ is its covariance matrix. For an integer $n \geq 1$, let $X_1, \dots, X_n$ be independent copies of X, and define the empirical second-moment matrix

$$\widehat{\Sigma}_n := \frac{1}{n}\sum_{i=1}^{n} X_i X_i^T.$$

For a symmetric matrix M, write

$$\|M\|_{\mathrm{op}} := \max_{|v|=1} |Mv|$$

for its operator norm. The next result is a matrix version of Bernstein's inequality. Its proof is outside the scope of this course.

Theorem 11.11 (Second-moment concentration)**.** *With the notation above, suppose that $|X|^2 \leq C$ with probability one for some $C > 0$. Then, for every $t > 0$,*

$$\mathbb{P}\left(\|\widehat{\Sigma}_n - \Sigma\|_{\mathrm{op}} > t\right) \leq 2d \exp\left(-\frac{nt^2}{2C(\|\Sigma\|_{\mathrm{op}} + t/3)}\right).$$

For population PCA, we assume that the population mean is known to equal zero. The matrices in the theorem are then the population and empirical covariance matrices. If the mean is unknown, the observations must first be centred by their sample mean; that case requires an additional error term. The theorem itself does not need centring because it concerns second moments.

The empirical matrix approaches the population matrix in operator norm when n is large. Weyl's theorem converts this bound into simultaneous control of the ordered eigenvalues.

Theorem 11.12 (Weyl's theorem)**.** *Let Σ and E be symmetric $d \times d$ matrices. Order the eigenvalues of Σ as $\lambda_1 \geq \cdots \geq \lambda_d$ and those of $\Sigma + E$ as $\widehat{\lambda}_1 \geq \cdots \geq \widehat{\lambda}_d$. Then*

$$\max_{1 \leq j \leq d} |\widehat{\lambda}_j - \lambda_j| \leq \|E\|_{\mathrm{op}}.$$

Proof. We first record the needed form of the min–max identity. If M is symmetric with eigenvalues $\mu_1 \geq \cdots \geq \mu_d$, then

$$\mu_j = \max_{\substack{S \subseteq \mathbb{R}^d \\ \dim(S)=j}} \min_{\substack{v \in S \\ |v|=1}} v^T M v.$$

To verify this identity, take an orthonormal eigenbasis $e_1, \ldots, e_d$. Every j-dimensional subspace S contains a nonzero vector in $\operatorname{span}\{e_j, \ldots, e_d\}$, whose Rayleigh quotient is at most μ_j. For $S = \operatorname{span}\{e_1, \ldots, e_j\}$, every unit vector has Rayleigh quotient at least μ_j.

For every unit vector v,

$$v^T \Sigma v - \|E\|_{\mathrm{op}} \leq v^T(\Sigma + E)v \leq v^T \Sigma v + \|E\|_{\mathrm{op}}.$$

Apply the min–max identity to the upper inequality to obtain $\widehat{\lambda}_j \leq \lambda_j + \|E\|_{\mathrm{op}}$. Interchanging Σ and $\Sigma + E$, and replacing E by $-E$, gives $\lambda_j \leq \widehat{\lambda}_j + \|E\|_{\mathrm{op}}$. This proves the result for every $1 \leq j \leq d$. □

Small covariance error does not by itself control an individual eigenvector when population eigenvalues are tied. For example, let

$$\Sigma = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \qquad E = \begin{bmatrix} 0 & \epsilon \\ \epsilon & 0 \end{bmatrix}.$$

Every unit vector is an eigenvector of Σ. For every $\epsilon > 0$, the ordered eigenvectors of $\Sigma + E$ point in the directions $(1,1)^T/\sqrt{2}$ and $(1,-1)^T/\sqrt{2}$. An eigenvalue gap is therefore needed to identify a stable individual eigendirection. When a group of eigenvalues is tied, the corresponding invariant subspace, rather than a chosen basis of eigenvectors, is the meaningful object.

11.7 Advanced: population reconstruction risk

Let k be an integer with $1 \leq k \leq d$, and let

$$\mathcal{P}_k := \{P : \mathbb{R}^d \to \mathbb{R}^d : P \text{ is an orthogonal projection of rank } k\}.$$

For a centred random vector X satisfying $\mathbb{E}\,[|X|^2] < \infty$, define

$$\mathcal{R}(P) := \mathbb{E}\,[|X - PX|^2], \qquad P \in \mathcal{P}_k.$$

Choose a population minimizer P_k^*. Every such minimizer projects onto a subspace spanned by eigenvectors corresponding to the k largest eigenvalues of Σ. The minimizing subspace need not be unique when the kth eigenvalue is tied with the next one. This follows from the proof of Definition 11.5, with the empirical sums of squared coordinates replaced by their expectations.

Given independent copies $X_1, \dots, X_n$, let $\widehat{P}_k$ minimize the empirical reconstruction risk

$$\widehat{\mathcal{R}}_n(P) := \frac{1}{n}\sum_{i=1}^{n} |X_i - PX_i|^2$$

over $P \in \mathcal{P}_k$.

Equivalently, $\widehat{P}_k$ projects onto a leading k-dimensional eigenspace of $\widehat{\Sigma}_n$. Define the excess risk by

$$\mathcal{E}_k := \mathcal{R}(\widehat{P}_k) - \mathcal{R}(P_k^*).$$

Lemma 11.13 (Deterministic excess-risk bound)**.** *With the notation above,*

$$0 \le \mathcal{E}_k \le 2k\|\widehat{\Sigma}_n - \Sigma\|_{\mathrm{op}}.$$

Proof. Write $\Delta = \widehat{\Sigma}_n - \Sigma$. Since $\mathcal{R}(P) = \mathbb{E}\left[|X|^2\right] - \operatorname{tr}(P\Sigma)$ and $\widehat{P}_k$ maximizes $\operatorname{tr}(P\widehat{\Sigma}_n)$ over $\mathcal{P}_k$,

$$\begin{aligned}
\mathcal{E}_k &= \operatorname{tr}(P_k^*\Sigma) - \operatorname{tr}(\widehat{P}_k\Sigma) \\
&\le |\operatorname{tr}(P_k^*\Delta)| + |\operatorname{tr}(\widehat{P}_k\Delta)|.
\end{aligned}$$

If P is a rank-k orthogonal projection with orthonormal basis $w_1, \dots, w_k$ for its range, then

$$|\operatorname{tr}(P\Delta)| = \left|\sum_{j=1}^{k} w_j^T \Delta w_j\right| \le k\|\Delta\|_{\mathrm{op}}.$$

Apply this bound to both projections. □

Combining this lemma with Definition 11.11 gives the following finite-sample guarantee.

Theorem 11.14 (PCA excess-risk bound)**.** *Suppose the hypotheses of Definition 11.11 hold. Then, for every $\epsilon > 0$,*

$$\mathbb{P}(\mathcal{E}_k > \epsilon) \le 2d \exp\left(-\frac{n\epsilon^2}{8Ck^2(\|\Sigma\|_{\mathrm{op}} + \epsilon/(6k))}\right).$$

Proof. The deterministic lemma shows that

$$\{\mathcal{E}_k > \epsilon\} \subseteq \left\{\|\widehat{\Sigma}_n - \Sigma\|_{\mathrm{op}} > \frac{\epsilon}{2k}\right\}.$$

Substitute $t = \epsilon/(2k)$ in Definition 11.11. □

11.8 Bibliographic notes

The presentation of SVD and dimensionality reduction is based in part on [BlHo]. The covariance concentration bound follows from matrix Bernstein inequalities; see [WW, Corollary 6.20]. For sharper results on PCA reconstruction error, see [ReWa].

11.9 Exercises

Exercise 11.15 (A random-sign projection). *Let $v \in \mathbb{R}^d$ be fixed and let $F(v) = k^{-1/2}Uv$, where the entries of the $k \times d$ matrix U are independent and uniform on $\{-1, 1\}$.*

1. *Prove directly that $\mathbb{E}[|F(v)|^2] = |v|^2$.*
2. *Use NumPy to generate $n = 200$ points in $\mathbb{R}^{100}$ once, and then regard them as fixed. For $k = 5, 20, 50$, draw one matrix U and compute the relative error of every nonzero pairwise distance. Use Matplotlib to show the three error distributions.*
3. *Repeat the experiment with a separately drawn matrix for every point. Explain why this second procedure is not the common linear map required by Definition 11.2.*

Exercise 11.16. *Use (11.1) and orthonormality to prove that $A^T u_j = \sigma_j v_j$ for $1 \leq j \leq r$. Conclude that u_j is an eigenvector of AA^T with eigenvalue σ_j^2.*

Exercise 11.17 (A rank-two example). *Let*

$$A = \begin{bmatrix} 3 & 0 \\ 0 & 2 \\ 0 & 0 \end{bmatrix}.$$

Find a reduced SVD of A. For $k = 1, 2$, compute the reconstruction A_k and verify the tail-sum formula in Definition 11.5. Describe all one-dimensional best-fitting subspaces if the upper-left entry of A is changed from 3 to 2.

Exercise 11.18 (Power-method diagnostics). *Construct a symmetric positive semidefinite matrix with eigenvalues $1, 0.8, 0.2$. Implement the power method with NumPy and plot, against the iteration number, the sign-invariant error $\min\{|q_t - v_1|, |q_t + v_1|\}$. Repeat with eigenvalues $1, 0.99, 0.2$. Explain the effect of the eigenvalue gap, and give an initial vector for which the method fails to find the leading eigenspace.*

Exercise 11.19 (PCA of handwritten digits). *Use `sklearn.datasets.load_digits`. Ignore the target labels: PCA is fitted to the image features alone. Fit a full `sklearn.decomposition.PCA` model, as well as models with $k = 2, 10, 20$ components.*

1. *Plot the cumulative explained-variance ratio as a function of k.*
2. *Reconstruct the same ten images for each of the three choices and display the originals and reconstructions with Matplotlib.*
3. *Compute the squared Frobenius reconstruction error directly and compare it with the appropriate tail sum of squared singular values. State whether the implementation uses the $1/n$ or $1/(n-1)$ convention when it reports explained variance.*

Exercise 11.20 (Lloyd's algorithm and initialization). *For the six points in Definition 11.6, start instead with centres $(-2, 0)$ and $(-1, 0)$. Carry out the assignment and centre-update steps by hand until the assignment is fixed, using the tie rule in the chapter. Compare the final objective with the value $8/3$ obtained in the example. Does the descent property in Definition 11.9 imply that the two final objectives must agree?*

Exercise 11.21 (PCA followed by K-means). *Use `make_blobs` from `sklearn.datasets` to generate a reproducible data set with at least five features and three groups. Standardize the features using `StandardScaler` from `sklearn.preprocessing`. Compare `KMeans` on the standardized features with K-means after retaining two components with `PCA`. Use several initializations for each fit. Plot the two-dimensional PCA scores, coloured by the fitted assignments, and report the objective in the space in which each model was fitted. Explain why objectives computed in different spaces are not directly comparable.*

Exercise 11.22 (Advanced: covariance perturbation). *Let Σ and E satisfy the hypotheses of Definition 11.12. Use NumPy to generate several symmetric perturbations E and verify the eigenvalue bound numerically. Then construct a two-dimensional example with a repeated leading eigenvalue in which $\|E\|_{\mathrm{op}}$ is small but a selected leading eigenvector changes substantially. Explain why this does not contradict Weyl's theorem.*

Chapter 12

Complexity and a priori bounds∗

Course guide

Scheduled lecture(s): Optional reference material
Required for those lectures: None
Prerequisites: Chapter 2, Section 2.6.5; Chapter 3, Section 3.1; Chapter 4, Section 4.1; Chapter 9, Section 9.1; and basic linear algebra
Optional or advanced: This entire chapter is optional and is not used in the current version of the course.

This optional chapter develops guarantees that do not use an independent test set. The first section treats a finite collection of classifiers by Hoeffding's inequality and the union bound. The later sections introduce the growth function and VC dimension. They are advanced and are included as reference material. The presentation follows [PTPR, SLT].

12.1 Optional: finite classifier classes

Let (X, Y) be an observation with $X \in \mathbb{R}^d$ and $Y \in \{0, 1\}$. For a fixed classifier $\phi : \mathbb{R}^d \to \{0, 1\}$, the 0–1 risk and its empirical version are

$$R(\phi) := \mathbb{P}(Y \neq \phi(X)), \qquad \widehat{R}_n(\phi) := \frac{1}{n}\sum_{i=1}^{n} \mathbb{1}_{\{Y_i \neq \phi(X_i)\}}.$$

Here $(X_1, Y_1), \ldots, (X_n, Y_n)$ are independent copies of (X, Y). We assume that $\phi(X)$ is a random variable, so every event in the display is defined.

An affine classifier has the form

$$\phi_{w,b}(x) = \begin{cases} 1, & w^T x + b > 0, \\ 0, & w^T x + b \leq 0, \end{cases} \qquad w \in \mathbb{R}^d, \quad b \in \mathbb{R}.$$

The collection of all affine classifiers is infinite. We first consider a finite collection because the union bound then controls every candidate at once.

For a concrete example, take $d = 1$ and fix the three threshold rules

$$\phi_t(x) = \mathbb{1}_{\{x>t\}}, \qquad t \in \{0, 1, 2\}.$$

After observing the training data, select a rule with the smallest empirical risk. Each rule is fixed before the data are observed, even though the selected rule is random.

Let

$$\mathcal{M} = \{\phi_1, \dots, \phi_N\}$$

be a finite collection fixed before the training sample is observed. Define $\widehat{\phi}$ to be the classifier with the smallest index among those that minimize empirical risk:

$$\widehat{R}_n(\widehat{\phi}) = \min_{\phi \in \mathcal{M}} \widehat{R}_n(\phi). \tag{12.1}$$

The tie rule makes $\widehat{\phi}$ unambiguous.

Theorem 12.1 (Generalization for a finite classifier class)**.** *Let $n, N \in \mathbb{N}$ with $n, N \geq 1$. Suppose that $(X_1, Y_1), \dots, (X_n, Y_n)$ are independent copies of (X, Y), where $Y \in \{0, 1\}$. Let $\mathcal{M} = \{\phi_1, \dots, \phi_N\}$ be fixed before these data are observed, and assume that every $\phi_j(X)$ is a random variable. If $\widehat{\phi}$ is defined by (12.1), then, for every $\epsilon > 0$,*

$$\mathbb{P}\left(R(\widehat{\phi}) > \min_{\phi \in \mathcal{M}} R(\phi) + \epsilon\right) \leq 2Ne^{-n\epsilon^2/2}.$$

Proof. Since $\mathcal{M}$ is finite, choose $\phi^* \in \mathcal{M}$ satisfying

$$R(\phi^*) = \min_{\phi \in \mathcal{M}} R(\phi).$$

On the event

$$\max_{1 \leq j \leq N} \left| \widehat{R}_n(\phi_j) - R(\phi_j) \right| \leq \frac{\epsilon}{2},$$

the definition of $\widehat{\phi}$ gives

$$\begin{aligned} R(\widehat{\phi}) &\leq \widehat{R}_n(\widehat{\phi}) + \frac{\epsilon}{2} \\ &\leq \widehat{R}_n(\phi^*) + \frac{\epsilon}{2} \\ &\leq R(\phi^*) + \epsilon. \end{aligned}$$

Therefore,

$$\begin{aligned} &\left\{ R(\widehat{\phi}) > R(\phi^*) + \epsilon \right\} \\ &\quad \subseteq \bigcup_{j=1}^{N} \left\{ \left| \widehat{R}_n(\phi_j) - R(\phi_j) \right| > \frac{\epsilon}{2} \right\}. \end{aligned} \tag{12.2}$$

For a fixed j, the losses $\mathbb{1}_{\{Y_i \neq \phi_j(X_i)\}}$, $i = 1, \dots, n$, are independent and belong to $[0, 1]$. Hoeffding's inequality gives

$$\mathbb{P}\left(\left| \widehat{R}_n(\phi_j) - R(\phi_j) \right| > \frac{\epsilon}{2} \right) \leq 2e^{-n\epsilon^2/2}. \tag{12.3}$$

The union bound in (12.2) now yields

$$\mathbb{P}\left(R(\widehat{\phi}) > R(\phi^*) + \epsilon \right) \leq 2Ne^{-n\epsilon^2/2}. \tag{12.4}$$

$\square$

Equivalently, for every $\alpha \in (0,1)$, with probability at least $1-\alpha$,

$$R(\widehat{\phi}) \le \min_{\phi\in\mathcal{M}} R(\phi) + \sqrt{\frac{2\log(2N/\alpha)}{n}}.$$

This is an **a priori bound**. It is available without reserving a test set, but it compares the selected rule only with the best rule in the chosen class. The term involving N is the price of selecting among several candidates.

The assumption that $\mathcal{M}$ is fixed before training is essential to this proof. If the same observations are used to construct a new list of candidates, the fixed-class bounds in (12.3) need not apply. An independent test set can evaluate an arbitrary fitted classifier as in Chapter 9. VC theory instead controls certain infinite classes by restricting the label patterns they can produce.

12.2 Optional: empirical measures and uniform deviation

Let $Z_1,\dots,Z_n$ be independent copies of a random variable Z taking values in a set $\mathcal{Z}$. If $A \subseteq \mathcal{Z}$ is an event, define its **empirical measure** by

$$\nu_n(A) := \frac{1}{n}\sum_{i=1}^n \mathbb{1}_{\{Z_i\in A\}}, \qquad \nu(A) := \mathbb{P}(Z\in A).$$

Thus $\nu_n(A)$ is the observed fraction of the sample in A, while $\nu(A)$ is the probability assigned to A by the model.

For a nonempty class $\mathcal{M}$ of classifiers, write

$$\Delta_n(\mathcal{M}) := \sup_{\phi\in\mathcal{M}} |\widehat{R}_n(\phi) - R(\phi)|.$$

Lemma 12.2 (Empirical risk minimization)**.** *Suppose that an empirical risk minimizer $\widehat{\phi}\in\mathcal{M}$ exists. Then*

$$R(\widehat{\phi}) - \inf_{\phi\in\mathcal{M}} R(\phi) \le 2\Delta_n(\mathcal{M}).$$

Proof. For every $\eta>0$, choose $\phi_\eta\in\mathcal{M}$ such that

$$R(\phi_\eta) \le \inf_{\phi\in\mathcal{M}} R(\phi) + \eta.$$

By the definition of $\Delta_n(\mathcal{M})$ and the minimizing property of $\widehat{\phi}$,

$$\begin{aligned} R(\widehat{\phi}) &\le \widehat{R}_n(\widehat{\phi}) + \Delta_n(\mathcal{M}) \\ &\le \widehat{R}_n(\phi_\eta) + \Delta_n(\mathcal{M}) \\ &\le R(\phi_\eta) + 2\Delta_n(\mathcal{M}). \end{aligned}$$

The result follows by letting η decrease to zero. □

The lemma is deterministic: it holds for every realized data set. A probability bound for $\Delta_n(\mathcal{M})$ therefore gives an excess-risk bound for empirical risk minimization.

12.3 Optional: VC theory

For a classifier $\phi : \mathbb{R}^m \to \{0,1\}$, define its error set in the observation space $\mathcal{Z} = \mathbb{R}^m \times \{0,1\}$ by

$$A_\phi := \{(x,y) \in \mathbb{R}^m \times \{0,1\} : y \neq \phi(x)\}, \qquad \mathcal{A} := \{A_\phi : \phi \in \mathcal{M}\}. \tag{12.5}$$

Every set A_ϕ is assumed to be an event. With $Z_i = (X_i, Y_i)$, we have

$$R(\phi) = \nu(A_\phi), \qquad \widehat{R}_n(\phi) = \nu_n(A_\phi).$$

Consequently,

$$\mathbb{P}\left(\sup_{\phi\in\mathcal{M}} |\widehat{R}_n(\phi) - R(\phi)| > \epsilon\right) = \mathbb{P}\left(\sup_{A\in\mathcal{A}} |\nu_n(A) - \nu(A)| > \epsilon\right), \tag{12.6}$$

whenever the two suprema are random variables.

If $\mathcal{A}$ is finite, Hoeffding's inequality and the union bound give

$$\mathbb{P}\left(\sup_{A\in\mathcal{A}} |\nu_n(A) - \nu(A)| > \epsilon\right) \le 2|\mathcal{A}|e^{-2n\epsilon^2}.$$

For an infinite class, its cardinality is not useful. The growth function below counts only the subsets that the class can select from a finite set of points.

Definition 12.3 (Trace and growth function)**.** *Let $\mathcal{A}$ be a nonempty collection of subsets of $\mathcal{Z}$. For a finite set $S \subseteq \mathcal{Z}$, the* **trace** *of $\mathcal{A}$ on S is*

$$\mathcal{A}|_S := \{A \cap S : A \in \mathcal{A}\}.$$

Set $s(\mathcal{A}, 0) := 1$. For $k \in \mathbb{N}$, the **growth function***, or shattering number, is*

$$s(\mathcal{A}, k) := \max_{z_1,\dots,z_k\in\mathcal{Z}} \#\left(\mathcal{A}|_{\{z_1,\dots,z_k\}}\right).$$

In particular, $1 \le s(\mathcal{A}, k) \le 2^k$.

Definition 12.4 (Shattering)**.** *The class $\mathcal{A}$* **shatters** *a finite set S if*

$$\mathcal{A}|_S = 2^S,$$

where 2^S is the collection of all subsets of S.

Example 12.5 (Thresholds on the real line)**.** *Let*

$$\mathcal{H} := \{(t,\infty) : t \in \mathbb{R}\}.$$

For distinct points $x_1 < \cdots < x_k$, a threshold selects one of the $k+1$ suffixes

$$\emptyset,\ \{x_k\},\ \{x_{k-1}, x_k\}, \dots, \{x_1, \dots, x_k\}.$$

Hence $s(\mathcal{H}, k) = k + 1$. One point is shattered, but two points are not.

The classifier $\phi_t(x) = \mathbb{1}_{\{x>t\}}$ has error set

$$A_{\phi_t} = \big((-\infty, t] \times \{1\}\big) \cup \big((t,\infty) \times \{0\}\big).$$

On any labelled sample, changing from the prediction pattern to the error pattern only flips the coordinates whose observed label is 1. It does not change the number of patterns.

Definition 12.6 (VC dimension)**.** *The* **VC dimension** *of a nonempty set class $\mathcal{A}$ is*

$$\mathcal{V}_{\mathcal{A}} := \sup\{k \in \{0, 1, 2, \ldots\} : s(\mathcal{A}, k) = 2^k\}.$$

If sets of every finite size are shattered, then $\mathcal{V}_{\mathcal{A}} = \infty$.

Thus the threshold class in Definition 12.5 has VC dimension 1. The next result gives the dimension of the affine classifiers introduced in Section 12.1.

Theorem 12.7 (VC dimension of affine classifiers)**.** *Let $m \in \mathbb{N}$ with $m \geq 1$. The class of affine classifiers on $\mathbb{R}^m$ has VC dimension $m + 1$. Its associated class of error sets in $\mathbb{R}^m \times \{0, 1\}$ has the same VC dimension.*

Proof. Choose affinely independent points $x_1, \ldots, x_{m+1}$. The square matrix with rows $(x_i^T, 1)$ is invertible. For any signs $r_1, \ldots, r_{m+1} \in \{-1, 1\}$, there are $w \in \mathbb{R}^m$ and $b \in \mathbb{R}$ such that $w^T x_i + b = r_i$ for every i. Therefore these points are shattered.

Now take any $m + 2$ points. They are affinely dependent, so there are numbers $\lambda_1, \ldots, \lambda_{m+2}$, not all zero, such that

$$\sum_{i=1}^{m+2} \lambda_i = 0, \qquad \sum_{i=1}^{m+2} \lambda_i x_i = 0.$$

Some coefficients are positive and some are negative. Assign label 1 when $\lambda_i > 0$ and label 0 when $\lambda_i \leq 0$. If an affine function $f(x) = w^T x + b$ realized these labels, then

$$0 = \sum_{i=1}^{m+2} \lambda_i f(x_i) > 0,$$

because every term is nonnegative and the terms with positive coefficient are strictly positive. This is impossible. Hence no set of $m + 2$ points is shattered.

On a labelled sample whose feature values are distinct, the map from a prediction pattern to its error pattern flips a fixed collection of coordinates. This map is a bijection. Repeated feature values cannot increase the number of prediction patterns. Taking the maximum over samples therefore shows that the prediction class and its error-set class have the same growth function. □

The VC dimension controls the entire growth function.

Lemma 12.8 (Sauer–Shelah lemma)**.** *Let $\mathcal{A}$ have finite VC dimension d. Then, for every $N \in \mathbb{N}$,*

$$s(\mathcal{A}, N) \leq \sum_{i=0}^{\min\{d,N\}} \binom{N}{i}.$$

Proof. If $d = 0$, no one-point set is shattered. Thus all members of $\mathcal{A}$ agree at every point, so $s(\mathcal{A}, N) = 1$. We may therefore assume that $d \geq 1$.

The case $N = 0$ is immediate. We prove the bound by induction on N. Fix $S = \{z_1, \ldots, z_N\}$ and write $S' = S \setminus \{z_N\}$. Separate the traces on S according to whether they contain z_N. After removing z_N from the second group, let $\mathcal{F}_0$ and $\mathcal{F}_1$ be the two resulting families of subsets of S'. Then

$$\#(\mathcal{A}|_S) = |\mathcal{F}_0 \cup \mathcal{F}_1| + |\mathcal{F}_0 \cap \mathcal{F}_1|.$$

If $d \geq N$, the desired conclusion is the trivial bound 2^N. Suppose instead that $d < N$. The family $\mathcal{F}_0 \cup \mathcal{F}_1$ has VC dimension at most d. If $\mathcal{F}_0 \cap \mathcal{F}_1$ is empty, its cardinality contributes zero. Otherwise, this intersection has VC dimension at most $d-1$: if it shattered d points, the original trace family would shatter those points together with z_N. The induction hypothesis and Pascal's identity give

$$\begin{aligned}\#(\mathcal{A}|_S) &\leq \sum_{i=0}^{d} \binom{N-1}{i} + \sum_{i=0}^{d-1} \binom{N-1}{i} \\ &= \sum_{i=0}^{d} \binom{N}{i}.\end{aligned}$$

Taking the maximum over S proves the lemma. $\square$

Lemma 12.9 (Polynomial form of Sauer–Shelah). *If $1 \leq d \leq N$, then*

$$\sum_{i=0}^{d} \binom{N}{i} \leq \left(\frac{eN}{d}\right)^d.$$

Proof. First suppose $d \leq N/2$ and put $x = d/N$. For $0 \leq i \leq d$,

$$x^i(1-x)^{N-i} \geq x^d(1-x)^{N-d}.$$

The binomial theorem therefore gives

$$1 \geq x^d(1-x)^{N-d} \sum_{i=0}^{d} \binom{N}{i},$$

and hence

$$\sum_{i=0}^{d} \binom{N}{i} \leq \left(\frac{N}{d}\right)^d \left(1 + \frac{d}{N-d}\right)^{N-d} \leq \left(\frac{eN}{d}\right)^d.$$

If $N/2 < d \leq N$, the left-hand side is at most 2^N. The function $u \mapsto u(1 - \log u)$ is increasing on $[1/2, 1]$, and its value at $1/2$ exceeds $\log 2$. With $u = d/N$, this gives $2^N \leq (eN/d)^d$. $\square$

We next prove the probability bound. To keep the argument within the stated prerequisites, assume in the remainder of this section that $\mathcal{A}$ is finite or countably infinite and enumerate it if necessary. Then every supremum below is a random variable. General uncountable classes require an additional measurability or separability argument, which is not part of this course.

Let $Z_1', \ldots, Z_n'$ be independent copies of $Z_1, \ldots, Z_n$, independent of the original sample, and let ν_n' be their empirical measure. This second sample is called a ghost sample.

Lemma 12.10 (Symmetrization). *Let $\epsilon > 0$. If $n\epsilon^2 \geq 2$, then*

$$\mathbb{P}\left(\sup_{A \in \mathcal{A}} |\nu_n(A) - \nu(A)| > \epsilon\right) \leq 2\,\mathbb{P}\left(\sup_{A \in \mathcal{A}} |\nu_n(A) - \nu_n'(A)| > \frac{\epsilon}{2}\right).$$

Proof. On the event on the left, choose the first set A^* in the enumeration satisfying $|\nu_n(A^*) - \nu(A^*)| > \epsilon$; outside this event, choose the first set in the enumeration. Conditional on the original sample, A^* is fixed. Since $\nu_n'(A^*)$ is the average of n independent Bernoulli random variables,

$$\mathbb{V}(\nu_n'(A^*)) = \frac{\nu(A^*)(1-\nu(A^*))}{n} \le \frac{1}{4n}.$$

Chebyshev's inequality and $n\epsilon^2 \ge 2$ give

$$\mathbb{P}\left(|\nu_n'(A^*) - \nu(A^*)| < \frac{\epsilon}{2} \;\middle|\; Z_1, \dots, Z_n\right) \ge 1 - \frac{1}{n\epsilon^2} \ge \frac{1}{2}.$$

When both displayed deviations occur, the triangle inequality gives $|\nu_n(A^*) - \nu_n'(A^*)| > \epsilon/2$. Taking expectations proves the result. $\square$

Lemma 12.11 (Conditional Rademacher bound)**.** *Let $a_1, \dots, a_n \in [-1, 1]$ be fixed, and let $\sigma_1, \dots, \sigma_n$ be independent random variables satisfying $\mathbb{P}(\sigma_i = 1) = \mathbb{P}(\sigma_i = -1) = 1/2$. Then, for every $\epsilon > 0$,*

$$\mathbb{P}\left(\left|\frac{1}{n}\sum_{i=1}^n \sigma_i a_i\right| > \frac{\epsilon}{2}\right) \le 2e^{-n\epsilon^2/8}.$$

Proof. The variables $\sigma_i a_i$ are independent, have mean zero, and belong to $[-1, 1]$. The two one-sided forms of Hoeffding's inequality give the stated bound. $\square$

Lemma 12.12 (Random-sign symmetrization)**.** *For every $\epsilon > 0$,*

$$\mathbb{P}\left(\sup_{A \in \mathcal{A}} |\nu_n(A) - \nu_n'(A)| > \frac{\epsilon}{2}\right) \le 2s(\mathcal{A}, 2n)e^{-n\epsilon^2/8}.$$

Proof. Independently for each i, either keep the pair (Z_i, Z_i') in its original order or swap it. Because the two observations in every pair have the same law, this does not change the joint distribution. If σ_i records the choice, then $\nu_n(A) - \nu_n'(A)$ has the same distribution as

$$\frac{1}{n}\sum_{i=1}^n \sigma_i \big(\mathbb{1}_{\{Z_i \in A\}} - \mathbb{1}_{\{Z_i' \in A\}}\big).$$

Conditional on the $2n$ observations, the vectors of indicator values as A varies have at most $s(\mathcal{A}, 2n)$ distinct values. Apply the union bound and Definition 12.11 to these values, and then remove the conditioning. $\square$

Theorem 12.13 (VC uniform-deviation bound)**.** *Let $n \in \mathbb{N}$ with $n \ge 1$, and let $Z_1, \dots, Z_n$ be independent copies of Z. Let $\mathcal{A}$ be a finite or countably infinite collection of events. If $\epsilon > 0$ and $n\epsilon^2 \ge 2$, then*

$$\mathbb{P}\left(\sup_{A \in \mathcal{A}} |\nu_n(A) - \nu(A)| > \epsilon\right) \le 4s(\mathcal{A}, 2n)e^{-n\epsilon^2/8}.$$

Proof. Combine Definitions 12.10 and 12.12. $\square$

Corollary 12.14 (VC generalization for classifiers)**.** *Let $\mathcal{M}$ be a finite or countably infinite class of classifiers, and let $\mathcal{A}$ be its error-set class from* (12.5)*. Suppose that every $A \in \mathcal{A}$ is an event. If $n\epsilon^2 \geq 2$, then*

$$\mathbb{P}\left(\sup_{\phi \in \mathcal{M}} |\widehat{R}_n(\phi) - R(\phi)| > \epsilon\right) \leq 4s(\mathcal{A}, 2n)e^{-n\epsilon^2/8}.$$

If $\mathcal{A}$ has VC dimension d and $1 \leq d \leq 2n$, the right-hand side is at most

$$4\left(\frac{2en}{d}\right)^d e^{-n\epsilon^2/8}.$$

If an empirical risk minimizer $\widehat{\phi}$ exists, the same two bounds control

$$\mathbb{P}\left(R(\widehat{\phi}) - \inf_{\phi \in \mathcal{M}} R(\phi) > 2\epsilon\right).$$

Proof. The first claim follows from (12.6) and Definition 12.13. Apply Definitions 12.8 and 12.9 with $N = 2n$ to obtain the polynomial bound. The last claim follows from Definition 12.2. □

For an implementation using finitely many representable parameter values, the corollary applies directly. Applying it to the uncountable class of all affine parameters requires the additional measurability step noted before Definition 12.10. This qualification does not change the geometric statement in Definition 12.7.

12.4 Optional: concentration of the minimum training error

Uniform convergence controls the difference between empirical and population risk. A different question is whether the minimum training error is close to its own expectation. The following bounded-differences result answers this question.

Theorem 12.15 (McDiarmid's inequality)**.** *Let $n \in \mathbb{N}$ with $n \geq 1$. Let $W_1, \ldots, W_n$ be independent random variables in one of the discrete or joint-density settings of Chapter 2. Let $F = f(W_1, \ldots, W_n)$ be an integrable random variable. Suppose that there are constants $c_1, \ldots, c_n \geq 0$, with at least one $c_i > 0$, such that for every i and every two possible input tuples that differ only in coordinate i, the corresponding values of f differ by at most c_i. Then, for every $\epsilon > 0$,*

$$\mathbb{P}(|F - \mathbb{E}[F]| > \epsilon) \leq 2\exp\left(-\frac{2\epsilon^2}{\sum_{i=1}^n c_i^2}\right).$$

The discrete-or-density assumption keeps the conditional expectations in the proof within the framework of Chapter 2. The standard measure-theoretic version holds more generally.

Proof. Define

$$M_i := \mathbb{E}[F \mid W_1, \ldots, W_i], \qquad i = 0, \ldots, n,$$

with $M_0 = \mathbb{E}[F]$ and $M_n = F$. Put $D_i = M_i - M_{i-1}$. After $W_1, \ldots, W_{i-1}$ are fixed, averaging over the future coordinates gives a function $g_i(W_i)$ whose range has length at most c_i. Therefore $D_i = g_i(W_i) - \mathbb{E}[g_i(W_i) \mid W_1, \dots, W_{i-1}]$ has conditional mean zero and lies in an interval of length at most c_i.

The proof of Hoeffding's lemma, applied after the first $i-1$ coordinates are fixed, gives

$$\mathbb{E}[e^{\lambda D_i} \mid W_1, \dots, W_{i-1}] \le e^{\lambda^2 c_i^2/8}.$$

Iterating the tower property yields

$$\mathbb{E}\left[e^{\lambda(F-\mathbb{E}[F])}\right] \le \exp\left(\frac{\lambda^2}{8}\sum_{i=1}^{n} c_i^2\right).$$

Markov's inequality and optimization over $\lambda > 0$ give the upper-tail bound. Apply the same argument to $-F$ and use the union bound. $\square$

Theorem 12.16 (Minimum training error)**.** *Let $(X_1, Y_1), \dots, (X_n, Y_n)$ be independent copies of (X, Y). Assume that each pair is in one of the discrete or joint-density settings required by Definition 12.15. Let $\mathcal{M}$ be any nonempty fixed class of classifiers for which*

$$T_n := \inf_{\phi \in \mathcal{M}} \widehat{R}_n(\phi)$$

is a random variable. Then, for every $\epsilon > 0$,

$$\mathbb{P}(|T_n - \mathbb{E}[T_n]| > \epsilon) \le 2e^{-2n\epsilon^2}.$$

Proof. Replacing one observation changes $\widehat{R}_n(\phi)$ by at most $1/n$ for every ϕ. Taking the infimum preserves this bound, so changing one observation changes T_n by at most $1/n$. Apply Definition 12.15 with $c_i = 1/n$ for every i. $\square$

This result concerns stability around $\mathbb{E}[T_n]$, not generalization. For every fixed ϕ, $\mathbb{E}[\widehat{R}_n(\phi)] = R(\phi)$, and hence

$$\mathbb{E}[T_n] = \mathbb{E}\left[\inf_{\phi \in \mathcal{M}} \widehat{R}_n(\phi)\right] \le \inf_{\phi \in \mathcal{M}} R(\phi).$$

The inequality can be strict because the same sample is used to select the classifier and compute its training error. Concentration of T_n around its expectation does not by itself show that T_n is close to the best population risk. For that conclusion one needs a uniform-deviation result such as Definition 12.14.

12.5 Optional: bounded losses on a finite grid

The preceding classification results use losses with values 0 and 1. The same set argument extends without integration when a loss takes finitely many values. Let $K \in \mathbb{N}$ with $K \ge 1$, let $Z_i = (X_i, Y_i)$ be independent copies of $Z = (X, Y)$, and suppose

$$\ell(y, g(x)) \in \left\{0, \frac{1}{K}, \dots, \frac{K-1}{K}, 1\right\}, \qquad g \in \mathcal{M}.$$

Assume that $\ell(Y, g(X))$ is a random variable for every $g \in \mathcal{M}$. For $g \in \mathcal{M}$, define

$$R(g) := \mathbb{E}[\ell(Y, g(X))], \qquad \widehat{R}_n(g) := \frac{1}{n}\sum_{i=1}^{n} \ell(Y_i, g(X_i)).$$

The uniform deviation is

$$\sup_{g\in\mathcal{M}} |\widehat{R}_n(g) - R(g)|. \tag{12.7}$$

Associate with the loss class the threshold sets

$$\mathcal{A} := \{\{(x,y) : \ell(y,g(x)) \geq r/K\} : g \in \mathcal{M},\ r = 1,\dots,K\}. \tag{12.8}$$

Write

$$A_{g,r} := \{(x,y) : \ell(y,g(x)) \geq r/K\}.$$

For every allowed loss value,

$$\ell(y,g(x)) = \frac{1}{K}\sum_{r=1}^{K} \mathbb{1}_{\{\ell(y,g(x))\geq r/K\}}.$$

Therefore, for each g,

$$\begin{aligned}|\widehat{R}_n(g) - R(g)| &\leq \frac{1}{K}\sum_{r=1}^{K} |\nu_n(A_{g,r}) - \nu(A_{g,r})| \\ &\leq \sup_{A\in\mathcal{A}} |\nu_n(A) - \nu(A)|.\end{aligned}$$

Taking the supremum over g shows that (12.7) is bounded by the same set-class deviation used in Section 12.3. Thus Definition 12.13 applies when the threshold-set class satisfies its hypotheses.

For losses taking every value in $[0,1]$, the finite sum is replaced by the layer-cake integral over all thresholds. That extension requires integration and additional measurability hypotheses, so it is not assumed in this chapter.

12.6 Bibliography

Finite-class empirical risk minimization is covered in [PTPR, Chapter 4]. VC theory and its extensions to real-valued losses are developed in [PTPR, SLT].

12.7 Exercises

Exercise 12.17 (Sample size for a finite class)**.** *Let $\mathcal{M} = \{\phi_1,\dots,\phi_N\}$ be fixed before training. Find a sufficient sample size n that makes the excess-risk term following Definition 12.1 at most ϵ with probability at least $1-\alpha$, where $\epsilon > 0$ and $0 < \alpha < 1$. Describe how the sufficient sample size changes when N is doubled.*

Exercise 12.18. *Verify the two identities before (12.6) directly from (12.5). Then prove the equality in (12.6) for a finite classifier class, without appealing to any measurability result for suprema.*

Exercise 12.19 (Threshold patterns)**.** *Let $\mathcal{H} = \{(t,\infty) : t \in \mathbb{R}\}$ as in Definition 12.5.*

1. *List all traces on a fixed set of three distinct real numbers and verify that the growth function equals 4 at $k = 3$.*
2. *Give a two-element set that is not shattered, and explain why no two-element set can be shattered.*

3. *Add the reverse thresholds* $(-\infty, t)$ *to the class. Determine the VC dimension of the enlarged class and justify both the lower and upper bounds.*

Exercise 12.20 (A conditional finite-class experiment)**.** *Use NumPy to generate independent observations from the following fixed model in* $\mathbb{R}^5$*. Let* $Y \sim \mathrm{Bernoulli}(1/2)$*, let* Z *be a spherical Gaussian independent of* Y*, and set*

$$X = (2Y - 1)(1, 1, 0, 0, 0)^T + Z.$$

Import `DecisionTreeClassifier` *from* `sklearn.tree`*. Predeclare a finite list of maximum depths and fix* `random_state`*. In each repetition, generate three independent samples: a construction sample, a selection sample of size* n*, and a large test sample. Before inspecting the selection sample, fit one tree of each predeclared depth to the construction sample. Conditional on that sample, the fitted classifiers form a fixed finite class.*

1. *Select the classifier with the smallest error on the selection sample and use the test sample to approximate the risk of every candidate.*

2. *Fix* $0 < \alpha < 1$ *and compute the corresponding excess-risk threshold following Definition 12.1. Repeat the whole experiment many times and use Matplotlib to display the approximate excess risks. Compare the fraction exceeding the threshold with* α*, and explain the extra approximation caused by using a finite test sample for the population risks.*

3. *Explain why fitting or tuning the candidates on the selection sample would violate the fixed-class hypothesis used by that bound.*

Exercise 12.21 (Inspecting a VC bound)**.** *Fix* $0 < \alpha < 1$ *and a positive integer* d*. Use NumPy to evaluate the upper bound in Definition 12.14 over a range of* n *and* ϵ*. Plot with Matplotlib the smallest* ϵ *for which the bound is at most* α*, restricting attention to values satisfying* $n\epsilon^2 \geq 2$ *and* $d \leq 2n$*. Explain why replacing the probability bound by its minimum with* 1 *does not change the theorem but improves the plot.*

Exercise 12.22 (A bounded grid of losses)**.** *Prove the indicator decomposition used after* (12.8) *for every value in* $\{0, 1/K, \ldots, 1\}$*. Derive the displayed uniform-deviation inequality, keeping the supremum over* g *outside until the final step. Identify the event and countability hypotheses needed before applying Definition 12.13.*

Chapter 13

Group Assignments

Course guide

Scheduled lecture(s): Group assessment after the corresponding lectures
Required for those lectures: All three assignment sections
Prerequisites: Assignment 1: Chapters 1–3; Assignment 2: Chapters 2 and 4–7; Assignment 3: Chapters 7 and 9–11
Optional or advanced: No new theory is introduced. The computational parts use NumPy, Matplotlib, and, where stated, scikit-learn.

This chapter is itself a collection of assignments, so it does not have a separate exercises section. For every theoretical problem, state the result used and show the main steps. For every simulation, use a fixed random-number generator seed, report the number of repetitions, and distinguish an empirical observation from a mathematical guarantee.

13.1 Group Assignment 1

1. Let A and B be independent events. Prove that A^c and B^c are independent.
2. A family has exactly three children. Suppose that each child has brown hair with probability $1/4$ and that the three hair-colour indicators are mutually independent.
   (a) Given that at least one child has brown hair, what is the probability that at least two children have brown hair?
   (b) Given that the oldest child has brown hair, what is the probability that at least two children have brown hair?

   Explain why the two conditioning events lead to different answers.
3. Let (X, Y) have joint density $1/\pi$ on
$$D := \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 \leq 1\}$$
   and density 0 outside D. Set $R = \sqrt{X^2 + Y^2}$. Find the distribution function and density of R, including their values outside the interval $[0, 1]$.
4. Repeatedly toss a fair coin, and assume that the tosses are mutually independent. Stop when Heads first appears, and let X be the number of tosses used. Find the probability mass function and expected value of X.

5. Let $n \geq 1$, let $0 < \alpha < 1$, and let $X_1, \ldots, X_n$ be i.i.d. Bernoulli(p) random variables, where $p \in [0, 1]$. Define

$$\widehat{p}_n := \frac{1}{n}\sum_{i=1}^{n} X_i, \qquad \varepsilon_n := \sqrt{\frac{1}{2n}\log\left(\frac{2}{\alpha}\right)}, \qquad I_n := [\widehat{p}_n - \varepsilon_n, \widehat{p}_n + \varepsilon_n].$$

   (a) Use Hoeffding's inequality to prove that $\mathbb{P}(p \in I_n) \geq 1 - \alpha$. Explain why intersecting I_n with the parameter space $[0, 1]$ does not change the coverage event.

   (b) Take $\alpha = 0.05$ and $p = 0.4$. For $n = 10, 100, 1000, 10000$, use NumPy to generate at least 10,000 independent data sets and estimate $\mathbb{P}(p \in I_n)$. Plot the estimated coverage against n with Matplotlib, together with a horizontal line at 0.95.

   (c) Plot the length $2\varepsilon_n$ against n. If you truncate the interval to $[0, 1]$, also plot its simulated mean length and explain why that length is random.

   (d) Suppose the data-generating parameter changes from $p_0 = 0.4$ to $p_1 = 0.5$. Generate the data using p_1. For the same four sample sizes, estimate both $\mathbb{P}(p_1 \in I_n)$ and $\mathbb{P}(p_0 \in I_n)$. The second probability is the probability that the interval still contains the obsolete value. Explain why this second probability tends to zero as n increases, whereas the coverage of the new true value remains at least the guaranteed level.

13.2 Group Assignment 2

1. Let $(x_1, y_1), \ldots, (x_n, y_n)$ be observed training pairs with $x_i \in \mathbb{R}^d$ and $y_i \in \{0, 1, 2, \ldots\}$. Suppose that, conditional on the feature vectors, the responses are independent and

$$\mathbb{P}(Y_i = y_i \mid X_i = x_i) = \frac{\lambda(x_i)^{y_i} e^{-\lambda(x_i)}}{y_i!}, \qquad \lambda(x) := \exp(\alpha^T x + \beta),$$

   where $\alpha \in \mathbb{R}^d$ and $\beta \in \mathbb{R}$. Follow the calculation in Section 4.2.1 to derive a loss whose minimizers, when they exist, are maximum likelihood estimators of (α, β). State which terms may be removed from the loss and why.

2. Let $n \geq 1$, let $\theta > 0$, and let $X_1, \ldots, X_n$ be i.i.d. Uniform($[0, \theta]$) random variables. Define $\widehat{\theta} = \max\{X_1, \ldots, X_n\}$. Find the distribution function of $\widehat{\theta}$, including the cases $x < 0$ and $x \geq \theta$. Then compute

$$\mathsf{bias}_\theta(\widehat{\theta}), \qquad \mathsf{se}_\theta(\widehat{\theta}), \qquad \mathsf{MSE}_\theta(\widehat{\theta}).$$

3. Consider the density

$$f(x) = \frac{1}{2}\cos x, \qquad -\frac{\pi}{2} < x < \frac{\pi}{2},$$

   with $f(x) = 0$ outside this interval.

   (a) Find the distribution function F on all of $\mathbb{R}$.

   (b) Find the quantile function $F^{-1}(u)$ for $0 < u < 1$.

(c) For accept–reject sampling, use the density of Uniform($[-\pi/2, \pi/2]$) as the proposal. Find the smallest constant M for which $f(x) \leq Mg(x)$ for every x, and find the resulting acceptance probability.

(d) Implement both the inversion sampler and the accept–reject sampler with NumPy. Compare histograms of their outputs with the density f using Matplotlib, and compare the empirical acceptance rate with its theoretical value.

4. Let $Y_1, Y_2, \ldots$ be i.i.d. random variables with

$$\mathbb{P}(Y_i = 0) = 0.1, \quad \mathbb{P}(Y_i = 1) = 0.3, \quad \mathbb{P}(Y_i = 2) = 0.2, \quad \mathbb{P}(Y_i = 3) = 0.4.$$

Set $X_0 = 0$ and $X_n = \max\{Y_1, \ldots, Y_n\}$ for $n \geq 1$. Prove that $(X_n)_{n \geq 0}$ is a time-homogeneous Markov chain on $\{0, 1, 2, 3\}$. Find its transition matrix, with rows and columns ordered as $0, 1, 2, 3$.

5. Let $X_1, \ldots, X_n$ be i.i.d. with an unknown distribution function F. For $0 < p < 1$, define the population quantile and its empirical counterpart by

$$q_p := \inf\{x \in \mathbb{R} : F(x) \geq p\}, \qquad \widehat{F}_n^{-1}(u) := \inf\{x \in \mathbb{R} : \widehat{F}_n(x) \geq u\}.$$

Fix $0 < \alpha < 1$ and set

$$\varepsilon_n := \sqrt{\frac{\log(2/\alpha)}{2n}}.$$

Assume that $\varepsilon_n < \min\{p, 1-p\}$. Use Definition 5.15 to prove that

$$\left[\widehat{F}_n^{-1}(p - \varepsilon_n), \widehat{F}_n^{-1}(p + \varepsilon_n)\right]$$

contains q_p with probability at least $1 - \alpha$. Express both endpoints in terms of order statistics. No continuity or density assumption on F may be used.

13.3 Group Assignment 3

1. Consider a Markov chain on $\{1, 2, 3\}$ with transition matrix

$$P = \begin{pmatrix} 0.5 & 0.5 & 0 \\ 0.5 & 0 & 0.5 \\ 0.5 & 0 & 0.5 \end{pmatrix}.$$

(a) Draw the transition diagram.

(b) Find the stationary distribution π and verify directly that $\pi P = \pi$.

(c) Given that the chain is in state 1 at time 1, find the probability that it is in state 2 at time 4.

(d) Starting in state 1, find the expected number of transitions until the first visit to state 3.

(e) Find the period of every state and justify the answer from the transition diagram.

2. Let $g : \mathbb{R}^d \to \{0, 1\}$ be a fixed classifier, trained without using the i.i.d. test observations $(X_1, Y_1), \ldots, (X_n, Y_n)$. Assume $\mathbb{P}(g(X) = 1) > 0$ and $\mathbb{P}(Y = 1) > 0$, and define

$$\begin{aligned} \text{precision}(g) &:= \mathbb{P}(Y = 1 \mid g(X) = 1), \\ \text{recall}(g) &:= \mathbb{P}(g(X) = 1 \mid Y = 1). \end{aligned}$$

   (a) Write the empirical precision and recall as ratios of sums of indicator variables. State when an empirical denominator can equal zero and how you would report that case.

   (b) Let $Y = 1$ mean that a battery has deteriorated. Running a confirmatory test after a false alarm costs $c_{\text{fp}} > 0$, while failing to test a deteriorated battery costs $c_{\text{fn}} > 0$. Define the per-battery cost random variable and show that its expected value is

$$c_{\text{fp}}\, \mathbb{P}(g(X) = 1)\big(1 - \text{precision}(g)\big) + c_{\text{fn}}\, \mathbb{P}(Y = 1)\big(1 - \text{recall}(g)\big).$$

   Explain why precision and recall alone do not determine this expected cost unless an additional base probability is known.

   (c) Derive a Hoeffding confidence interval for the expected cost from the independent test sample. For precision and recall, explain why the relevant sample sizes are the observed numbers of predicted and actual positives, respectively; compare with Definition 9.2.

   (d) Implement the empirical quantities in Python. Import `precision_score` and `recall_score` from `sklearn.metrics`, and use them to check your values on a small example. Do not silently replace an undefined ratio by zero.

3. Let $X = (X_1, \ldots, X_d)$ and $Y = (Y_1, \ldots, Y_d)$ be independent spherical Gaussian vectors; equivalently, all $2d$ coordinates are independent standard Gaussian random variables.

   (a) Show that $\mathbb{E}[X \cdot Y] = 0$ and $\mathbb{V}(X \cdot Y) = d$.

   (b) Use Chebyshev's inequality to prove that, for every $\epsilon > 0$,

$$\mathbb{P}\left(\frac{|X \cdot Y|}{d} \geq \epsilon\right) \leq \frac{1}{d\epsilon^2}.$$

   (c) Combine this result with concentration of $|X|/\sqrt{d}$ and $|Y|/\sqrt{d}$ to explain why the cosine $(X \cdot Y)/(|X||Y|)$ converges to 0 in probability. This is the precise sense in which the two vectors are nearly orthogonal.

   (d) For $d = 2, 10, 100, 1000$, simulate the cosine with NumPy and use Matplotlib to compare its empirical distributions.

4. Let $1 \leq r \leq n$, and let $u_1, \ldots, u_r \in \mathbb{R}^n$ be linearly independent unit vectors.

   (a) Prove that $u_i u_i^T$ has rank one. Find its null space and range.

   (b) Let

$$M := \sum_{i=1}^{r} u_i u_i^T.$$

   Prove that M has rank r.

(c) Must the vectors $u_1, \dots, u_r$ be right singular vectors of M? If not, give a two-dimensional counterexample.

(d) Suppose instead that $u_1, \dots, u_r$ are orthonormal. Determine all singular values of M, including their multiplicities. Explain in what sense the u_i may be chosen as singular vectors. When $r > 1$, explain why the repeated positive singular value means that an SVD algorithm need not return these particular vectors.

5. Fix an integer $d \geq 1$. Let $X \sim \text{Uniform}(B_1)$ in $\mathbb{R}^d$, in the sense of Definition 10.8, and define $Y = |X|$.

   (a) Find the distribution function and density of Y.

   (b) Find the distribution of $Z = \log(1/Y)$. The event $Y = 0$ has probability zero, so Z is defined with probability one.

   (c) Calculate $\mathbb{E}[\log(1/Y)]$ first from the density of Y and then from the distribution of Z.

   (d) Simulate Y using Definition 10.12 for several dimensions $d \geq 2$. Plot histograms of $\log(1/Y)$ and compare them with the density found in the previous part.

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Index